DOCTORAL THESIS



Light Scattering in Fiber-based Materials

A foundation for characterization of structural properties

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LULEÅ UNIVERSITY OF TECHNOLOGY

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A foundation for characterization of structural properties

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Abstract

Deeper knowledge of light propagation in fiber-based materials is fundamental in order to understand their optical appearance as well as for industrial applications. Light scattering measurements are appropriate in handling dynamic industrial environments and can provide information regarding structural properties. In general, on-line property measurements are best utilized by establishing an understanding of the underlying physics and using that knowledge in an optimal way to determine the parameters or properties sought after. Light scattering is affected by numerous parameters such as size, shape, concentration and refractive index of the scattering particles as well as the wavelength of the incident source. In addition, anisotropic light diffusion in media which have a directional-dependency, such as structured fiber-based materials, are neither well understood nor well investigated.

By approximating cellulose fibers as infinitely long, straigth cylinders it is possible to use an analytical solution to Maxwell's equations to describe the scattering characteristics such as phase functions and scattering efficiency. This makes it possible to utilize both the wave nature of light and structural properties of the fiber network when modelling multiple light scattering. The developed model solves the radiative transfer equation numerically using the Monte Carlo method resulting in a description of multiple scattering in a sphere-cylinder media.

The results show that scattering media consisting of infinite long, straigth, homogeneous or hollow cylinders scatter light very differently as compared with a media consisting of spherical particles. Both scattered intensity and the degree of depolarization are affected by a strong forward scattering behavior observed for cylindrical particles. This strong forward scattering behavior was also found to enhance lateral scattering in paper, and therefore predicts a larger extent of lateral light scattering than models using rotationally invariant single scattering phase functions. A strong relationship between anisotropic diffusion and to degree of in-plane fiber orientation was also observed using both measurements and simulations.

In conclusion, it was found that the approximation of cellulose fibers as infinitely long, straigth cylinders is reasonable when modelling scattering in paper. The findings indicate that parameters such as geometrical properties, particle composition, fiber orientation and fiber orientation variations can be measured by monitoring scattered light intensity. The obtained knowledge provides a base for further development of on-line sensing techniques that meet industrial requirements. Since the theory is general, it is likewise relevant and applicable to other areas of material science where imaging or remote sensing techniques are of interest.

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Tomas Linder
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Part I

Chapter 1

Thesis Introduction

1.1 Thesis Introduction

Paper products are an essential part of our daily lives, yet few know how large the industry behind them is. In 2012, the countries belonging to the Confederation of European Paper Industries (CEPI) employed over 180 000 people, produced close to a hundred million tons of paper, and had a turnover of 75 billion euros. The yearly consumption of paper in most European countries is about a few hundred kilos per person per year [1]. The paper industry in Sweden makes up for 30% of the pulp production and over 10% of the paper and board production in Europe making it one of the cornerstones in the Swedish economy [2]. Increasing demands on a highly competitive market forces manufacturers to seek ways to increase both production and quality as well as optimizing the energy consumption. On-line quality control is one of the key areas that is constantly being developed to meet these demands.

This thesis is devoted to light scattering in fiber-based materials such as paper and pulp. Light scattering is affected by numerous parameters such as size, shape, concentration and refractive index of the scattering particles as well as the wavelength of the incident source. Light scattering measurements have the benefit of being contact-free, making it suitable for on-line use in dynamic industrial environments, and can provide information regarding structural properties.

The strong scattering process in paper is directly related to the shape and structure of the fibers and fiber network. For example, an important parameter for papermaking is the fiber orientation in the finished paper-sheet. Light propagation in fiber-based materials, where the fibers are aligned, is known to have an anisotropic behavior. However, even though this is a known phenomenon it is not well investigated nor well understood. A way to better understand anisotropic diffusion and how it affect lateral light scattering in paper can be done using numerical modeling based on fundamental physical principles.

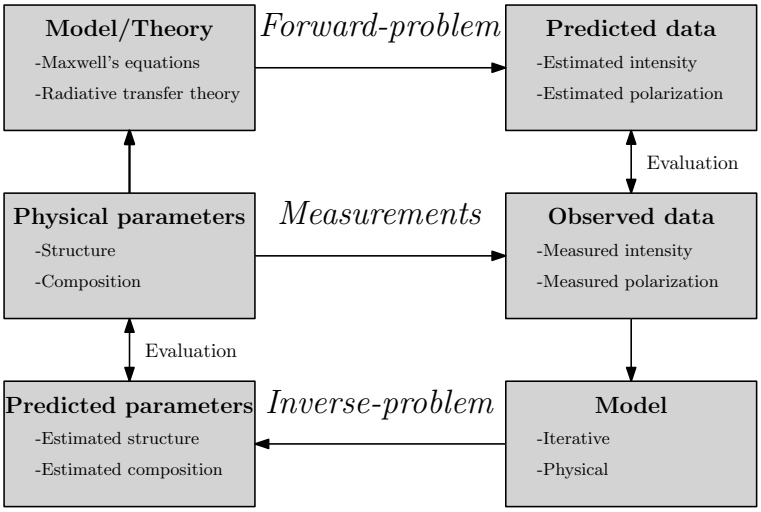


Figure 1.1: Overview of model-based approaches.

1.1.1 Measurement Science and Modeling

The fundamental requirement in any measurement setup is to have a physical quantity connected to measured data. The relation between the quantity sought for and the measured data is often of a very complex nature making this connection obscure. For light diffusion in paper this would correspond to a description of how shape, size, orientation and concentration of the fibers relate to transmitted and reflected intensity.

There are a few different ways to understand observations and its relation to a sampled parameter. A measurement-based method is when repeated measurements of a property are made as an estimation tool for future measurements. While this method works well and is easily implemented, there is a possibility that wrong conclusions can be made due to the lack of understanding of the underlying physics. It is also possible that a deeper understanding of the problem can provide knowledge on how to improve estimation techniques. A model-based approach is therefore an important tool that can be used for these purposes. A model does not necessarily need to be based on physics but so is often the case. Approximations are almost always required when modeling complex systems but they still provide deeper understanding of the nature of the problem. The first step using a model is to make an estimation of the physical parameters and then use these parameters in the model to obtain a prediction. The prediction can then be compared with the observed measurement data and an evaluation of the model can be done. This is often referred to as the *forward-problem* and it is illustrated at the top of Figure 1.1. However, the real task of interest for a measurement system is to convert the measured data into information about the sought-after sample property. The step from a measured quantity to the sample property is called the *inverse-problem*, and it is

illustrated at the bottom of Figure 1.1.

There are situations where the forward model is invertible and can relate to the physical quantity directly. However, this is rare if the problem is of a complex nature, and approximate methods and models usually are required. One possible method to address the inverse problem is to iteratively solve the forward-problem where the task is to find the best fit between the model and measurement data. This obviously requires a correct and robust forward model.

1.1.2 Thesis Objective

The work done in this thesis has focused on the forward-problem of how light scatter in fiber-based materials. The two main components regarding the theoretical modeling are analytical solutions to electromagnetic scattering by small particles and how these affect multiple scattering described by radiative transport theory.

In order to understand the light-matter interaction in a scattering medium a correct description of the scattering particles is required. Although a spherical approximation is used in many physical problems there exist many other situations where such an approximation is not valid. The geometrical shape of a cellulose fiber is one that clearly does not satisfy this criterion. A intuitive approximation would instead be to describe the fiber as a cylinder. This is convenient as infinitely long, straight, homogeneous cylinders and also a infinitely long, straight, coated cylinders are two of the handful of geometries where analytical solutions to Maxwell's equations can be derived. The first research question is therefore stated as:

- **Is it a valid approximation to simplify scattering by a cellulose fiber as an infinitely long, straight, homogeneous cylinder?**

Further, if scattering is assumed to reflect the properties of a infinitely long cylinder, then how will this affect multiple scattering? By solving the the radiative transfer equations numerically using the Monte Carlo method it is possible to model the influence of particle level light scattering. This means that both particle composition, and for example the influence from fiber orientation, can be considered. This in turn leads to two additional research questions:

- **Can a model using infinitely long cylinders as scatterers describe light diffusion in paper better than current models using rotationally-symmetric scattering?**
- **Can this model also be used to estimate particle compostion in pulp and anisotropic diffusion in paper?**

The ultimate aim of this work is to provide a foundation from which to address the inverse-problem. Hopefully this can lead to a more efficient measurement technique for estimating the fiber structure and fiber orientation in paper.

1.1.3 Thesis Outline

This thesis is composed of two parts. The first part consists of an overview of both the papermaking process and the physics behind light-matter interaction. It also describes the contributions of the scientific papers with a wider perspective and provide a base for better understanding the model used in the thesis. The second part consists of the five appended scientific papers. The thesis chapters can be summarized as follows:

Chapter 2 will provide an overview of fiber properties along with a description of the sheet forming stages that affect the structure of the finished paper-sheet. A discussion regarding the importance of fiber formation and fiber orientation along with an overview of different fiber orientation measurement techniques are also provided.

Chapter 3 provides the background physics of light-matter interaction such as scattering by small particles and radiative transfer theory. The purpose of this chapter is to provide a fundamental understanding of the interations in light scattering media.

Chapter 4 describes in detail how the radiative transfer equation is solved using the Monte Carlo method. It also describes the essential parts of how multiple scattering by cylinders affects light propagation.

Chapter 5 describes various aspects of light scattering in paper. The effects of lateral and anisotropic diffusion along with scattering in cellulose suspensions are discussed in this chapter.

Chapter 6 provides a summary of the appended papers and lists additional academic work not included in the thesis.

Chapter 7 contains conclusions and suggestions for future work.

Chapter 2

Papermaking and Fiber Properties

2.1 Papermaking

The word paper (or "papyrus") originated in Egypt some 5000 years ago when the first sheets of paper where created from the stems of the papyrus plant. The origin of the modern papermaking process, as we know it today, where the paper is created with individual fibers from the pulp rather than a lamination of plants as the Egyptians did, is said to have been invented in China by Ts'ai Lun roughly 2000 years ago [3]. The basic principle is to break down the raw material into a pulp with the addition of water, then to form it into a sheet, and subsequently to dry it. The papermaking process later spread through the middle east and reached Europe in the 12th century. The first papermaking machine was patented by Louis Robert in 1798 and from there papermaking transformed into a an industrial process. Over the last 200 years it has become more and more an automated process where a modern paper mill today is controlled by several hundreds of sensors and control loops.

Papermaking is a very complex process involving both mechanics and chemistry. As the majority of the work done in this thesis has been focused on determining the structure of the finished product, only small relevant parts are covered here, and further reading on the subject could be found, in for example, [1, 3].

2.2 Paper

Paper is a network of fibers where the strength of the structure originates from the physical properties of the fibers and the properties of the contact points between the fibers. Fibers can make hydrogen bonds between the surfaces, so they bond without the need of adhesives [1]. An illustration of what a fiber network inside a paper-sheet looks like is shown in Figure 2.1.

Depending on the purpose of the end product, the desired properties of the paper

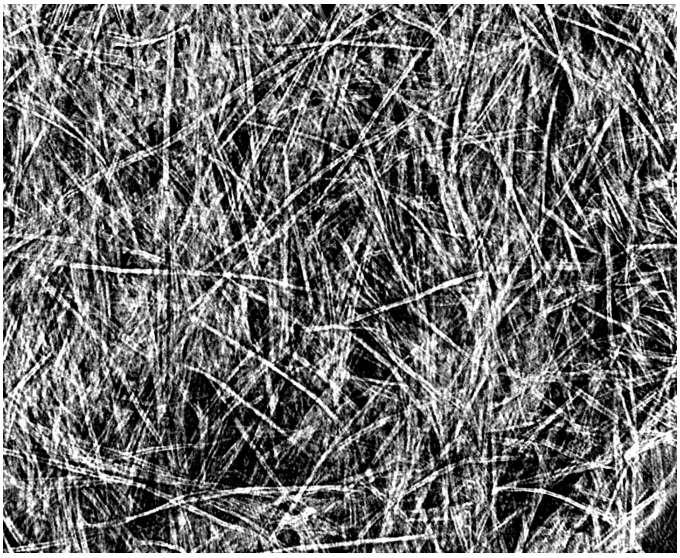


Figure 2.1: High resolution $3.0 \times 2.4\ mm^2$ image of the fiber network in a paper-sheet were fiber properties such as orientation, diameter and wall thickness are easily determined. The image is reconstructed from x-ray computed tomography, see **Paper D**.

are optimized during the papermaking process, see [4] for a review. For example, the paper this thesis is printed on has properties which are optimized for printing and visual appearance. Fluorescent agents, coatings and other filler materials are often added to make the appearance of the print more pleasant to the eye. Other products, such as tissues or diapers, are made to absorb moisture. Sheet strength is also important for a wide variety of paper products, especially in packaging. The strength of a paper-sheet depends heavily on an even distribution of fibers along with their orientation. However, other products, for example paper bags, would get weaker if their fibers were aligned along one specific direction.

In order to better understand the fiber structure in a paper-sheet the essential building block, the cellulose fiber, has to be considered.

2.2.1 Fiber

The cellulose fiber is the fiber that builds up plants and trees. It can vary quite a lot depending on its origin, but it is generally roughly cylindrical, with a hollow in the center. The hollow center of a cellulose fiber is called lumen and the surrounding outer

Table 2.1: Fiber dimensions for Swedish birch and pine [1].

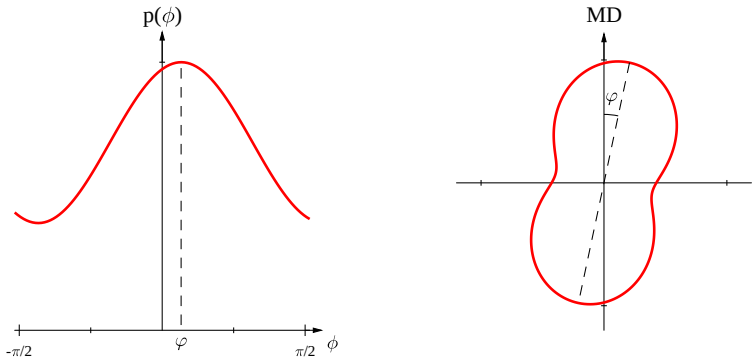
	Birch	Pine	
		Early-wood	Late-wood
Mean width (μm)	22	35	25
Lumen diameter (μm)	16	30	10
Mean length (mm)	1.1	2.7	3.0

edge is called fiber wall. The fiber wall in turn has a layered structure with different compositions and orientation of its smaller building block microfibrils or nanofibrils [5].

Paper is most often manufactured from wood fibers. However, in some parts of the world other raw materials are used, such as rice, straw, bamboo or cotton [1]. Fibers used in the Swedish paper industry usually originate from either softwood (spruce and pine) or hardwood (birch). Size parameters for Swedish pine and birch are shown in Table 2.1. Softwood fibers, which are longer than hardwood fibers, are very suitable for products which require high sheet strength such as kraftliner or newsprint. Birch fibers are often used to make fine papers for prints as short and thin fibers give a good and uniform distribution of fibers in the sheet. Fiber geometries not only vary depending on the species but also within species as the fiber wall thickness depends on season. Fibers from early-wood or spring-wood have a thin fiberwall, they are located in the paler parts of the growth rings in the trunk, and their main function is to transport water. Late-wood or summer-wood fibers, found in the darker parts of the growth rings, have the main function to give stability to the tree and thereof have a thicker fiberwall [3]. There are also differences in chemical composition between different cellulose fibers, but that is beyond the scope of this work and will not be discussed here.

There are two different methods in which fibers are separated: chemical and mechanical. The mechanical separation usually damages the fibers, decreases their strength, and creates small fiber fragments. Fibers from the chemically separated pulp are usually longer and undamaged, which makes them suitable for paper products that require high sheet strength. Further treatment of the fibers involves beating and refining to increase their flexibility and cause micro-fibrils to bow out from the cell wall, which increases binding strength between fibers. During the stages of separation of the raw material, smaller particles such as short fibers or fragments of fibers are also mixed into the pulp, they are called fines. They are typically defined as particles of sizes less than 200 μm and aggregate in the inter-fiber spaces in the paper-sheet. Fines have a large influence on the finished paper where they can improve sheet strength and reduce shrinkage [6]. The properties of the produced paper-sheets are hence greatly influenced by the properties of the fibers and the particle composition of the pulp.

Needless to say the structure of paper can be quite different depending on all factors mentioned above. Another important property regarding the structure is how the fibers are aligned in the fiber-network.



(a) Fiber orientation distribution (b) Orientation ellipse

Figure 2.2: Illustrations of a fiber orientation probability distribution $p(\phi)$ and the corresponding orientation ellipse.

2.3 Fiber Formation and Fiber Orientation

Formation is a term that describes the uniformity of the paper structure and the regularity of the fiber orientation. The term formation can be defined as the small-scale basis weight variation of the ready paper-sheet. Irregular formation means that there are a lot of thick and thin spots in the sheets while regular formation means that the paper has a uniform fiber distribution and thereof an even quality. The importance of formation in the structure of the paper-sheet has been extensively reviewed in for example [7–9].

Fiber orientation most often refers to the misalignment of the overall fiber orientation at a small-scale location in the sheet in relation to the machine direction (MD). The degree of fiber orientation, or fiber anisotropy, is usually referred to as the ratio between the machine direction and the cross direction (CD), either by tensile strength testing or by ultrasonic techniques. This is referred to as the MD/CD-ratio. The fiber orientation main axis does not always coincide with the MD-direction. It is therefore convenient to sometimes define the angle φ between the main fiber orientation axis and the direction of manufacture [10]. A typical probability density function $p(\phi)$ of the fiber orientation in the plane of a paper-sheet can be viewed in Figure 2.2.

The fluid dynamics of the suspension in which the wood fibers are dispersed in (the pulp) and the jet-to-wire interaction is a very complex process. An initial fiber orientation is achieved in the headbox jet where, for example, turbulence will create a more random fiber orientation, while an accelerating flow increases fiber alignment. In addition, the jet-to-wire speed ratio also greatly affects the resulting fiber orientation in the paper sheet. The jet-to-wire ratio is often set to not equal one, because the induced shear stresses on the suspension when it lands on the wire breaks up flocs and enhances the

formation. A strong fiber orientation in the MD direction also reduces the chances of web break during production. However, a too strong fiber orientation is not always wanted in the end product, and higher degrees of fiber orientation can also increase tendencies for the sheet to curl and twist during moisture variations. It is therefore often a trade off between good formation and preferred fiber orientation. Devices for measuring fiber formation and fiber orientation are essential to be able to produce the desired product.

2.4 Measurement Techniques

There are a few different techniques to determine fiber formation and fiber orientation, e.g., image analysis, ultrasonic propagation, tensile strength measurements, x-ray diffraction, light diffraction and light diffusion. On-line sensors for feed-back control in the production are desirable to maintain even quality. However, there are different advantages and drawbacks with each of these techniques.

2.4.1 Mechanical and Ultrasonic Measurement Techniques

A standard test method to measure mechanical properties that relate to the fiber orientation are tests of tensile strength [1]. However, it is a method that is highly unsuitable for use in moving industrial environments and therefor limited to laboratory use only.

An alternative method of measuring mechanical properties is to use ultrasound. It was introduced in the early 1980s [11] and is today a standard laboratory technique [10]. Ultrasonic techniques have the benefit of being able to measure both mechanical parameters and structural inhomogeneities. However, the large wavelength of ultrasound used in in-plane testing is many times greater than the thickness of the paper and of the size of individual fibers. This prevents ultrasonic measurements from detecting variations in the thickness direction and rather measures exclusively in the plane of the sheet. Ultrasonic techniques are also sensitive to how fiber bonds and to the stresses that arise during drying. Sheets with the same structure but different mechanical properties can therefore exhibit different measurement results. Additionally, as ultrasonic techniques require contact with the surface of the paper it also makes it unsuitable for on-line measurements. However, interferometric techniques have been modified to detect wave propagation in paper-sheets, and this opens up for possible on-line measurements based on ultrasonic wave propagation [12, 13].

2.4.2 Optical Measurement Techniques

One of, if not the first, technique to determine formation on-line was done by looking at the absorption of beta radiation. Beta radiation is convenient for this purpose as its scattering is weak while the absorption is proportional to the density of the paper-sheet [1]. Drawbacks are that it is affected quite a lot by moisture content and that the technique only measures formation and not fiber orientation.

However, fiber orientation can be determined using diffraction of laser light from paper-sheets, though it is limited to very thin papers with just a few layers of fibers [14]. Anisotropic light diffusion, or light scattering through paper, has also been suggested as a possible method to determine fiber orientation and formation [15]. This was briefly reviewed in [7], however, little work on this technique has been done since the 80s, and it is not a well understood phenomenon. There are also other optical-based measurements where a polarized incident beam illuminates a sheet at oblique angles with several strategically positioned light detectors above the sheet. By processing the intensity and depolarization of the diffused light along with the reflectance at the surface it is possible to determine the fiber orientation in the depth of the sheet [16].

2.4.3 Image Analysis

One of the earliest methods to determine fiber orientation was to mix colored fibers into the pulp before sheet forming and then statistically evaluate their positions and orientations on the surface of the finished paper-sheet through image analysis. With present technology, the same principle can be used in sensing and computational power on-line [17, 18]. Note that this kind of sensing only senses the surface of the paper-sheet, and the fiber orientations often differ between layers in the thickness direction. There are also sheet splitting techniques that enable image analysis between layers, see for example [19] and references there in. As these techniques require a splitting of the paper-sheet makes them inappropriate for on-line use although they can provide valuable information from laboratory investigations.

There are also commercial devices that monitor the flow direction profile of the jet before it reaches the wire [20]. This is prior to the induced formation and fiber orientation generated by the jet-to-wire speed ratio and therefore ignores the effect it has on the finished paper-sheet.

There is also a possibility to determine fiber formation and fiber orientation on finished paper-sheets using 3D x-ray computed tomography images [21]. Note that scanning and reconstruction of a paper sample roughly 10 mm^3 in size takes several hours making this technique inappropriate for on-line use. However, it does provide an excellent tool to verify other measurement techniques as it provides a “true” measure on the position and orientation of each fiber in the paper-sheet.

Chapter 3

Light-Matter Interaction

The perception of light is one of the most crucial aspects in our everyday interaction with the environment around us. The human eye has three color sensors, red, green and blue, and the brain combines these and recreates images with the whole color spectrum. A modern camera works in principle in the same way, but we can today detect beyond what our eyes can see at all fronts, pixels, pixel depth, wavelengths and frames per second. Optical sensing techniques are therefore used successfully in a wide variety of fields.

There are a few different reasons to why objects visually appears as they do. It is usually not enough to only consider the surface of an object to understand its appearance but also the interactions of the electromagnetic wave inside the material needs to be considered. The detection of light that has propagated inside a bulk material can provide useful information about the composition and structure of the material. Therefore, it is important in a wide variety of fields including, for example, atmospheric physics [22], biomedical imaging [23,24], and remote sensing for industriall applications [25]. Electromagnetic scattering and absorption are two properties often used to retrieve information about structural properties of matter. Scattering is also a cornerstone in the optical appearance of materials making the understanding of it very important in fields like computer rendering and printing [26, 27]. A key part in any measurement device is to understand what is measured, and how to use its full potential.

The underlying physics of electromagnetic interaction with matter is a strong classical field that primarily has originated from atmospheric physics and astrophysics. There are numerous books on the subject both considering single scattering [28–31] and multiple scattering [23, 32–34].

3.1 Electromagnetic Wave Interaction in Matter

Matter is composed of neutrons and charged particles: electrons and protons. When an electromagnetic wave interacts with a solid particle, molecule or atom; the electric charges are set into oscillation by the electromagnetic wave. This excitation of energy within the particle will change the electromagnetic field in the environment compared to

that of the incident field. The accelerated electric charges radiate electromagnetic waves in all directions, and part of the incident wave is scattered. In addition, the excited charges within the particle can also transfer into other forms of energy, as for example thermal, a process called absorption.

The fundamentals of electromagnetic wave theory are described by Maxwell's equations [35]. This chapter will briefly show the derivation of the wave equation from Maxwell's equations for light-matter interaction. They will later be used as the fundamental concepts in order to understand scattering and absorption by small particles.

The macroscopic Maxwell equations can be derived by averaging the microscopic equations, in which all electromagnetic interactions are attributed to charged particles. This means that the starting point of the derivations, as follows, is one level above interactions between clouds of electrons bound to atoms and molecules. Optical phenomena such as propagation, scattering, absorption and refraction, can all be quantified by material parameters at the macroscopic level. In addition it usually is variations on the macroscopic scale that are the measured quantities in most optical experiments.

The Maxwell macroscopic equations, also known as Maxwell's equations in matter, can be expressed as

$$\nabla \cdot \mathbf{D} = \rho \tag{3.1}$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \tag{3.2}$$

$$\nabla \cdot \mathbf{B} = 0, \tag{3.3}$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \tag{3.4}$$

where $\mathbf{D}$ is the electric displacement, $\mathbf{E}$ is the electric field, $\mathbf{B}$ the magnetic induction, $\mathbf{H}$ the magnetic field, ρ is the charge density and $\mathbf{J}$ the current density. Further, the electric displacement and magnetic fields are defined as

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P} \tag{3.5}$$

$$\mathbf{H} = \frac{\mathbf{B}}{\mu_0} - \mathbf{M} \tag{3.6}$$

where $\mathbf{P}$ is the electric polarization, $\mathbf{M}$ the magnetization, ϵ_0 is the electric permittivity and μ_0 is the magnetic permeability of free space.

Equations 3.1 - 3.6 are still not enough to describe the interaction between a wave in free space and matter. The so-called *constitutive relations* relate the electromagnetic fields to the material properties of the interacting media and are described as

$$\mathbf{J} = \sigma \mathbf{E} \tag{3.7}$$

$$\mathbf{P} = \epsilon_0 \chi_\epsilon \mathbf{E} \tag{3.8}$$

$$\mathbf{M} = \chi_\mu \mathbf{H} \tag{3.9}$$

which in turn together with Equations 3.5 and 3.6 lead to

$$\mathbf{B} = \mu_0(1 + \chi_\mu)\mathbf{H} = \mu_0\mu_r\mathbf{H} = \mu\mathbf{H} \tag{3.10}$$

$$\mathbf{D} = \epsilon_0(1 + \chi_\epsilon)\mathbf{E} = \epsilon_0\epsilon_r\mathbf{E} = \epsilon\mathbf{E} \tag{3.11}$$

where σ is conductivity, μ permeability, $\chi_\epsilon = \epsilon_r - 1$ is the electric susceptibility and $\chi_\mu = \mu_r - 1$ is the magnetic susceptibility, all of which depend on the medium under consideration. Together with Maxwell's equations, the *constitutive relations* completely describe the electromagnetic fields interacting in matter.

The next step is to decouple the electric and magnetic fields and derive the wave equation. The curl of Equations 3.2 and 3.4, together with Equation 3.4 and 3.10, leads to the expressions

$$\nabla \times (\nabla \times \mathbf{E}) = -\mu\frac{\partial(\nabla \times \mathbf{H})}{\partial t} = -\mu\frac{\partial}{\partial t}(\mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}) \tag{3.12}$$

$$\nabla \times (\nabla \times \mathbf{H}) = \nabla \times \mathbf{J} + \nabla \times \frac{\partial \mathbf{D}}{\partial \mathbf{t}}. \tag{3.13}$$

The left hand side of Equations 3.12 and 3.13 can be rewritten in the form

$$\nabla \times \nabla \times \mathbf{E} = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2\mathbf{E} = -\nabla^2\mathbf{E} \tag{3.14}$$

$$\nabla \times \nabla \times \mathbf{H} = \nabla(\nabla \cdot \mathbf{H}) - \nabla^2\mathbf{H} = -\nabla^2\mathbf{H} \tag{3.15}$$

where it can be realized from Equations 3.2 and 3.4 that the first term equals zero for both $\mathbf{E}$ and $\mathbf{H}$. If we now take Equations 3.12 and 3.13 and the expressions for $\mathbf{D}$ and $\mathbf{J}$ from Equations 3.7 and 3.11; with Equations 3.2 and 3.10 and insert them into 3.12 and 3.13, we get

$$\nabla^2\mathbf{E} = \sigma\mu\frac{\partial \mathbf{E}}{\partial t} + \epsilon\mu\frac{\partial^2\mathbf{E}}{\partial t^2} \tag{3.16}$$

$$\nabla^2\mathbf{H} = \sigma\mu\frac{\partial \mathbf{H}}{\partial t} + \epsilon\mu\frac{\partial^2\mathbf{H}}{\partial t^2}. \tag{3.17}$$

The decoupled electric and magnetic fields can now be considered as simple time harmonic wave functions $\mathbf{E} = \mathbf{E}_0(\mathbf{r})e^{-i\omega t}$ and $\mathbf{H} = \mathbf{H}_0(\mathbf{r})e^{-i\omega t}$, respectively. By solving the time derivatives we obtain the following differential equations for the $\mathbf{E}_0$ and $\mathbf{H}_0$ fields

$$\nabla^2\mathbf{E}_0 = -i\omega\sigma\mu\mathbf{E}_0 - \omega^2\epsilon\mu\mathbf{E}_0 \tag{3.18}$$

$$\nabla^2\mathbf{H}_0 = -i\omega\sigma\mu\mathbf{H}_0 - \omega^2\epsilon\mu\mathbf{H}_0. \tag{3.19}$$

With the relation of the angular free $\omega = 2\pi c/\lambda$ and the wave number $k_0 = 2\pi/\lambda$ this becomes

$$\nabla^2\mathbf{E}_0 + k_0^2c^2\epsilon\mu(1 - \frac{\sigma}{\omega\epsilon}i)\mathbf{E}_0 = 0 \tag{3.20}$$

$$\nabla^2\mathbf{H}_0 + k_0^2c^2\epsilon\mu(1 - \frac{\sigma}{\omega\epsilon}i)\mathbf{H}_0 = 0. \tag{3.21}$$

where the expression for the complex refractive index now has appeared and can be defined as

$$m = c\sqrt{\epsilon\mu(1 - \frac{\sigma}{\omega\epsilon}i)}. \tag{3.22}$$

It is often convenient to use the refractive index rather than the conductivity, permittivity and permeability parameters when considering light-matter interactions. Scattering of electromagnetic waves by small particles is usually stated to be due to the mismatch of refractive index between particle and surrounding medium. However, it should not be forgotten that the refractive index in turn is induced by the electric and magnetic properties of the materials. The complex part of the refractive index causes absorption of energy from the electromagnetic wave and transfers the energy into other forms. This behavior is, even though it is not clearly shown here, dependent of wavelength.

The scalar wave equation, also known as the Helmholtz equation, for the $\mathbf{E}_0$ and $\mathbf{H}_0$ fields, can now be written as

$$\nabla^2\mathbf{E}_0 + k^2\mathbf{E}_0 = 0 \tag{3.23}$$

$$\nabla^2\mathbf{H}_0 + k^2\mathbf{H}_0 = 0 \tag{3.24}$$

where $k = k_0 m = 2\pi m/\lambda$. We now have the mathematical expressions showing that electromagnetic energy propagates as waves during light-matter interaction. The electric and magnetic field vectors are perpendicular to the propagation direction of the wave as well as to each other.

In order to fully describe scattering we also need to consider the Poynting vector $\mathbf{S}$. It specifies the magnitude and direction of the rate of transfered electromagnetic energy; it therefore has a fundamental importance in problems regarding electromagnetic propagation. It is useful to consider the average power flow over time as the Poynting vector may vary rapidly. The time-averaged Poynting vector is given by

$$\langle\mathbf{S}\rangle = \frac{1}{2}Re\{\mathbf{E}_0 \times \mathbf{H}_0^*\}. \tag{3.25}$$

The magnitude of $\langle\mathbf{S}\rangle$ is often refereed to as the irradiance I where its dimensions are energy per unit area $[Wm^{-2}]$. However, it is convenient to use the radiance L in scattering theory instead of irradiance. The radiance L with dimensions $[Wm^{-2}sr^{-1}]$ of a source is a property that can be described as

$$L = r_s^2\frac{dI}{dA} = \frac{dI}{d\omega} \tag{3.26}$$

where $d\omega$ is the solid angle which describes the flux through cross section dA at distance r_s from the source. Radiance does not depend on the distance to the observer, but it does depend on the direction and is therefore conveniently used for scattering particles viewed from the far-field.

3.2 Scattering

Scattering is the physical process that occurs when an electromagnetic wave encounters an inhomogeneity in the medium it propagates in. This inhomogeneity can, from a macroscopic perspective, be a particle with a different refractive index than the surrounding. The particle will be the subject of an oscillating electric field and these oscillations will cause some of the energy to "scatter" and propagate into new directions.

Some materials that scatter light can also induce a change in wavelength, and this is often referred to as inelastic scattering. Raman scattering (fluorescence) and Mandelstam-Brillouin scattering are examples of this phenomenon. As these effects usually are much weaker than elastic scattering, they are not considered in this work.

The total field in the presence of an object $\mathbf{E}$ can be represented as the sum of the incident field $\mathbf{E}_i$ and the scattered field $\mathbf{E}_s$ such as

$$\mathbf{E} = \mathbf{E}_i + \mathbf{E}_s. \tag{3.27}$$

Hence, energy is transfered from the incident field $\mathbf{E}_i$ into the scattered field $\mathbf{E}_s$ where the energy is radiated in all directions. The fundamental problem in a scattering and absorbing medium is to understand this transformation of energy. The treatment of scattering and absorption by small particles is therefore a first important step in order to understand a scattering medium.

3.3 Scattering by Small Particles

A particle that generates the distortion of the wave is often referred to as a scatterer. This applies for all the previously mentioned particles ranging from atoms and molecules to macroscopic objects.

When considering a scattering and absorbing particle it is often convenient to consider the total extinction of energy evoked by the particle. The scattering cross section σ_s can be defined as the total power removed from the incident wave as a result of scattering. The scattering cross section σ_s in turn relates to the geometrical area of the particle σ_g as

$$\sigma_s = Q_s \sigma_g \tag{3.28}$$

where Q_s is the scattering efficiency. Analogously, the absorption cross section σ_a is the power removed from the incident wave as a result of absorption by the particle and can be described as

$$\sigma_a = Q_a \sigma_g \tag{3.29}$$

where Q_a is the absorption efficiency. The total amount of energy lost can then be described by the total extinction

$$\sigma_e = \sigma_s + \sigma_a. \tag{3.30}$$

Hence, the total extinction efficiency of energy by the particle is $Q_e = Q_s + Q_a$. Note that both the scattering and absorption efficiency depend on the relation between wavelength and particle size.

In addition to the amount of scattered energy it is also important to consider the direction of scattering. The phase function $p(\mathbf{s}, \mathbf{s}')$ specifies the amount of light scattered from incident direction $\mathbf{s}$ into the direction $\mathbf{s}'$. It describes the far-field Poynting vector $\mathbf{S}$, or radiance L, of the scattered field in the direction $\mathbf{s}'$. The phase function is usually normalized so that the integration over all solid angles equals one

$$\int_{4\pi} p(\mathbf{s}, \mathbf{s}') d\Omega = 1 \tag{3.31}$$

this makes it possible to regard the phase function as a probability distribution function as well as a measure of the scattered field strength. The distribution in which the light is scattered depends on the size and shape of the particles as well as the wavelength of the incident light and the incidence angle of the light toward the particles. The simplest kind of scattering is isotropic where the phase function is

$$p(\mathbf{s}, \mathbf{s}') = \frac{1}{4\pi} \tag{3.32}$$

in all directions. However, large particles usually scatter in an anisotropic manner, where the phase function most often shows a strong forward tendency. This makes the phase function one of the crucial components when modeling and developing an understanding of scattering.

3.4 Scattering Regimes

The main parameters that govern the scattering by a particle aside from the refractive index is the particle size in relation to the wavelength. It is common for spherical particles to define an optical size parameter, or the so called *size factor* as

$$x = \frac{2\pi r}{\lambda} \tag{3.33}$$

where r is the particle radius and λ the wavelength. Depending on the size factor x, particles can be placed in different scattering regimes.

Particles with sizes of a fraction of the wavelength scatter a insignificant amount of energy from the incident wave. A simple example is radio waves ($\lambda \sim m$) that can propagate through the atmosphere without being scattered or "damped" by molecules or water clouds. The large size difference between the wavelenght and the particle $x \approx 0$ causes an insignificant amount of interaction. This scattering is therefore said to be negligible.

When a particle is a few orders of magnitude smaller than the wavelength $x \ll 1$ it scatters light isotropically in all directions. The electromagnetic energy oscillates with the frequency of the incident light. However as the particle is small the phase relations between the scattered waves are going to be little affected. This cause little variation in the scattering phase function for the different directions. This type of scattering is often referred to as Rayleigh scattering [36]. The most classical example of Rayleigh

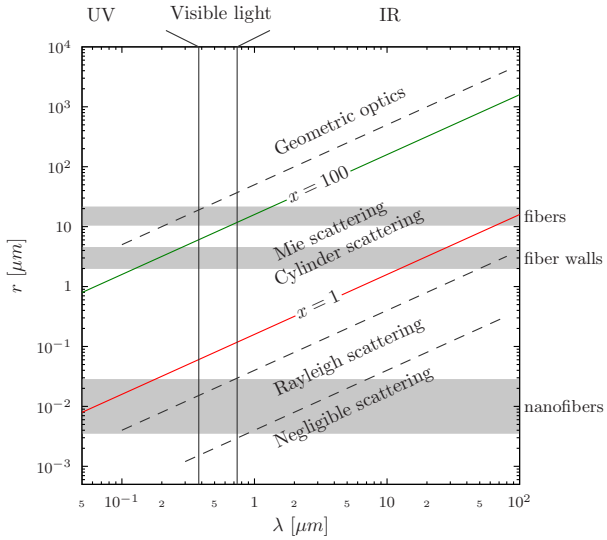


Figure 3.1: Scattering regimes in relation to wavelength (x-axis) and particle size (y-axis). The size parameters of cellulose fibers such as the outer radius, fiber wall, and with the smaller building block nanofibers are marked in the figure. It can be seen that the outer fiber radius in relation to the visible spectrum lie in the region between Mie scattering and geometrical optics. Note, however, that the width of the fiber wall clearly belongs to the Mie scattering regime. Note also that nanofibers in the visible region belongs in the Rayleigh regime.

scattering is the interaction between sunlight and the molecules in our atmosphere. The dependency between the wavelength and the scattering cross section σ_s is proportional to $\sim \lambda^{-4}$ which means that more energy from the shorter wavelengths will be scattered. This in turn makes the sky appear blue to any observer not looking directly at the sun. If the sun is close to the horizon, where the path to the observer is longer, the blue wavelengths have already been scattered and we see more of the longer wavelengths and the sky will appear yellow or red.

As the particle size increases to be in the same order of magnitude as the wavelength $x \approx 1$ an increasing number of mutual interferences between scattered waves in the particles will occur. This will cause peaks and valleys in the scattering pattern, and the number of these will increase with particle size. This is commonly refereed to as Mie theory, or Mie scattering and describes how plane electromagnetic waves are scattered by homogeneous spherical objects [37]. Mie theory is based upon quite complex math and invloves summations of large numbers of Bessel functions and Legendre polynomials that obscure the underlying physics. In contrast to Rayleigh scattering, larger particles

that follow Mie theory are roughly independent of wavelength and usually scatter light strongly in the forward direction. This explains for example why clouds, which consist of large water droplets roughly 10–20 μm in diameter, appear white and not blue as the sky. Further it also implies that scattering media, consisting of particles of sizes comparable with the visible spectrum, will not change color due to scattering. Any change in color is instead attributed to absorption of specific wavelengths in the visible spectra, inelastic scattering such as fluorescence, or other wave phenomenon. This also applies to cellulose fibers, making pigments and fluorescences agents important when considering the optical appearance of the paper.

Scattering from many kinds of particles in addition to water droplets in the atmosphere, such as droplets in emulsions in milk or biological cells, can often be described by Mie theory. Analytical solutions similar to Mie theory are limited to particles with simple geometries. Along with the spherical particles there are also analytical solutions available to spheroids, coated spheres, infinite cylinders and infinite coated cylinders [30].

Further approximations can be done when the particle size is orders of magnitude larger than the electromagnetic radiation $x \gg 1$. This is when the scattered fields within the particles are simple enough that geometrical optics, or raytracing, can be applied. However, geometrical optics is an approximation that ignores wave effects. It can be debated whether or not scattering by a cellulose fiber can be approximated using geometrical optics. The outer diameter of a cellulose fiber lies close to where this approximation would be a valid alternative in the visible region. If the size of the fiberwall is included, which usually is in the magnitude of a few visible wavelengths, then caution should be used with this approximation. However, it must be mentioned that Saarinen et al. [38] have shown that the overall behaviors in scattering and depolarization from a cellulose fiber can be modelled using geometrical optics.

Figure 3.1 illustrates different sizes of cellulose particles and where they lie in relation to the different scattering regimes.

3.5 Analytical Solutions to Scattering by Small Particles

As previously mentioned scattering properties from particles with simple geometrical shapes can be derived analytically from Maxwell's equations. Three different solutions have been considered under the progress of this work: spheres, infinite homogeneous cylinders and infinite hollow cylinders.

The most common of these solutions is scattering by an homogeneous spherical particle which was suggested by Gustav Mie over a century ago [37]. However, the main focus has been on the less known solutions for cylindrical particles as they obviously resemble a fiber to a higher degree than for example a spheroid. As wood fibers often are a lot longer than they are wide it can be a valid assumption that they scatter light in a fashion similar to infinitely long cylinders. The analytical solution to infinite homogeneous cylinders was first solved by Waits in the 1950s [39], then outlined by van de Hulst [28]

and Kerker [29]. More recent literature also contains computer code for the solution at normal incidence [30] and at oblique incidences [40].

For two concentric cylinders, or a coated/hollow cylinder, Kerker and Matijevic presented the original solution at perpendicular incident light to the cylinder axis [41]. G. A. Shah later extended the solution to oblique incidences [42]. Even though these solutions were done long ago (in the 1960s) little work has been published using them since. A reasonable explanation for this is that the computational capability required has not existed until more recent years. This becomes quite apparent by looking at the work done by Yousif et al. from 1994 [43] were they used a computer cluster to calculate solutions that today can be done on any home computer within minutes.

The fundamental task to describe scattering by a particle is to solve the Maxwell equations, i.e., the wave equations 3.23 and 3.24, both inside and outside the particle. The requirement is that the tangential components of the electromagnetic field are continuous across the boundary separating two media where the material parameters are discontinuous. The boundary conditions between two media 1 and 2 with the refractive index m_1 and m_2, respectively, can be described as follows

$$[\mathbf{E}_2 - \mathbf{E}_1] \times \hat{\mathbf{n}} = 0 \tag{3.34}$$

$$[\mathbf{H}_2 - \mathbf{H}_1] \times \hat{\mathbf{n}} = 0 \tag{3.35}$$

where $\hat{\mathbf{n}}$ is the unit normal to the surface. This means that for a scattering particle the field within the particle $\mathbf{E}_1$ needs to be continuous with both the incident $\mathbf{E}_i$ and scattered $\mathbf{E}_s$ fields.

When constructing a solution it is possible to consider two plane monochromatic waves of two orthogonal polarization states. The superposition principle then enables any arbitrary polarization state to be calculated for the given direction of propagation. The electric field vector $\mathbf{E}$ can therefore be decomposed into two components in the transverse plane

$$\mathbf{E} = \mathbf{E}_\parallel + \mathbf{E}_\perp. \tag{3.36}$$

These components are often refereed to as the perpendicular component $\mathbf{E}_\perp$, also known as transverse electric TE or s-wave, and the parallel component $\mathbf{E}_\parallel$, also known as transverse magnetic TM or p-wave. Any arbitrary phase relations can then be quantified with for example the Stokes formalism. The Stokes vector is

$$S = \begin{pmatrix} I \\ Q \\ U \\ V \end{pmatrix} = \begin{pmatrix} \mathbf{E}_\parallel \mathbf{E}_\parallel^* + \mathbf{E}_\perp \mathbf{E}_\perp^* \\ \mathbf{E}_\parallel \mathbf{E}_\parallel^* - \mathbf{E}_\perp \mathbf{E}_\perp^* \\ \mathbf{E}_\parallel \mathbf{E}_\perp^* + \mathbf{E}_\perp \mathbf{E}_\parallel^* \\ i(\mathbf{E}_\parallel \mathbf{E}_\perp^* - \mathbf{E}_\perp \mathbf{E}_\parallel^*) \end{pmatrix} \tag{3.37}$$

where I is the intensity, Q the degree of linear polarization, U the degree of linear polarization tilted 45 degrees and V is the degree of circular polarization.

Consider an arbitrary particle illuminated by a plane monochromatic wave from the z-direction. It can be shown that the solution in the far-field region is a linear function of

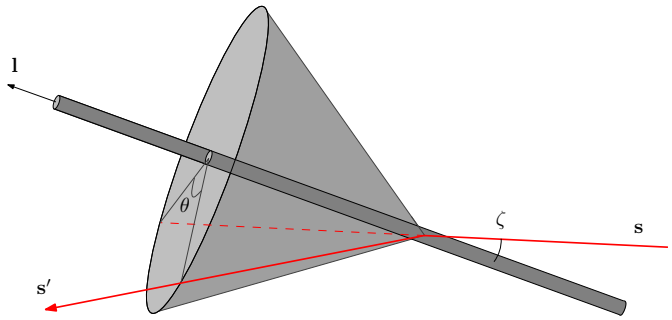


Figure 3.2: Light propagating from direction $\mathbf{s}$ is scattered into direction $\mathbf{s}'$ by a cylinder. The scattered light is distributed in a cone around the longitudinal axis $\mathbf{l}$ of the cylinder and has half-angle ζ.

the incident fields [30]. Once the fields have been calculated it is possible to again look at the amplitude in the far-field. The relationship between incident and scattered fields can be written as

$$\begin{pmatrix} E_{\parallel s} \\ E_{\perp s} \end{pmatrix} = \frac{e^{ik(r-z)}}{-ikr} \begin{pmatrix} S_2 & S_3 \\ S_4 & S_1 \end{pmatrix} \begin{pmatrix} E_{\parallel i} \\ E_{\perp i} \end{pmatrix} \tag{3.38}$$

where the scattering elements S_1, S_2, S_3 and S_4 depend on the new propagation direction and describe how the amplitude and polarization of the incident field changes. The scattering elements are usually quite complex and require a lot of computational time for summarizing series in, e.g., Mie theory.

The properties that can be derived from these solutions are the phase function $p(\mathbf{s}, \mathbf{s}')$, Mueller matrix M, and extinction parameters Q_s and Q_a. These parameters, togheter with the density of scatterers, is enough to describe light propagation in a multiple scattering medium.

3.5.1 Scattering by a Infinitely Long Cylinder

The primary particle used in this thesis is the infinitely long, straight, homogeneous cylinder. The scattering characteristics of a cylinder are quite different from the rotationally invariant scattering by a spherical particle as described in traditional Mie theory. Scattering by a cylinder strongly depends on the incident angle ζ towards the cylinder. Light scatter in a conical shape around the longitudinal axis of the cylinder $\mathbf{l}$ and the phase function describe the intensity distribution in the cone, see illustration in Figure 3.2. This means that the phase function can be written as $p(\mathbf{s}, \mathbf{s}') = p(\zeta, \theta)$ where θ describes the angular distribution around the cylinder. The solution of the wave equations of the scattered field generated by a cylinder can be given as [30]:

$$\begin{pmatrix} \mathbf{E}_{\parallel s} \\ \mathbf{E}_{\perp s} \end{pmatrix} = e^{i3\pi/4} \sqrt{\frac{1}{\pi k r \sin\zeta}} e^{ik(r\sin\zeta - z\cos\zeta)} \begin{pmatrix} T_1 & T_4 \\ T_3 & T_2 \end{pmatrix} \begin{pmatrix} \mathbf{E}_{\parallel i} \\ \mathbf{E}_{\perp i} \end{pmatrix} \tag{3.39}$$

where scattering elements T_1–T_4 are

$$\begin{aligned}
T_1 &= \sum_{-\infty}^{\infty} b_{nI} e^{-in\Theta} = b_{0I} + 2\sum_{n=1}^{\infty} b_{nI}\cos(n\Theta) \\
T_2 &= \sum_{-\infty}^{\infty} a_{nII} e^{-in\Theta} = a_{0II} + 2\sum_{n=1}^{\infty} a_{nII}\cos(n\Theta) \\
T_3 &= \sum_{-\infty}^{\infty} a_{nI} e^{-in\Theta} = -2i\sum_{n=1}^{\infty} a_{nI}\sin(n\Theta) \\
T_4 &= \sum_{-\infty}^{\infty} b_{nII} e^{-in\Theta} = -2i\sum_{n=1}^{\infty} b_{nII}\sin(n\Theta)
\end{aligned} \tag{3.40}$$

$$\begin{aligned}
a_{nI} &= \frac{C_n V_n - B_n D_n}{W_n V_n + iD_n^2} \\
a_{nII} &= -\frac{A_n V_n - iC_n D_n}{W_n V_n + iD_n^2} \\
b_{nI} &= \frac{W_n B_n - iD_n C_n}{W_n V_n + iD_n^2} \\
b_{nII} &= -i\frac{C_n W_n + A_n D_n}{W_n V_n + iD_n^2}
\end{aligned} \tag{3.41}$$

$$\begin{aligned}
A_n &= i\xi[\xi J_n'(\eta)J_n(\xi) - \eta J_n(\eta)J_n'(\xi)] \\
B_n &= \xi[m^2\xi J_n'(\eta)J_n(\xi) - \eta J_n(\eta)J_n'(\xi)] \\
C_n &= n\cos\zeta\eta J_n(\eta)J_n(\xi)(\frac{\xi^2}{\eta^2} - 1) \\
D_n &= n\cos\zeta\eta J_n(\eta)H_n^{(1)}(\xi)(\frac{\xi^2}{\eta^2} - 1) \\
W_n &= i\xi[\eta J_n(\eta)H_n^{(1)\prime}(\xi) - \xi J_n'(\eta)H_n^{(1)}(\xi)] \\
V_n &= \xi[m^2\xi J_n'(\eta)H_n^{(1)}(\xi) - \eta J_n(\eta)H_n^{(1)\prime}(\xi)]
\end{aligned} \tag{3.42}$$

$$\begin{aligned}
\Theta &= \pi - \theta \\
\xi &= x\sin\zeta \\
\eta &= x\sqrt{m^2 - \cos^2\zeta} \\
x &= ka
\end{aligned} \tag{3.43}$$

Note that a is the radius of the cylinder, m its refractive index, and the functions J_n and H_n are Bessel and Hankel functions of the first kind and order n. The prime notation is used to denote when it is the derivative of the Bessel and Hankel functions. The summation $n \to \infty$ also needs to be truncated in order to reach a solution at some proper accuracy.

3.5.2 Phase Functions and Mueller Matrix

It is now possible to use the Stokes formalism in addition to the Mueller matrix to calculate the scattering properties without using the **E**–fields directly. The Muller matrix for a cylinder is

$$M(\zeta,\theta) = \frac{2}{\pi kr \sin\zeta} \begin{pmatrix} T_{11} & T_{12} & T_{13} & T_{14} \\ T_{21} & T_{22} & T_{23} & T_{24} \\ T_{31} & T_{32} & T_{33} & T_{34} \\ T_{41} & T_{42} & T_{43} & T_{44} \end{pmatrix} \tag{3.44}$$

where the Mueller elements are descibed as follows

$$\begin{aligned}
T_{11} &= \tfrac{1}{2}(|T_1|^2 + |T_2|^2 + |T_3|^2 + |T_4|^2) \\
T_{12} &= \tfrac{1}{2}(|T_2|^2 - |T_1|^2 + |T_4|^2 - |T_3|^2) \\
T_{13} &= Re\{T_2T_3^* + T_1T_4^*\} \\
T_{14} &= Im\{T_2T_3^* - T_1T_4^*\} \\
T_{21} &= \tfrac{1}{2}(|T_2|^2 - |T_1|^2 - |T_4|^2 + |T_3|^2) \\
T_{22} &= \tfrac{1}{2}(|T_2|^2 + |T_1|^2 - |T_4|^2 - |T_3|^2) \\
T_{23} &= Re\{T_2T_3^* - T_1T_4^*\} \\
T_{24} &= Im\{T_2T_3^* + T_1T_4^*\} \\
T_{31} &= Re\{T_2T_4^* + T_1T_3^*\} \\
T_{32} &= Re\{T_2T_4^* - T_1T_3^*\} \\
T_{33} &= Re\{T_1T_2^* + T_3T_4^*\} \\
T_{34} &= Im\{T_2T_1^* + T_4T_3^*\} \\
T_{41} &= Im\{T_2^*T_4 + T_3^*T_1\} \\
T_{42} &= Im\{T_2^*T_4 - T_3^*T_1\} \\
T_{43} &= Im\{T_1T_2^* - T_3T_4^*\} \\
T_{44} &= Re\{T_1T_2^* - T_3T_4^*\}.
\end{aligned}$$

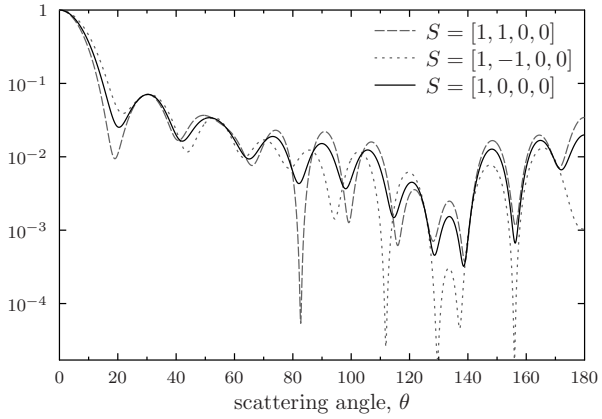


Figure 3.3: Scattering phase functions from a cylinder illuminated at normal incidence.

After each scattering event a new state of polarization can now be calculated using the Stokes vector for the scattered field

$$S_s = M(\zeta, \theta)R(\psi)S_i \tag{3.45}$$

where S_i is the incident Stokes vector and $R(\psi)$ is a rotational matrix used to keep track of the reference plane for the linear polarization. Note that the the Mueller matrix depends on both ζ and θ meaning that it will be different for each incident angle. From the incident Stokes vector $S_i = [I, Q, U, V]$ the phase function can also be computed as

$$p(\zeta, \theta) = T_{11}(\zeta, \theta)I + T_{12}(\zeta, \theta)Q + T_{13}(\zeta, \theta)U + T_{14}(\zeta, \theta)V. \tag{3.46}$$

This means that the phase function depends on the polarization of the incident light. The phase function of a cylinder with $x = 10.0$ and $m = 1.5$ illuminated at normal incidence can be seen in Figure 3.3.

3.5.3 Scattering Efficiency

The last calculation required to describe scattering by a cylinder is that of the scattering efficiency Q_s and extinction efficiency Q_e. Their parallel and perpendicular components

can be calculated using the coefficients from Equation 3.41 as follows

$$\begin{aligned} Q_{\parallel s} &= \frac{2}{x}\left[|b_{0I}|^2 + 2\sum_{n=1}^{\infty}(|b_{nI}|^2 + |a_{nI}|^2)\right] \\ Q_{\parallel e} &= \frac{2}{x}Re\left\{b_{0I} + 2\sum_{n=1}^{\infty} b_{nI}\right\} \end{aligned} \tag{3.47}$$

and

$$\begin{aligned} Q_{\perp s} &= \frac{2}{x}\left[|a_{0II}|^2 + 2\sum_{n=1}^{\infty}(|a_{nII}|^2 + |b_{nII}|^2)\right] \\ Q_{\perp e} &= \frac{2}{x}Re\left\{a_{0II} + 2\sum_{n=1}^{\infty} a_{nII}\right\}. \end{aligned} \tag{3.48}$$

Figure 3.4 shows the scattering efficiency changes with the size factor x for both spherical particles and cylinders. Note that the scattering efficiency for a cylinder depends on ζ, hence $\overline{Q}_{e,cyl}$ is the average scattering efficiency for all incident angles. It can be seen from Figure 3.4 that, e.g., $Q_{e,sph}$ approaches a limit value as the size factor increases

$$\lim_{x\to\infty} Q_e(x) = 2. \tag{3.49}$$

Hence, a large particle scatters twice the energy that is geometrically incident upon it meaning that the incident wave is going to be influenced beyond the physical boundary of the particle. This puzzling property is called the *extinction paradox* and can be considered as an interference phenomenon. The effect is explained as the interference between the incident wave and the far-field scattered wave in the forward scattering direction [28, 30, 44, 45].

3.5.4 Other Scattering Calculation Methods for Particles

Many materials, such as tissue or paper, contain a wide number of different scatterers, with different shapes, sizes and refractive indices. For such materials it if often convenient to apply approximate phase functions such as the widely used Henyey-Greenstein phase function [46]. It is an approximation of scattering by a sphere but is often used to average the combined scattering behavior of many different particles into one phase function

$$p(\theta) = \frac{1}{4\pi}\frac{1-g^2}{(1+g^2-2g\cos\theta)^{3/2}}. \tag{3.50}$$

It is controlled by adjusting the asymmetry parameter g, which is the average cosine of the scattering angle

$$g = \langle\cos\theta\rangle. \tag{3.51}$$

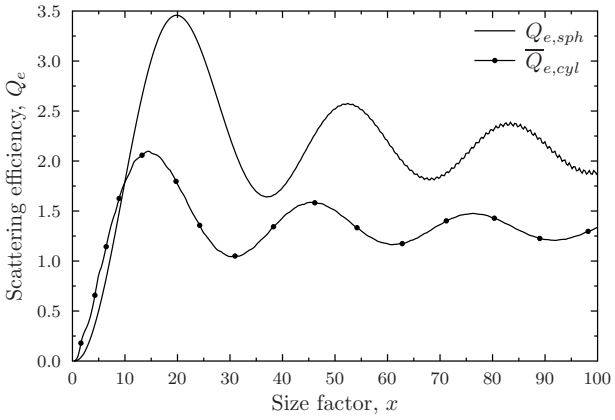


Figure 3.4: The scattering efficiency Q_e for spheres and randomly oriented cylinders.

The value of g can vary between -1 and 1. If $g = 0$ it means that scattering is isotropic; $g = 1$ means a strong forward scattering and $g = -1$ means a strong backward scattering. As g can be solved directly with Mie theory it is possible to use the Henyey-Greenstein phase function as a direct approximation for many types of scatterers.

There are numerous other methods, both exact and approximate, to calculate scattering by particles of various sizes and shapes. Some examples of these are the discrete dipole approximation, finite-difference time-domain method and the T-matrix method. None of these methods have explicity been used in this thesis, a good summary of different calculation methods can be found in [31].

3.6 Multiple Scattering by Collections of Particles

It is unusual to encounter scattering events from isolated particles in nature, and this obviously also applies for paper and pulp. Most materials are bounded together as either liquids, gases, or solid matter, which makes scattering effects much more complex to understand.

To solve multiple scattering problems it is convenient to start by considering the scattering particles using a statistical description. The concentration of scattering particles can therefore be described by defining the number density N_s $[m^{-3}]$. It describes the total number of particles per unit volume in the scattering media. The scattering coefficient can therefore be defined as

$$\mu_s = \sigma_s N_s \tag{3.52}$$

where it describes the probability of interaction with a particle per unit length traveled. If we consider a medium that, in addition to scattering, also absorbs energy the total energy loss can be described with the extinction coefficient

$$\mu_e = \sigma_e N_s = (\sigma_s + \sigma_a) N_s. \tag{3.53}$$

The irradiance loss for a plane electromagnetic wave propagating the distance Δs in direction $\mathbf{s}'$ due to scattering and absorption is therefore

$$dL = -\mu_e \Delta s L. \tag{3.54}$$

If it can be assumed that no additional energy is added to the system, i.e., insignificant additinal energy is scattered into direction $\mathbf{s}'$, we can directly from Equation 3.54 get the expression

$$L = L_0 e^{-\mu_e \Delta s}. \tag{3.55}$$

This is a form of the well known Beer-Lambert law and describes the exponential decay of energy for a wave propagating in a scattering and absorbing medium. Equation 3.55 can also be considered as a probability of interaction per unit length propagation distance, this will be furter discussed in the following chapter.

Another important property to consider is the average distance the scattered field propagates before it interacts with another particle, the *mean free path*, and it is defined as

$$\ell = \frac{1}{\mu_s}. \tag{3.56}$$

If the mean free path is a lot longer than the wavelength, it can be assumed that the wave front of the scattered field will be equivalent to a plane wave at the next scattering event. If the mean free path is smaller than the wavelength, the interaction is in the near-field, and the theory of independent single scattering breaks down. Particles are electromagnetically coupled, and this coupling becomes stronger the closer the particles are togheter. A very common simplification is to assume that each particle is sufficiently far away from each other that the coupling between them become insignificant. Under this assumption the fields scattered by individual particles are summed and the scattering problem poses no more analytical challange than scattering from single isolated particles. In literature, the treatment of scattering by single isolated particles is referred to as independent scattering where as its counterpart is called dependent scattering.

3.6.1 Radiative Transfer

Even though Lord Reyligh formulated his theory for single particle scattering particles as early as 1871, it took a long time before the problem of multiple scattering to be correctly formulated. Radiative Transfer Theory (RTT) describes the interaction of electromagnetic radiation in a medium that emits, scatters and absorbs. The modern formulation of the RTT is often credited to Arthur Schuster's formulation from 1905 [47]. The Radiative Transfer Equation (RTE) is a differential equation that describes these interactions

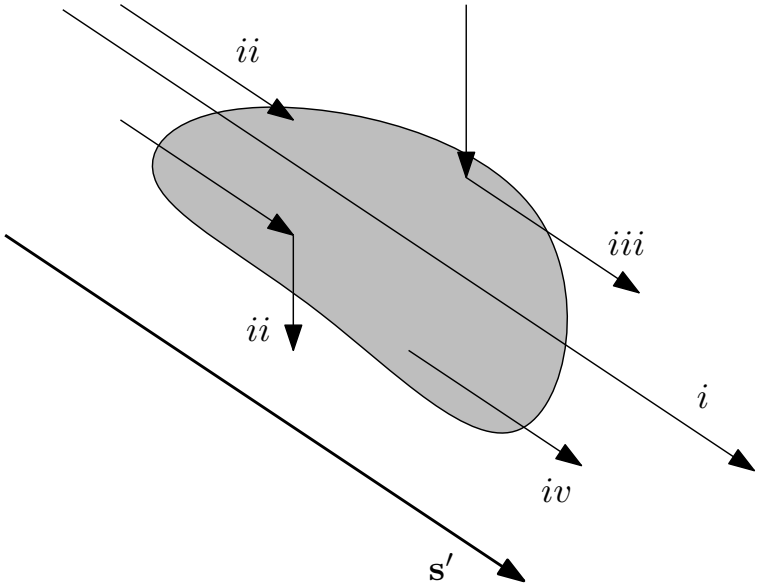


Figure 3.5: Principle of radiative transfer theory.

mathematically in a small control volume, heuristically derived using energy conservation [32]. However, it has recently shown that the RTE can be calculated directly from Maxwell's equations [33, 48, 49].

To formulate the RTE, the radiance $L(\mathbf{r}, \mathbf{s}')$, where the position is denoted by the vector $\mathbf{r}$ and the unit vector $\mathbf{s}'$ specifies the direction, needs to be considered. By using the optical properties previously discussed in this chapter, μ_e and $p(\mathbf{s}, \mathbf{s}')$, the following formulation of the RTE can be stated

$$\frac{n}{c}\frac{\partial L(\mathbf{r}, \mathbf{s}')}{\partial t} = -\mathbf{s}' \cdot \nabla L(\mathbf{r}, \mathbf{s}') - \mu_e L(\mathbf{r}, \mathbf{s}') + \mu_s \int_{4\pi} p(\mathbf{s}, \mathbf{s}') L(\mathbf{r}, \mathbf{s}') d\Omega + q(\mathbf{r}, \mathbf{s}') \qquad (3.57)$$

where the terms on the right hand side describe:

- (***i***) $-\mathbf{s}' \cdot \nabla L(\mathbf{r}, \mathbf{s}')$ - Radiance lost by crossing the border of the volume
- (***ii***) $-\mu_e L(\mathbf{r}, \mathbf{s}')$ - Radiance lost due to scattering and absorption
- (***iii***) $\mu_s \int_{4\pi} p(\mathbf{s}, \mathbf{s}') L(\mathbf{r}, \mathbf{s}') d\Omega$ - Radiance gained from light scattered from direction $\mathbf{s}$ into $\mathbf{s}'$
- (***iv***) $q(\mathbf{r}, \mathbf{s}')$ - Radiance gained from a source inside the volume

These four sources of radiance change are illustrated in Figure 3.5.

There are numerous different methods used to find solutions to the RTE. One possible way of solving the RTE is by expanding the radiance into series of N spherical harmonics [50]. Another is to use the discrete ordinates method in which the radiance is discretized into discrete fluxes [32, 51], a method also used in paper optics [52]. The systems of equations produced by these methods are both complicated and computationally demanding. The RTE is usually hard to solve analytically and is therefore often solved numerically using the Monte Carlo technique or approximated using, e.g., the diffusion equations. The following subsections will briefly discuss the last two mentioned techniques, Monte Carlo and the diffusion equations, as they are of most relevant for this thesis.

3.6.2 Diffusion Theory

By making appropriate assumptions about the behavior of the scattering medium, the number of independent variables of the RTE can be reduced. This can, e.g., lead to the so called diffusion theory, or diffusion approximation, and the diffusion equations. A more detailed description on the subject can be found in reference [23]. Diffusion theory is going to be briefly covered here as it is an alternative method of modeling anisotropic diffusion media.

The diffusion theory is a special case of the spherical harmonics method where only the first term is used $N = 1$, and its essential equation is the reduced scattering coefficient

$$\mu_s' = \mu_s(1 - g). \tag{3.58}$$

This means that μ_s and g are inseparable using the diffusion approximation and this is often referred as the *similarity relation*. Equation 3.58 implies that the anisotropic behavior of a forward scattering medium $g > 0$ will be approximated into a medium with isotropic scattering $g = 0$ with a larger mean free path between scattering events. However, this limits the validity of the diffusion equations for different scattering media. The most commonly recognized limitations are that the media has to be weakly absorbing, $\mu_s' \gg \mu_a$ and that it can be inaccurate close to the light source. Diffusion theory is primarily used to model semi-infinite turbid media even though it is possible to use it when modelling scattering through a slab. This is because another limitation is that diffusion theory does not hold when the thickness of the slab is in the same order of magnitude as the mean free path [53].

3.6.3 Monte Carlo

The Monte Carlo method is a stochastic technique that has become a golden standard in light scattering literature. It is based on tracking a large amount of possible particle paths through the medium and is an exact solution to the RTE when a infinite number of particles are traced. Monte Carlo will be covered in more detail in the following chapter.

3.7 Anisotropic Scattering Media

A scattering media having asymmetric particles, where their orientations exhibit directional alignment, will have directional-dependent scattering. This means that light diffusion will vary depending on the propagation direction in the structure. There are two main reasons for why anisotropic diffusion occurs for aligned structures: the scattering cross section of the particles and the scattering phase function can both be directional-dependent. This has already been stated regarding the cylindrical particle which has a directional-dependent scattering efficiency $Q_s(\zeta)$ and a phase function $p(\zeta, \theta)$ which scatters in a conical shape around the cylinder.

There are primarily two methods to model anisotropic diffusion, either using the Monte Carlo method or using anisotropic diffusion theory. For fiber-based materials over the last decade there has been a growing interest to utilize the Monte Carlo method and phase functions from infinite cylinders. It was originally suggested by Kienle et al. [54], and materials such as dentin [55,56], muscles [57], tendons [58], softwood [59], and textiles [60] have been covered since. Other anisotropic scattering media where the particle shape often is somewhere in between that of a sphere and a cylinder have traditionally been estimated using anisotropic diffusion theory. Examples are aligned liquid crystals [61,62], etched gallium phosphide [63], wood [64], plastics [65] and granular materials [66]. It has been debated whether or not the anisotropic diffusion theory provides a correct solution to the RTE [67–69]. However, Alerstam have recently shown that anisotropic light diffusion works equally well as the Monte Carlo method, within the limits of diffusion theory, when applying correct boundary conditions [70]. This makes diffusion theory an excellent option for many anisotropic structures.

The work in this thesis has mainly focused on the Monte Carlo method rather than the anisotropic diffusion equations. The primary reason for this is that the diffusion equations can become inaccurate for very thin turbid medias such as, e.g., paper-sheets. As cellulose fibers resemble infinitely long cylinders, it is also likely that they also produces realistic phase functions. Both the phase function and orientation distributions of cylinders are very intuitively implemented with the Monte Carlo method making it the most suitable option.

3.7.1 The Different Degrees of Anisotropy

The term anisotropy can be found in various places within scattering literature, including this thesis. Especially when anisotropic scattering media, such as fiber-based materials are introduced, it can become confusing to which kind of anisotropy that is referred to.

Anisotropic scattering, or directional-dependent scattering, describes how the light scatters at each scattering event. The Henyey-Greenstein phase function is one example of this where an asymmetry factor of $g = 0$ describes isotropic scattering while any $g \neq 0$ is referred to as anisotropic scattering. Similarly, particles in the Rayleigh regime scatter isotropically, i.e., with equal probability in all directions while larger particles described by Mie theory scatter anisotropically.

Anisotropic scattering media, or anisotropic diffusion, refers to to the scattering medium itself which exhibits directional-dependent properties. Note that directional-dependent scattering by single particles in a random structure would not cause anisotropic diffusion. Both an asymmetric shape of the particles and a non-random alignment of the structure is required.

Anisotropic properties of the scatterer can also be considered. Peng et al. [60], for example, used analytical solutions to scattering from cylinders with different refractive indexes along the longitudinal axis and radial direction. **Paper E** in this thesis considers fibers that can be regarded as anisotropic since the refrective index of the mantle and core regions of the cylinder differ.

Chapter 4

The Monte Carlo Method

The Monte Carlo method is a numerical method that relies on sampling of stochastic variables to estimate solutions to a complex problem. The Monte Carlo method may be applied whenever it is possible to establish equivalence between the desired result and the expected behavior of a stochastic system. While the RTE is hard to solve analytically for most light diffusion problems, information regarding the behavior of individual scattering events is still, as shown in **Chapter 3**, fairly easy to describe. This makes the Monte Carlo method an excellent alternative for finding exact solutions to the RTE. It has in fact become the standard method in light diffusion problems, and this method it is often used to evaluate other approximate-solution methods to the RTE. The basic principle of this simulation technique is to regard the electromagnetic waves as particles, where their behavior is controlled by various probability functions driven iteratively by random numbers. It basically reduces the RTE to a problem were bouncing particles are tracked until they leave the considered scattering medium.

4.1 Monte Carlo Simulations of Electromagnetic Wave Propagation

The Monte Carlo method originated during the 1940s within the Manhattan project were Metropolis and Ulam introduced it when solving neutron diffusion problems related to research involving the hydrogen bomb [71]. Since then the Monte Carlo method has spread to many different fields in both physics and mathematics [72]. For electromagnetic wave propagation the principle was already used in the 60s to describe multiple scattered light in the atmosphere [73,74]. However, it was not until Prahl [75] and Wang et al. [76] introduced the simulation technique in biomedical optics that it started to spread to other fields outside atmospheric physics. Their work, based on the Henyey-Greenstein phase function, has served as a base for numerous Monte Carlo algorithms that exists today. It is also possible to use phase functions calculated using, e.g., Mie theory or Rayleigh scattering and then apply the Stokes formalism to track the state of polarization

[77–79]. Different methods have been suggested to track polarization and intensity for both transmission [80] and backscattering [81]. A good summary of the different methods to keep track of the state of polarization using the Monte Carlo method can be found in work done by Ramella-Roman et al. [82, 83]. Monte Carlo simulations using cylinders as scattering particles was introduced by Kienle et al. as a possible way to describe anisotropic diffusion [54, 55]. A Monte Carlo algorithm for both scattered intensity and polarization was later presented by Yun et al. [84].

The basic principle is to reformulate the radiative transfer theory in Equation 3.57 and consider the irradiance L as transport of scalar particles. By then applying the Monte Carlo method the problem is reduced to tracking particles through the medium. The Monte Carlo method relies on random sampling from well-defined probability distributions. Probability density functions for both random walk, i.e., the Beer-Lambert law and scattered energy, i.e., the phase functions, are used to describe the behavior of the electromagnetic field.

By tracking a large number of particles the statistical nature of the probability density functions will eventually converge and quantities such as total and spatially resolved transmittance and reflectance, distributed absorption and degree of polarization can be simulated. If the number of tracked particles, N, is allowed to approach infinity an exact solution to the RTE will be acquired. However, an finite number of particles is always used in practice, and it will therefore be statistical noise in the solutions. If the number of particles N is large enough their behavior can be considered a good estimate for the entire problem, and the noise can be considered insignificant.

There are often misinterpretations about what these simulated particles really are and how they should be regarded in these kinds of simulations. They are often referred to as photons though they have nothing to do with photons or any physical particle at all. Rather than a particle nature of light it should be viewed as a quantization of the electromagnetic field. Further, the Monte Carlo method is also only a mathematical way of solving the RTE and not a real physical process where particles bounce around. A detailed description regarding these misconceptions, along with a few others, can be found in work done by Mishchenko [85].

4.2 Random Numbers and Probability Distributions

"Any one who considers arithmetical methods of producing random digits is, of course, in a state of sin."

John von Neumann

The most essential part in the Monte Carlo method is the sampling of stochastic events from known probability distributions. In practice most random generation is deterministic, generated by arithmetic methods and usually referred to as *pseudo-random numbers*. This means that if the period of the generated random sequence is short, it can potentially start repeating itself during a simulation. In order to get good qualitative results in a Monte Carlo simulation, $10^4 - 10^9$ tracked particles are usually required.

Each tracked particle in turn often requires in the order of some tens to hundreds of random numbers to generate a particle path. This makes the series of random numbers required potentially high, and some consideration should be taken on the pseudo-random generation routine. Many computer languages have built-in random number generators that in many cases are far from optimal. For example, the standard random number generator in the Fortran programing language was made in the 1970s when requirements where quite different from today [86]. The period of its random sequence is therefore very short, which makes it unsuitable for large Monte Carlo simulations. The performance of most modern pseudo-random number generators far exceeds the requirements of this kind of simulations, but some consideration regarding the pseudo-random number generator should be taken.

The following subsection will describe how to use pseudo-random numbers to achieve parameters such as step-size or scattering angles from the essential probability density functions in Monte Carlo simulations of electromagnetic wave propagation.

4.2.1 Density Functions

A *probability density function* $P(\chi)$ describes how a variable χ varies stochastically. It is a non-negative function in an interval $[a, b]$ that specifies the probability that a random sample ξ will be within an infinitively small interval of χ.

It is, as we shall see, more convenient to use the normalized *cumulative density function* defined as

$$F(\chi) = \int_a^{\chi} P(\chi')d\chi' \tag{4.1}$$

as it is a direct measure of probability. It is a constantly increasing function in the interval $[a, b]$ were the value of $F(a) = 0$ and $F(b) = 1$. The value $F(\chi)$ hence represents the probability that a random sample ξ will assume a value between a and χ.

Step-size

Recall the decay of energy of a propagating field in a scattering and absorbing medium is described by the Beer-Lambert law in Equation 3.55. Assume that this equation is a probability density function

$$P(\Delta s) = L_0 e^{-\mu_e \Delta s} \tag{4.2}$$

where, if it is assumed that $L_0 = 1$, it describes the probability that a particle will propagate the distance Δs. By analytically calculating the cumulative density function it is now possible to generate a variable step size with a uniform distribution of random numbers. Set a random number ξ in the interval $[0, 1)$ equal the cumulative density function such that

$$\xi = \int_0^{\tau} e^{-\tau'} d\tau' e^{-\tau'} d\tau' = [-e^{-\tau'}]_0^{\tau} = 1 - e^{-\tau} \tag{4.3}$$

where $\tau = \mu_e \Delta s$ is the dimensionless optical thickness. Equation 4.3 can now be rewritten as

$$\tau = \mu_e \Delta s = -\ln(1 - \xi) \tag{4.4}$$

and the variable step-size become

$$\Delta s = \frac{-\ln(1 - \xi)}{\mu_e}. \tag{4.5}$$

This expression directly describes the step-size generated by a random number based on the probability that a particle will interact in a given propagation distance. This is, together with the treatment of absorption and the generation of new propagation directions, one of the essential equations in Monte Carlo simulations of electromagnetic wave propagation.

Absorption

The generated step-size in Equation 4.5 describe the interaction from both scattering and absorption. A standard method, at each interaction site, is to check if the particle will scatter or get absorbed. This can be done by generating a random number to decide that the particle will be scattered if

$$\xi < \frac{\mu_s}{\mu_a + \mu_s} \tag{4.6}$$

and absorbed otherwise.

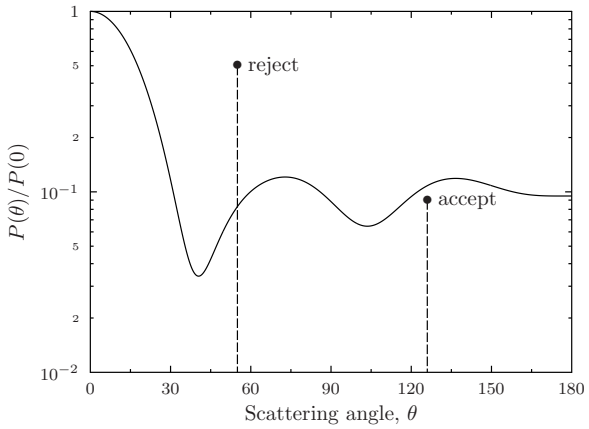
An alternative, more efficient way, is to use variance reduction techniques for the absorption. The principle is to introduce a dimensionless energy or "weight" of the particle and then reduce this weight at each interaction event using

$$\Delta W = \frac{\mu_a}{\mu_a + \mu_s} W. \tag{4.7}$$

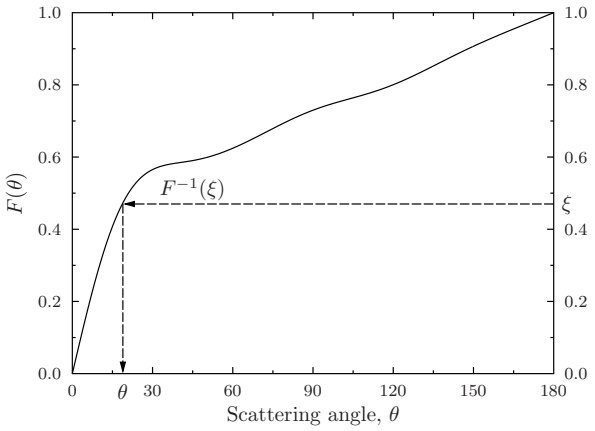
where W describes the current weight. This is in principle equivalent to having many particles propagating over the same path where the absorption now is distributed to each scattering event. Rather than terminating a particle after a significant amount of computational time spent propagating in the medium, it enables the particle instead to leave the medium with reduced energy. This technique is called *implicit capture* and was originally described by Witt [87] to reduce the variance in Monte Carlo simulations, and it is a standard method found in a majority of Monte Carlo algorithms today. An alternative method in time-resolved simulations is to add absorption afterwards using the Beer-Lambert law as the time, or distance, each particle has propagated is known [88–90].

Scattering direction

The next step is to consider the new scattering direction $\mathbf{s}'$. The phase function $p(\mathbf{s}, \mathbf{s}')$ is now reinterpreted as the probability density function P of the new scattering direction.



(a) Probability density function



(b) Cumulative density function

Figure 4.1: A probability density function in (a) illustrates how the rejection method is used to generate a scattering angle. The scattering angle can also be retried using the corresponding cumulative density function illustrated in (b).

It is usually not a given case that you will end up with an analytical expression as in Equation 4.5. For example, using phase functions from analytical solutions of Maxwell's equations does not provide a simple analytical expression but rather a numerical estimation to an infinite series of expressions. Stochastic methods such as the rejection method or numerical solutions to the cumulative density function instead can be used to generate the wanted parameters.

The rejection method also relies on sampling of random numbers and is, in fact, a Monte Carlo–based solution method. If we consider a phase function that only depends on one scattering angle, e.g., the deflection angle for spherical particles or the angle that describe the conically shaped scattering around a cylinder, we can define the probability density function over this angle θ. Consider two random variables, a scattering angle generated uniformly with the random number ξ_1 and a second random number ξ_2 generated between 0 and the maximum value of $P(\theta)$. If this value now lies below the value of the probability density function for the scattering angle such that

$$\xi_2 < P(\xi_1) \tag{4.8}$$

then ξ_1 is accepted as the new scattering angle. Otherwise the process needs to be repeated using new randomized values of ξ_1 and ξ_2 until the condition is satisfied. An illustration of a phase function and the rejection method can be viewed in Figure 4.1(a). Note that while the rejection method is exact, it will be slow for phase functions that have a sharp peak because the majority of the generated values are discarded.

An alternative method is to numerically calculate the cumulative density function for the scattering angle $F(\theta)$ and to invert it so that $F^{-1}(\xi) = \theta$. A random integer ξ between the first and last entry of the array can therefore serve as an index, which selects a new scattering angle θ directly. Figure 4.1(b) illustrates how a random number or random index generates a scattering angle θ. Note, however, that there can be discretization issues using this technique, and some caution should be taken. There are also issues if polarization is considered as the phase function change for different polarization states. The rejection method is therefore, even though it can be a lot slower, still often preferred because it is more robust and easier to implement.

4.3 Basics of the Monte Carlo Method

Even though more detailed descriptions of MC algorithms can be found in many references [75, 76, 82, 84], a small summary will be presented here. A simple illustration of the process of tracing different paths with different outcomes can be found in Figure 4.2. The four most essential step are listed below.

Initializing the particle

The initialization part depends on the size and shape of the light source according to the RTE source term $q(\mathbf{r}, \mathbf{s}')$ in Equation 3.57. The initial position and the propagation direction of the particle need to be defined along with its Stokes vector. The most

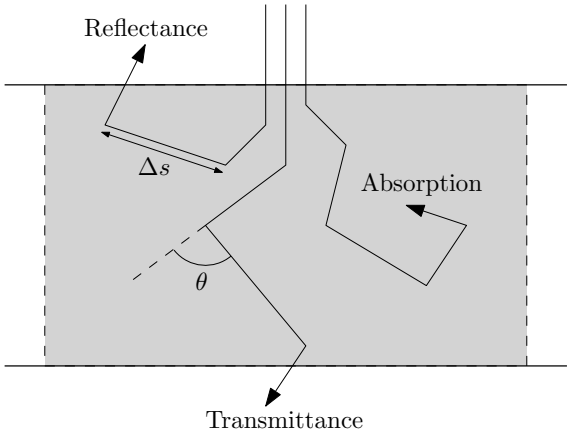


Figure 4.2: Illustration of particle propagation using the Monte Carlo method for three different paths with three different outcomes. Both the step-size Δs and scattering angle θ are also illustrated.

commonly used sources includes a infinitively narrow beam, also known as a pencil beam or, e.g., Gaussian-shaped distributions generated to resemble the beam profile of a laser. Any source can easily be implemented into the Monte Carlo method as it is simply a question of generating positions and vectors in space that describe the intial propation.

Move the particle

The distance a particle propagates in the medium before interacting with a scatterer is described by the step-size, Δs, from Equation 4.5. If the particle cross the border of the current scattering medium it will either be detected or internally reflect depending on the refractive index mismatch at the boundary. Surface models are usually based on Fresnel's and Snell's laws as described in, e.g., [91]. For porous structures, where the background medium in the bulk is air, no internal reflections will occur and the particle can be detected directly. If the particle is still inside the medium it will loose energy according to Equation 4.7 before the new scattering direction is generated.

Scatter in new direction

After the particle has moved a step within the medium it will scatter according to the phase function $p(\mathbf{s}, \mathbf{s}')$. This is done differently depending on which kind of scatterer is considered. For spheres, or the Henyey-Greenstein phase function, this is done by generating a deflection angle θ from the phase function and a uniformly generated azimuth angle. For a cylinder it depends on the orientation of its longitudinal axis, where the phase function generates a direction in the scattering cone. After the new propagation

direction $\mathbf{s}'$ is set, another step inside the scattering medium is taken and the process of moving and scattering is repeated until the particle leaves the medium.

Detection

When the particle leaves the scattering medium it needs to be detected. This is, again, a very intuitive process since it is merely a question of detecting vectors in space. This makes it simple to simulate the positions of a camera and specify the detected solid angles for spatially resolved simulations. After each detected particle another one is initialized until the specified total number of particles has been simulated.

4.4 Multiple Scattering by Cylinders

A few additional aspects need to be considered if the scatterers are cylinders. As mentioned in **Chapter 3**, anisotropic diffusion is generated when the structure is anisotropic, i.e., directional-dependent. To describe this a model of the structure is obviously required. Kienle et al. [55] suggested that a Gaussian distribution of the fiber orientation around a main axis can model fiber alignment. Another alternative is to align all cylinders perfectly in one direction and add isotropic scatterers to match the measured results [58, 59].

It has previously been mentioned that anisotropic diffusion can be tracked back to two fundamental properties of the scattering medium. The direction-dependency is usually generated by a combination of the phase function $p(\mathbf{s}, \mathbf{s}')$ and the scattering efficiency $Q_s(\zeta)$. While the new scattering directions generated by the phase function can be considered quite easily, it can be a bit more obscure to consider what impact Q_s has. During a Monte Carlo simulation with aligned cylinders, once a new propagation direction is chosen, a vector which describes the orientation of the next cylinder with which to interact is generated. Hence, an update of the current scattering coefficient is required using the $Q_s(\zeta)$ generated by the propagation direction $\mathbf{s}$ and the orientation of the cylinder $\mathbf{l}$. The effect that having a distributed orientation of the fibers has on the scattering efficiency is therefore not directly apparent. To display the effect of a fiber orientation distribution G has on Q_s, a convolution between them can be computed

$$Q_s^* = Q_s * G. \tag{4.9}$$

Figure 4.3 illustrates how the effective Q_s^* change for different fiber distributions. Here σ is the standard deviation of a Gaussian distribution of the fiber orientations. Note that in this example, the angle ζ describes the angle between the main fiber axis and the propagation direction of the particle. A particle propagating parallel to the main axis, i.e., when $\zeta = 0°$, will have no chance to interact if all cylinders are aligned perfectly with the main axis. However Q_s^* becomes more and more leveled out as the anisotropy of the structure decreases using distributions following $\sigma = 10°$, $20°$ and $40°$. A completely random fiber distribution would result in a constant value of Q_s^* over all angles ζ. Note that the difference, if a particle propagates parallel to the main axis $\zeta = 0°$ and perpendicular to it $\zeta = 90°$, is still fairly large when $\sigma = 40°$. Thus, it is clearly shown that the

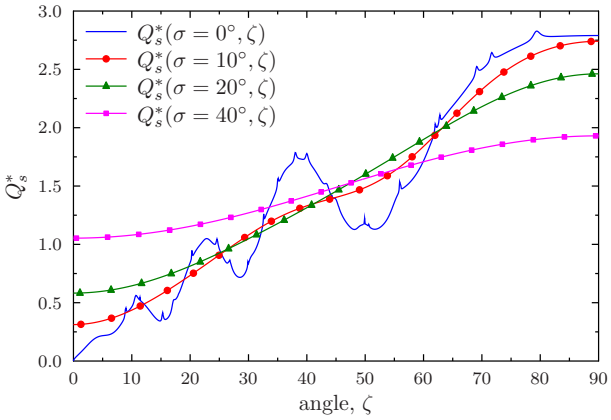


Figure 4.3: The behavior of the scattering efficiency Q_s^* in a medium with cylinder orientaitons following Gaussian distributions with different standard deviations σ.

anisotropic diffusion effect from the scattering efficiency, especially in structures with a predominant orientation axis, is significant.

4.4.1 Parameter Estimation

A key part in validating simulation models is to use correct parameters. Parameters such as μ_s, μ_a and $p(\mathbf{s}, \mathbf{s}')$ are often estimated iteratively to values of total transmitted and reflected intensity measured using an integrating sphere [91]. The simulations in **Paper C** used this principle from measurements done by Arney et al. [92].

Validation of a model can also be done by creating phantoms with controlled particle compositions. To validate scattering models with spherical particles different mixtures of uniformly sized latex spheres in water are usually created, and these scattering media are referred to as *mono-disperse* suspensions. Thus, the phase function and scattering efficiency are nearly invariant. The number density of the mixture can then be calculated as

$$N_s = \frac{\rho_s - \rho_{bg}}{V(\rho - \rho_{bg})} \tag{4.10}$$

where ρ_{bg} is the density of the background medium, ρ the density of the scattering medium, ρ_s and V are the density and volume of the spheres. The scattering coefficient can then be computed using Equation 3.28 and 3.52 as

$$\mu_s = Q_s N_s \pi r^2. \tag{4.11}$$

where r is the radius of the spheres.

This is more complicated for fiber materials as additional consideration needs to be taken on shape and structural parameters. However, by assuming a uniform particle shape, backward estimations from the density are also possible for cylinders. The benefits of using this approach is that that the scattering coefficient is directly calculated from the structure of paper, something few other models can do. Infinitely long, straight, homogeneous cylinders obviously cannot physically have a number density, and instead, the *cylinder density*, C_a, defined as the total length of cylinders per volume unit $[m^{-2}]$, can be applied. The cylinder density can hence be calculated as

$$C_a = \frac{\rho_s - \rho_{bg}}{A(\rho - \rho_{bg})} \tag{4.12}$$

where A is the average cross-sectional area of the fibers. The scattering coefficient can then be computed as

$$\mu_s(\zeta) = Q_s(\zeta) C_a d. \tag{4.13}$$

A rough estimate of the scattering coefficient can thereof be done by an rough assumption regarding the size distribution of the fibers; this was done in **Paper D**. Note that this is only possible for scattering media that are close to mono-disperse, i.e, it is not applicable to unknown mixtures of particles. The sheets used in **Paper D** where therefore custom made to contain primarely fiber of fairly uniform sizes. It must also be noted that the methods used to estimate the cylinder density both in **Paper C** and **Paper D** produced values in the same order of magnitude.

Chapter 5

Optical Properties of Paper

There are many optical properties that can be considered in a paper-sheet. As the optical appearance is one of the key aspects in many paper grades, it is also the field in which most research has been done. Optical properties of paper such as transparency or opacity, whiteness, color and gloss are all considered parameters. The interaction between ink and paper, the so-called Yule Nielsen effect, or optical dot gain, is another effect that is often studied [93]. A review on paper optics and the complexities regarding the appearance of paper can be found in [7].

Light scattering and absorption have traditionally been calculated by paper-makers using Kubelka-Munk theory [94]. The benefits of the the Kubelka-Munk theory are that it is quick and invertible, which makes it suitable for industrial applications. By knowing the optical properties of the materials added to the pulp it is possible to predict the optical properties of the paper-sheet. It must be noted that there are several shortcomings of the Kubelka-Munk theory because it is only a one-dimensional, two-parameter model that only considers isotropic light scattering. To fully understand and predict the appearance of paper, researchers have instead turned to radiative transfer theory [52, 95].

Paper is a complex scattering medium, especially when a lot of fines and filler materials are added. The amount of scattering depends heavily on the amount of air-fiber interfaces, or total surface area, in the structure where, e.g., pore sizes govern scattering for high density papers [96, 97]. It has been suggested that in this case paper can be regarded as a solid matrix where the pores act as scattering sites. On the other hand if we have a porous paper that contains primarily fibers, it is reasonable to assume that individual fibers will act as scatterers. Models for light scattering in paper using the Monte Carlo method have been used to describe time-resolved diffusion [98], optical coherence tomography [99, 100], dot gain [101], fluorescence [102], and spatially resolved transmittance [103]. All the above mentioned models use rotationally invariant single scattering phase functions, and none considers scattering by actual fibers.

As previously mentioned in **Chapters 3** and **4**, an intuitive way to address directional-dependent scattering is to include it through modeling the fiber alignment. This in turn requires models where particle-level scattering of the fibers is considered. Models which

rely on geometrical optics have this capability, see, e.g., Green et al. [104] and Coppel and Edström [105]. Obviously models using particle level scattering and infinitely long, straight, homogeneous cylinders can also directly be used for this purpose. They also have the benefit of not being limited by the ratio between particle size and wavelength. However, phase functions derived from analytical solutions of Maxwell's equations are limited to a few geometrical shapes and thereof lack the possibility to include, e.g., non-cylindrical shapes of the fiber cross-section.

5.1 Cylinders as Scatterers in Paper

It is safe to say that paper hardly can be regarded as an independently scattering medium because the fibers are in contact with each other. However, as paper consists of long fibers, where the distance between each contact-point usually is fairly large in comparison to the wavelength, such an approximation can still be a valid alternative. A reasonable assumption is that paper can be viewed as a material where a majority of the scattering will behave as independent scattering events. One can expect that any model using this assumption will decrease in accuracy when the density of scatterers increases. The addition of fines and filler materials will also make scattering more complex until the independent scattering approximation eventually breaks down as their volumetric conentration increases.

As mentioned, many optical properties can be considered for a paper-sheet. A model using particle level scattering by cylinders may not be optimal to address all aspects of paper optics directly, However, it is suitable for both lateral and anisotropic diffusion as they depend on the fiber alignment. Another area of application for such a model is when estimation of particle composition is of interest as, e.g., in pulp consistency measurements where fibers are mixed with smaller particles.

5.1.1 Lateral Scattering

Lateral scattering is of key interest when considering printing on paper. Printed dots appear larger due to lateral light propagation of the incident beam in the vicinity of the printed dot. This phenomenon is therefore referred to as optical dot gain, and is illustrated in Figure 5.1. It is not completely obvious how to model lateral diffusion and compare it to measurements. While a point source, or a *point spread function*, PSF, is easy to simulate, it is impossible to illuminate a sample using an infinitely small point source. Instead, different methods for measuring the *modulation transfer function*, MTF, in paper have been suggested, and a good review on both dot gain and MTF measurements can be found in [106]. One method for measuring lateral scattering is to use a knife-edge projection and detection of spatially resolved reflectance perpendicular to the edge, the so called *edge spread function*, ESF [92]. Note that Monte Carlo simulations of spatially extended sources, as for the ESF, do not necessarily need to require much more computational power than a PSF, since spatial convolutions of the PSF and the incident source can be used [107]. The derivative of the ESF can then be used to calculate

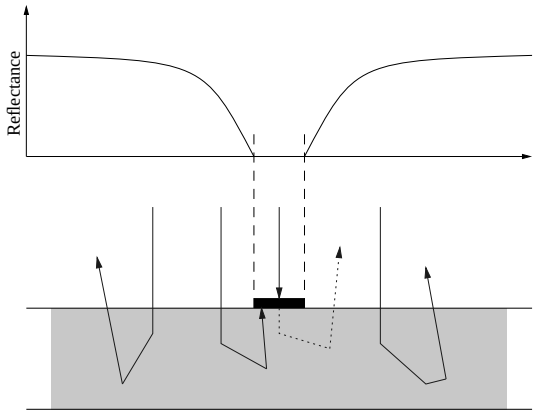


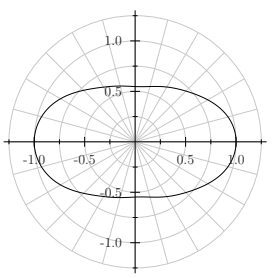
Figure 5.1: Illustration of optical dot gain generated by lateral scattering where the dot absorbs all light propagating into it. Incident light blocked by the dot gives rise to blurry edges in the vicinity of the dot, and this is illustrated with a dashed propagation path. The figure is a remake of a illustration found in [109].

the MTF. It has been shown that scattering models using forward single scattering phase functions can not accurately predict the amount of lateral diffusion in paper-sheets [108]. More accurate models, which consider the directional scattering of fibers are therefore of interest for better estimation techniques and increased understanding of optical dot gain.

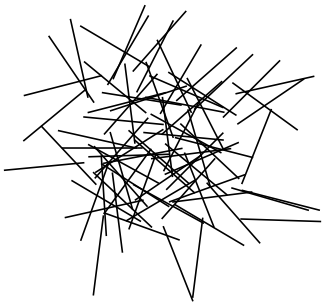
As the conically shaped scattering behavior of cylinders in aligned fiber structures enhances scattering along the main fiber axis, an enhancement is also expected in structures where fibers primarily are isotropically aligned in the plane. The topic of **Paper C** is to investigate how a model using cylinders can predict lateral scattering as compared with forward single scattering models using the Henyey-Greenstein phase function.

5.1.2 Anisotropic Diffusion

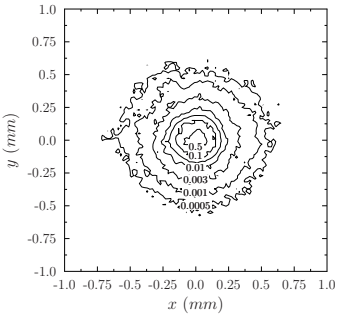
Anisotropic diffusion is analogous to lateral scattering in depending on how the fibers are aligned in the paper-sheet. However, in anisotropic diffusion the in-plane fiber alignment of the paper-sheet is also considered. In order to use anisotropic diffusion in on-line sensing a practical method is to use a point sources, rather than a knife-edge. The anisotropic shape of the point spread function can then directly be used as a measure of the fiber alignment. Even though a point source does not exist in practice, an incident beam width of some tens of μm can still easily be acquired. This can be done using an optical fiber to guide the incident source or focusing the beam using a lens system. Both **Papers B** and **D** investigate the effects of anisotropic diffusion in paper.



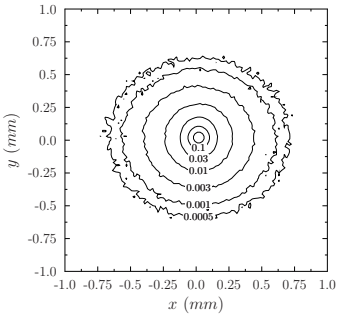
(a) Orientation ellipse



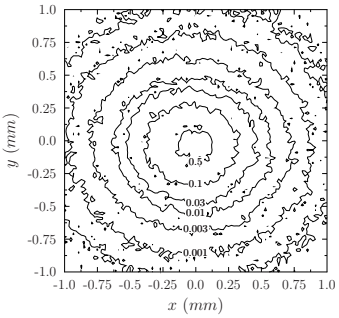
(b) Fibers



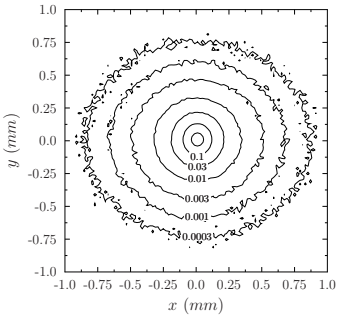
(c) Measured reflectance



(d) Simulated reflectance

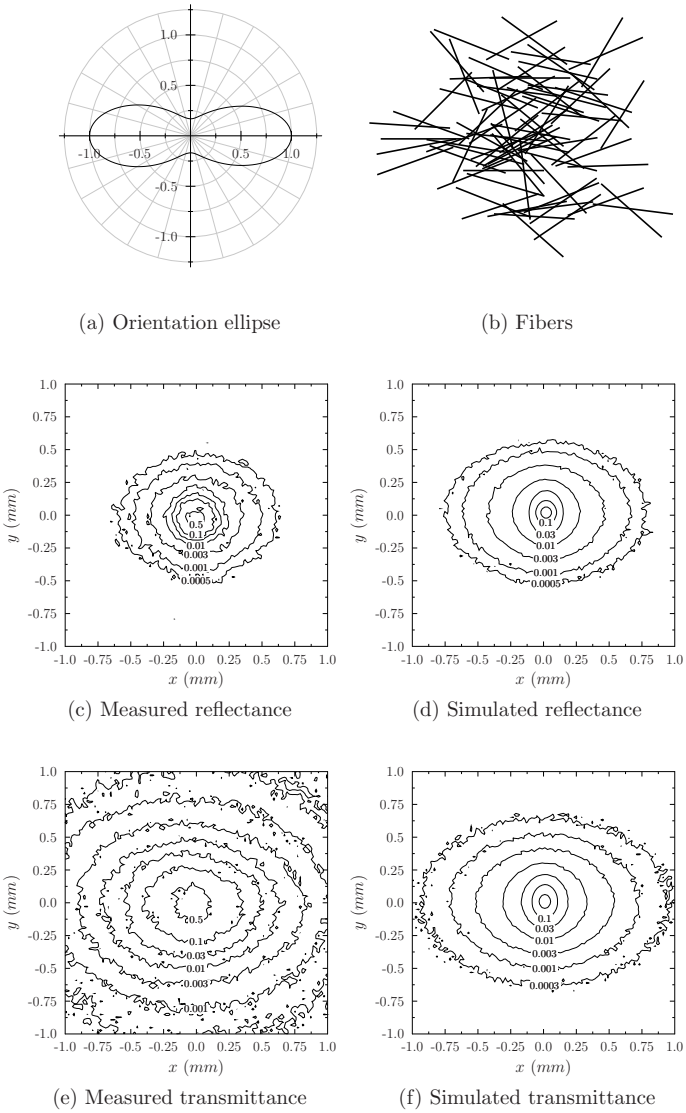


(e) Measured transmittance



(f) Simulated transmittance

Figure 5.2: Anisotropic diffusion illustrated for a paper-sheet with low degree of fiber alignment.



(a) Orientation ellipse

(b) Fibers

(c) Measured reflectance

(d) Simulated reflectance

(e) Measured transmittance

(f) Simulated transmittance

Figure 5.3: Anisotropic diffusion illustrated for a paper-sheet with high degree of fiber alignment.

Figures 5.2 and 5.3 illustrate anisotropic diffusion in paper-sheets using two different degrees of in-plane fiber alignment. These results are directly taken from paper-sheets investigated in **Paper D** to illustrate the effect of anisotropic diffusion in paper. The distribution functions of the fiber orientation in the plane of the paper-sheets were measured using x-ray computed tomography images and are illustrated using the orientation ellipses in Figures 5.2(a) and 5.3(a). They are then used to distribute 80 fibers in Figures 5.2(b) and 5.3(b) to further illustrate the fiber orientation differences in the sheets. Measurements and simulations of the spatially resolved reflectance and transmittance is shown in Figures 5.2(c)–(f) and 5.3(c)–(f). A fairly large difference can be observed in the spatially resolved patterns between the figures were a more profound elliptical shape can be seen in Figure 5.3. Note that the contour patterns use a logarithmic scale and that the elliptical shapes appear at low relative intensity levels.

Detection

In order to measure relative intensities of parts of thousands or even parts of tens of thousands, a good optical detection system is required. Rapid technological advancements of new imaging systems and sensor technology make optical measurement methods a field with a high development potential. Since sensing techniques are constantly being improved, this allows for faster, more accurate and less expensive measurement systems. Modern high-end CCD and CMOS sensors can today provide several millions of pixel resolution and dynamic ranges down to 16-bits of depth.

Single-photon technology is also a field in constant progress [110]. In recent years, single-photon avalanche diodes SPADs have revolutionized the potential dynamic range in optical measurements systems [111,112]. So far, measurements using SPADs have been shown to reach levels of dynamic range of about 8 decades [113]. Dynamic range of SPAD cameras [114, 115] have also been shown to be superior to CCD cameras, even though the acquisition times are orders of magnitude quicker. This development potentially provides significant improvements to the capability of the detection of both spatial and time-resolved diffused light.

Another consideration that needs to be mentioned is that background noise can be an issue, especially when trying to detect low-intensity levels. This obviouslly needs to be considered if low intensities are going to be used in an industrial environment. The camera setup used in the measurements in **Paper D** included a high perfomance bandpass filter with a sharp transmission peak matching the wavelength of the laser. This allowed for the measurements to be done in a lit laboratory environment as the wavelength of the lighting in the room was separated from the detected wavelength. The same principle can also be used in an industrial environment.

5.2 Scattering in Cellulose Suspensions

Aside from scattering in a finished paper-sheet it is also possible to determine pulp consistency using using optical sensors [116]. Scattering of light highly depends on the geometrical size and shape of the particles. Pulp consistency, i.e., the amount of fibers and fines, can therefore be correlated to both transmitted and reflected intensity. It is also possible to include linear polarization as fibers are known to depolarize light quicker than fines in normal pulp consistency ranges [117]. Even though these properties have been known for a long time, few or no light diffusion models consider, e.g., particles with different shapes and sizes. The differences in scattering and depolarization between cylinders and small spheres is therefore investigated in **Paper A**.

Aside from normal paper and pulp, another fairly new field of cellulose fibers has appeared. *Cellulose nanofibers*, CNF, and *cellulose nanocrystals*, CNC, have over the last decade become one of the most exciting future aspects for the wood and paper industry [118, 119]. CNF is manufactured by a mechanical separation of normal cellulose fibers into its smaller constituents. The process is often combined with chemical pretreatment to improve efficiency. Since CNF is a relatively new material, and little work on how to perform measurements for the manufacturing process have been reported.

It is not everyday that you get presented with an interesting optical phenomenon. At the Wood and Bionanocomposite Lab, Luleå University of Technology, the following observations has been made [120]: When they have an untreated pulp with fibers at low consistency it appears nearly transparent. Once the separation process of the pulp starts the mixture quickly become opaque. As the separation process progresses, and as the particles approach nano-scale sizes, the suspension again become transparent. Since particle level scattering using analytical solutions to Maxwell's equation is not limited by the size factor, it makes an excellent option to model this phenomenon using infinitely long, straight, homogeneous cylinders.

A simple explanation for this phenomenon is going to be discussed here along with simulation results supporting the hypothesis. Consider a 10 mm wide sample cell filled with water and cellulose fibers with a concentration by weight of 0.5%. A mono-disperse mix of the particles, or cylinders, will in the simulations span in radius from a few microns down to tens of nanometers. Simply put, the decrease in radius of the scattering cylinders is compensated by an increase in cylinder density C_a to maintain the concentration by weight. The cylinder orientation is assumed to be random and the wavelength of the incident light source is $\lambda = 633\ nm$. Absorption is also assumed to be insignificant and set to zero. Results from Monte Carlo simulations of the total transmitted light intensity of the size spectrum is shown in Figure 5.4. The same phenomenon, where the medium first is fairly transparent for particle sizes $r > 10\ \mu m$, opaque at an intermediate region $0.1\ \mu m < r < 10\ \mu m$ and transparent again as particles approach nano-scale, $r < 0.1\ \mu m$, can be observed.

The explanation to this can in large part be found in Figure 3.1. Large particles, which have a strong forward scattering phase function, combined with having the scattering material localized in large particles and thereof a small scattering coefficient, cause the

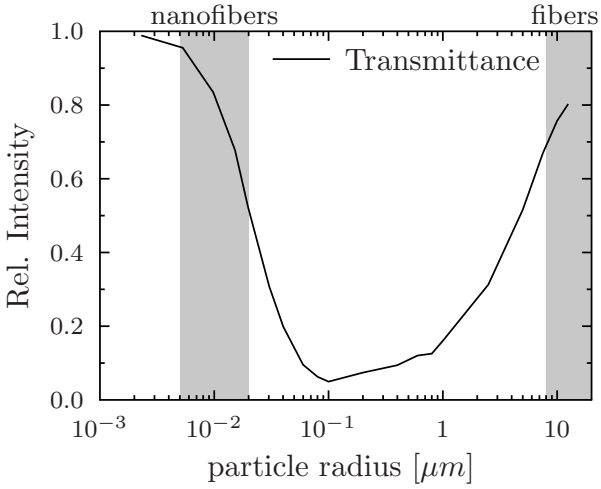


Figure 5.4: Simulation of transmitted light from scattering media consisting of randomly oriented cylinders where the volume density of particles is kept constant while the particle size decreases.

medium to be highly transmitting. As the particle size decreases, the scattering phase functions become more isotropic, but there is also an increase of the scattering coefficient. This is a known phenomenon that also can be realized by considering that the amount of scattering interfaces per volume unit increases as the particle size decreases. However, there is a turning-point at which the cylinder radius approach sizes of some tens of nanometers and become a lot smaller than the wavelength. Less of the light is going to be scatted as the scattering efficiency Q_s decreases rapidly as the size factor approaches the negligible scattering regime.

Even if these simulations are oversimplified, because they assume an unrealistic size decrease using mono-disperse particles, the turning-point occurs at a very promising region. This means that this effect could be used in an optical measurement system for process control of the manufacturing process of nano-cellulose. The proposed technique can be improved further by including polarization in both spatially resolved and time-resolved measurements.

Chapter 6

Summary of the Papers

6.1 Papers Included in the Thesis

This sections presents a short summary of the appended papers included in this thesis.

Paper A - Monte Carlo simulation of photon transport in a randomly oriented sphere-cylinder scattering medium

Summary

In this paper the composition of particles in a sphere-cylinder scattering medium is investigated numerically using the Monte Carlo method. The medium can be considered to represent a paper pulp suspension where the constituents are assumed to be microspheres, representing fiber fragments or fines, and infinitely long, straight, randomly oriented cylinders representing fibers.

In the simulations the total degree of transmittance and reflectance along with the associated degree of polarization for different concentrations of sphere and cylinder scatterers are investigated. The anisotropy factor is observed to vary little with the size of the cylinders for a random orientation. Instead, the dominant effect stems from the orientation-dependent characteristics of the phase function, where light scatters in a conical shape toward the cylinder longitudinal axis. The cones half-angle toward the longitudinal axis of the cylinder equals that of the incident light. This affects both transmittance and reflectance as light has a dominant tendency to scatter forward for most incident angles toward the cylinder. Cylinders were also found to depolarize light to a higher extent than spheres, even when the scattering anisotropy and size factor was similar.

In summary, a large difference between the scattering behavior of the investigated particles can be observed. The results give a good foundation for better understanding of optical particle consistency sensors.

Authors: Tomas Linder and Torbjörn Löfqvist

Published in: Applied Physics B: Volume 105, Issue 3 (2011), Page 659-664
Contribution: Work based on my simulations and analysis under the supervision of Torbjörn Löfqvist.

Paper B - Anisotropic light propagation in paper

Summary

In this paper light scattering by the anisotropic structure of a paper-sheet is investigated using both measurements and Monte Carlo simulations. Anisotropic diffusion was investigated by guiding light through an optical fiber to the surface of the paper-sheet. The model assumes that wood fibers can be represented as infinitely long, straight, homogeneous cylinders. The layer-like micro-structure and anisotropic orientation of the fibers in paper is considered using different orientation distributions of the cylindrical scatterers.

Measurements of spatially resolved transmittance of a standard kraft-liner product was found to have increased light propagation in the main fiber orientation direction. Elliptical iso-contour patterns of the spatially resolved transmittance was observed both experimentally and using Monte Carlo simulations. The directional-dependency of the scattering coefficient in the fiber structure is also investigated and found to depend on the fiber orientation anisotropy in the plane of the scattering medium. The conical scattering by cylinders along with the varying optical thickness is argued as the source of anisotropic scattering in paper.

It is concluded both using measurement and simulations that spatially resolved transmittance patterns contain measurable information of the in-plane fiber orientation. The observed anisotropic diffusion can therefore potentially be used as an on-line characterization technique of fiber orientation in paper.
Authors: Tomas Linder and Torbjörn Löfqvist
Published in: Nordic Pulp & Paper Research Journal. 27, 2, Page 500-506.
Contribution: Work based on my simulations, experiments and analysis under the supervision of Torbjörn Löfqvist.

Paper C - Anisotropic light propagation in paper

Summary

This paper investigates the influence of lateral light scattering from a scattering medium using infinitely long cylinders as scatterers. In total simulations of 22 different paper-sheets are compared to both measurements and corresponding simulations using rotationally invariant isotropic and strong forward phase functions of the modulation transfer function MTF. The layer-like structure of paper is considered by modeling the cylinder orientation isotropically in the plane and with a small orientation distribution in the thickness direction. The cylinder density C_a was estimated by matching measured values of total transmittance and reflectance values for each of the 22 samples. Simulations of the edge response from a sharp knife edge is then, via the line spread function, used to calculate the MTF.

It is found that the conical scattering by cylinders enhances lateral scattering compared with models using rotationally invariant single scattering phase functions. Compared with simulations using asymmetry factors $g = 0.0$ and $g = 0.8$, it is observed that the MTF has a more narrow shape when using cylinders, especially at low frequencies. The simulation results still slightly overestimate the measured values of the MTF, indicating that the model cannot fully describe the large lateral scattering in paper.

It is concluded that the cylinder phase function gives a good insight into how lateral light scattering in fibrous materials is generated. This further means that by using infinitely long cylinders as scattering particles optical dot gain in paper can better be estimated.

Authors: Tomas Linder, Torbjörnn Löfqvist, Ludovic G. Coppel, Magnus Neuman and Per Edström

Published in: Optics Express, Vol. 21, Issue 6, pp. 7835-7840 (2013)

Contribution: Most work in terms of simulations and writing was done by me. The idea of this paper originated as a continuation of previous work done by Ludovic G. Coppel, Magnus Neuman and Per Edström.

Paper D - Light scattering in fibrous media with different degrees of in-plane fiber alignment

Summary

This paper investigates the relation between in-plane fiber alignment and anisotropic light scattering. A set of paper-sheets was studied where the degree of fiber alignment in the plane varied from randomly oriented fibers, i.e., isotropic structural properties, to high fiber orientation along one axis.

Measurements of both spatially resolved transmittance and reflectance from a focused light source showed elliptically shaped intensity patterns where the main axis is aligned towards the fiber orientation. Light scattering was modeled using the Monte Carlo method where particle level scattering was considered using infinitely long, straight, homogeneous cylinders to represent the fibers. The micro-structure of the paper-sheets were modeled using distribution functions of the fiber orientations estimated using 3D X-ray computed tomography images. A strong correlation was found between the fiber orientation and elliptical shape of measured and simulated spatially resolved transmittance and reflectance iso-intensity patterns. In order to investigate this effect a least square fitting of an ellipse was used to match different relative intensity levels. The elliptical iso-conours of the measured and simulated scattering patterns were found to agree well. A close to linear relationship between the ellipse major and mino ratio and tensile strength ratio between the major and minor fiber orientation axis of the paper-sheets is observed.

These findings indicate that parameters such as fiber orientation and variations in fiber orientation can be measured on-line by monitoring scattered light in various fiber-based materials. As both spatially resolved transmittance and reflectance can be measured simultaneously it is also argued that it is possible to detect variations in the fiber

orientation between the top and bottom layers of the paper-sheet. It is also concluded that as the spatial resolved reflectance only starts to show an elliptical shape at very low relative intensity levels, none or very little influence from anisotropic diffusion should therefore be expected for optical dot gain in paper.
Authors: Tomas Linder, Torbjörn Löfqvist, Erik L. G. Wernersson, and Per Gren
Published in: Optics Express , Vol. 22, Issue 14, pp. 16829-16840 (2014)
Contribution: The idea and planing of the different experiments where made in cooperation between myself and Torbjörn Löfqvist. Per Gren aided in designing the experimental setup for the light scattering measurements, and the estimation of fiber orientation from the micro tomographed images was done by Erik L. G. Wernersson. The main part of the light scattering measurements, simulations and writing was done by me.

Paper E - Light scattering characteristics by infinitely long cylinders with a hollow core

Summary

When considering light scattering models utilizing particle level scattering it is obviously of interest to use a realistic description of the scattering behavior. Fibers rarely are of a completely homogeneous nature including, e.g., cellulose fibers that in most cases have a hollow core.

This paper investigates the scattering characteristics from an infinitly long, straight, hollow cylinder by solving Maxwell's equations analytically. The influence for the wall of the cylinder is analyzed and compared to infinitely long homogeneous cylinders. Differences in the phase function, extinction efficiency and Mueller matrix are all considered in this work.

It is found that the cylinder wall generates an increasing amount of resonance modes, especially when the particle size is significantly larger than the wavelength. It is also found that even though the phase function show a large extra peak induced by the cylinder wall, it still is not significantly different from a cylinder without a hollow core. However, the phase functions become more narrow as the cylinder wall gets thinner, and this effect is only prominent for extremely thin cylinder walls.

In general, the differences between the phase functions, scattering efficiency and Mueller matrix elements are small when a hollow is introduced in the cylinder. The only property that is affected greatly by the thickness of the cylinder wall is the amount of absorption.

It is concluded that, in most cases, the effects from a hollow core are small and only likely to be noticeable for a completely mono-disperse scattering medium. The infinitively long homogeneous cylinder can therefore be considered a valid approximation where multiple scattering is considered even though it has a hollow core.
Authors: Tomas Linder and Torbjörn Löfqvist
Published in: To be submitted.
Contribution: This work is based on my idea, analysis and writing under supervision of Torbjörn Löfqvist.

6.2 Other Work not Included in the Thesis

This section lists other academic work to which i have made contributions to but are out of the scope for this thesis. The open-source simulation tool that was created during the course of this work is also briefly mentioned in this section.

Analytical one-dimensional model for laser-induced ultrasound in planar optically absorbing layer

Authors: Erika Svanström, Tomas Linder,and Torbjörn Löfqvist.
Published in: Ultrasonics, Vol. 54, Nr 3, 2014, s. 888-893.

Combined physical and statistical modeling of laser induced ultrasound signals from thin light absorbing films

Authors: Erika Svanström, Tomas Linder and Johan E. Carlson.
Published in: IEEE International Ultrasonics Symposium: IUS 2013, Prague, Czech Republic; 21 - 25 July 2013 . Piscataway, NJ : IEEE, 2013. s. 2167-2170

FiberMC

Summary

The simulation tool used in this work has been published as an online open-source project under the GNU public license. It is a Monte Carlo code designed to solve the RTE numerically using particle level scattering. It includes three kinds of scattering particles, homogeneous spheres, infinitely long homogeneous cylinders and infinitely long concentric cylinders. As the particle level scattering is based upon analytical solutions to Maxwell's equations it is also possible to include polarization using the Stokes formalism. It can be used to calculate properties such as spatially resolved transmittance, reflectance, and the degree of polarization.
Author: Tomas Linder
Published in: http://sourceforge.net/projects/fibermc/
Last update: July 2014

Chapter 7

Conclusions and Future Work

This thesis investigates and utilizes modeling of light scattering in order to describe structural characteristics in fiber-based materials. It is focused on applications within the pulp and paper industry, but since the theory is general, it is likewise relevant and applicable to other fiber-based materials such as biological tissue, fiber composites and textiles.

7.1 Conclusions

The forward problem was investigated using particle level scattering from analytical solutions to Maxwell's equations for both spherical and cylindrical particles. The combination of these particles, spheres and cylinders, was then used to model the effect of particle composition on light scattering. By aligning the cylindrical particles it was also possible to address anisotropic diffusion in paper-sheets with anisotropic fiber alignment.

The three research questions, addressed in the thesis, are answered as follows:

- **Is it a valid approximation to simplify scattering by a cellulose fiber as an infinitely long, straight, homogeneous cylinder?**

  The answer to this question is simply yes. It can further be concluded as shown in **Paper E** that the additional influence of the hollow core does not significantly change the scattering behavior. It is hence probable that the geometric shape of infinitely long homogeneous cylinders induces the essential properties of scattering by a fiber. The model as it stands, using infinitely long, straight, cylinders to describe multiple scattering, therefore appears as a robust method that can provide useful predictions of the scattering behavior of various fiber-based materials.

- **Can a model using infinitely long cylinders as scatterers describe light diffusion in paper better than current models using rotationally-symmetric scattering?**

It is clearly shown in **Paper C** that a model using cylinders is able to better predict more realistic amounts of lateral scattering than models using rotational invariant single scattering phase functions. A model using cylinders as scatterers can therefore more realistically describe, e.g., optical dot gain in paper.

- **Can this model also be used to estimate particle composition in pulp and anisotropic diffusion in paper?**

  Both research questions can be addressed well considering the large amount of approximations involved. A model combining scattering by spheres and cylinders, as presented in **Paper A**, do possess the potential to be used to model scattering media consisting of different particles such as paper pulp. Anisotropic diffusion was found, as shown in **Papers B** and **D**, to agree well between model and measurements.

The results show that the approximation of using an infinitely long, straight, homogeneous cylinders to describe scattering in fiber-based materials has many benefits. It provides a simple, realistic and intuitive way of including scattering effects in fiber-based materials especially when the influence of the fiber alignment is of interest. However, there are limitations to the model as all scattering is assumed to be in the far-field and that it cannot include non-cylindrical shapes of the fiber cross-section.

Regarding the potential of using anisotropic diffusion for process control of the fiber orientations, several conclusions can be made. As shown in **Paper D** both spatially resolved reflectance and transmittance from a focused point source can be used. The elliptical shape of the iso-intensity patterns can then be used to both detect the direction, magnitude and variation, of the in-plane fiber alignment of the paper-sheet. Further, as both transmittance and reflectance can be measured simultaneously, it also opens up for detection of variations in the top and bottom layers of fibers in a paper-sheet. In order to address the inverse problem, iterative solutions for the forward problem are not feasible using the Monte Carlo method as it is very computationally demanding. A simple way of addressing the inverse problem can be to create a database of tabular values of, e.g., the elliptical shapes of the spatially resolved patterns. Doing so using repeated measurements covering all paper grades, including different thicknesses, densities, fiber types, and filler content, would be a extensive task. Using simulation results to create such a table, therefore, is a valid option.

7.2 Future Work

Even if the findings presented in this thesis present several promising aspects, there are still issues that need to be addressed and developed further.

This thesis has primarily investigated anisotropic diffusion using paper-sheets without fillers, fines, pigments and other additives, all of which needs to be considered. The influence from different kinds of cellulose fibers with varying sheet thicknesses and densities are also properties that can be investigated further. In order to fully determine the potential of a fiber orientation sensor, larger quantitative studies including different paper grades with different amounts of filler materials and fiber concentrations are required. Note also that the measurements presented in this thesis were conducted in a laboratory environment. To fully understand the potential in a new sensing technique, tests needs to be performed in an industrial environment. As has been mentioned several times in this thesis, the independent scattering approximation will eventually break down once the density of particles becomes to high. To explore the limitations of the model is therefore a subject for further studies.

It has also been suggested that the model presented in this thesis can be used to estimate scattering by nano-scale particles. As the manufacturing of nano-cellulose and nano-composites is a field of promising future aspects, it is of obvious interest. Investigations could, e.g., involve using light scattering as a tool to control the manufacturing process of nano-cellulose fibers or determine the structural characteristics of nano-composites.

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Part II

Paper A

Monte Carlo simulation of photon transport in a randomly oriented sphere-cylinder scattering medium

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Monte Carlo simulation of photon transport in a randomly oriented sphere-cylinder scattering medium

Tomas Linder and Torbjörn Löfqvist

Abstract

A Monte Carlo simulation tool for simulating photon transport in a randomly oriented sphere-cylinder medium has been developed. The simulated medium represents a paper pulp suspension where the constituents are assumed to be mono-disperse micro-spheres, representing dispersed fiber fragments, and infinitely long, straight, randomly oriented cylinders representing fibers. The diameter of the micro-spheres is considered to be about the order of the wavelength and is described by Mie scattering theory. The fiber diameter is considerably larger than the wavelength and the photon scattering is therefore determined by an analytical solution of Maxwell's equation for scattering at an infinitely long cylinder. By employing a Stokes-Mueller formalism, the software tracks the polarization of the light while propagating through the medium. The effects of varying volume concentrations and sizes of the scattering components on reflection, transmission and polarization of the incident light are investigated. It is shown that not only the size but also the shape of the particles has a big impact on the depolarization.

1 Introduction

Recently, there has been a growing interest in experimentally studying polarization in randomly oriented scattering media [1] to enhance image quality and improving the characterization of the media. The experimental studies are often backed up by Monte Carlo simulations to achieve a deeper understanding of the underlying physics. Monte Carlo simulations has over the years has proven to be an effective and accurate method for investigating light propagation in turbid media of various kinds. [2,3] The simulation technique has been used in studies of polarization in scattering media. Kattawar *et al*. [4] originally suggested that polarization from multiple scatterers could be calculated using Monte Carlo simulations. In the work by Bruscaglioni *et al.*, [5] the changes in polarization were investigated for light pulses that were transmitted through a turbid medium. Bartel and Hielsher [6] studied individual photons and introduced a method that uses local reference planes to keep track of the polarization of each photon through the simulation process. Different methods on how to track polarization based on previous mentioned models are thoroughly described by Ramella-Roman. [7,8]

Niemi [9] briefly touched upon the possibility to use the Monte Carlo technique on

paper pulp. However, this model lacked possibilities to model different particle shapes and materials as well as the wave nature of light and state of polarization. Considering paper, Monte Carlo modeling of light scattering has been used earlier. In Carlsson *et al.*, [10] time resolved optical scattering the paper is studied and simulations are validated through experiments. A similar model was used by Modric *et al.* [11] to better understand the origin of the optical dot gain. Kirillin *et al.* [12, 13] investigated the structure of paper using optical coherence tomography and created a model that could track the state of polarization using the Jones formalism. In Green [14] fibers were represented by a randomly generated fiber network and light scattering was calculated using Gaussian optics.

When modeling light scattering in paper pulp one has to consider the pulp to generally consist of two main classes of scatterers; wood fibers and fines. The wood fibers range in size from less than a millimeter to a few millimeters in length and their cross section is roughly cylindrical with a diameter of the order of 10 to 20 microns. The fines fraction consists of small fiber fragments as well as other small particles in which the size ranges from sub-micron scale to some tens of microns. Fiber length and diameter distributions depend mainly on wood species [15]. The fines fraction and its size distribution depend to a large extent on the processing of the pulp as well as the presence of different additives, such as pigments. Since the fines in general originate from fibers or other wood cells, the refractive index of both constituents are assumed to be the same. Thus, from a simplified modeling perspective, the paper pulp can be considered as a bimodal mixture of spheres and randomly oriented cylinders where the difference in sizes, and consequently the scattering mechanisms, has to be considered. The idealized medium used in this paper is modeled as a random mixture of infinitely long, straight, homogeneous, cylindrical fibers with isotropic material properties. The fine material is modeled as spheres with isotropic properties. In reality, the fibers in pulp generally have a non-cylindrical cross section, a hollow (lumen) and the cell wall is a layered structure with anisotropic properties. The fibers are finite in length and can be curved or have kinks. The fine material can have different shapes, ranging from slender wood fibrils to flake-shaped mineral particles. [15]

The scattering of light in Monte Carlo simulations are commonly described by either the Henyey-Greenstein phase function or by the Mie theory. Kienle *et al.* [16] introduced the scattering theory for infinitely long cylinders when modeling anisotropic light propagation in dentin. That model was later modified for biological tissue [17] and softwood [18] where cylinders acted as the main scattering component. The methodology for calculating multiple scattering by cylinders was recently described in detail. [19]

In a turbid medium, light scattering in general depends on the amount of scatterers per unit volume. This is true for paper pulp but one also has to consider the different scattering properties of the constituents. Zhao *et al.* [20–22] studied light scattering in paper pulp suspensions at different mass fractions of fibers and fines. The results indicate that fines exhibit almost isotropic scattering while fibers have a fairly strong forward scattering. Hence, the majority of the scattering from fines are probably close to the Rayleigh regime.

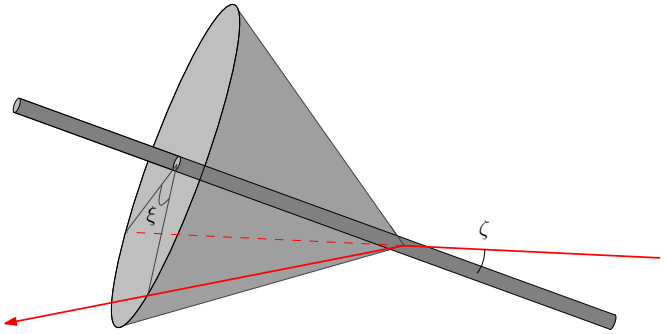


Figure 1: Light scattering characteristics by a cylinder.

The aim of this paper is to describe and evaluate a Monte Carlo simulation method developed to mimic the light scattering properties of paper pulp. A sphere-cylinder modeling approach for the Monte Carlo simulations is used. The impact on scattering from different volume fractions and distributions of the scattering components is investigated. Phase functions from cylindrical scatterers and from Mie theory are implemented to represent light scattering by wood fibers and small wood fiber filaments respectively. The main differences compared to previously published simulation models is the treatment of the size difference between light scattering constituents as well as the random orientation of the cylindrical scatterers. The simulations could potentially be applied in estimating the fiber and fines contents in pulp. The simulations of light scattering using the sphere-cylinder model could be compared to the light scattering from an illuminated pulp sample. A correlation of the results is then used to estimate the relative proportions of fibers and fines or other properties of the pulp.

2 Method

The principle of the Monte Carlo method for photon transport is to calculate a large number of photon trajectories as they propagate through the medium. Each photon will propagate between scattering events inside the medium until it either gets absorbed or leaves the medium. The sequential random walk between scattering events is determined by a probability density function based upon Beer-Lamberts law. Anisotropic light scattering is generally described by a scattering probability distribution. These distributions are commonly known phase functions and can be derived from Mie theory or approximated by, for example, the Henyey-Greenstein phase function. The phase functions for the cylindrical scatterers used in this study are obtained from an analytical solution of Maxwell's equations. [23] In addition, since we consider two different particles in our model, probability also decides on which particle the photon scatters at each event.

The angle between the direction of the incident photon and the longitudinal axis of the cylinder is termed ζ, see Fig. 1. Photons hitting the fiber at an incident angle ζ

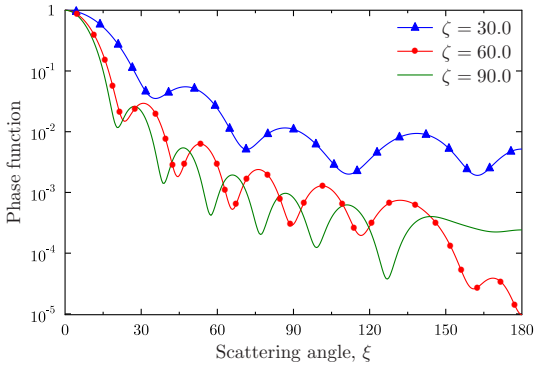


Figure 2: Phase functions over all scattering angles ξ for different incident angles, ζ.

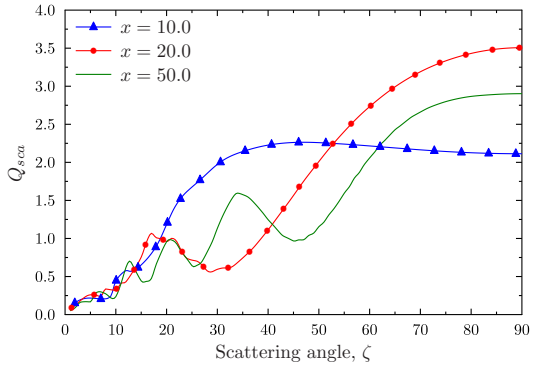


Figure 3: Scattering efficiency over all incident angles ζ for different size factors, x.

has a scattering probability function in a conical shape around the axis of the cylinder. The scattering angle will have the half angle to the fiber longitudinal axis. The phase function for a fiber is, in contrast to phase functions derived from Mie theory, orientation dependent. Therefore, a phase function has to be determined for each incident angle where each phase function is defined as the scattering around the cone by the scattering direction ξ.

As an example, the scattering intensity for a cylinder with a diameter of $d = 1.8$ μm, a refractive index inside and outside the cylinder of 1.5 and 1.36, respectively, and wavelength $\lambda = 625$ nm for three different incident angles ζ are shown in Fig. 2. For a perpendicular incident angle, $\zeta = 90°$, the light is scattered in a disc, rather than a cone, around the cylinder. This case is similar to scattering according to Mie theory.

In addition to the phase function, the scattering efficiency Q_{sca} has to be considered. It has to be noted that the phase function is a function of the photons incident angle toward the fiber which implies that the scattering efficiency also is a function of the incident angle. For example, a cylinder with its longitudinal axis close to parallel to that of the photon propagation direction will have a very small likelihood of impact. As an example, Fig. 3 shows the scattering efficiency for unpolarized light on three different size factors, $x = 10, 20, 50$ over all incident scattering angles, ζ.

The scattering coefficient of cylinders, $\mu_{s,cyl}$, is described as, [16, 24]

$$\mu_{s,cyl}(\zeta) = Q_{sca}(\zeta) d C_A \tag{1}$$

where d is the diameter and C_A is the density of cylinders (number per area). Q_{sca} is the scattering efficiency of the cylinder and depends on the incident angle ζ. Averaging the scattering coefficient over all directions is possible for randomly oriented structures but $Q_{sca}(\zeta)$ still has to be taken into consideration as a probability density function when choosing the scattering angle.

The single scattering from a cylinder is implemented into the Monte Carlo code to calculate the multiple-scattering characteristics of the mixture. The fibers are allowed to have any orientation; from axially orientated as in many kinds of tissue to random as for paper pulp.

The fines are assumed to have an anisotropic factor close to zero. The total extinction coefficient, μ_t, including both spherical and cylindrical scatterers is described as

$$\mu_t = \mu_a + \mu_{s,sph} + \mu_{s,cyl} \tag{2}$$

where μ_a is the absorption coefficient while $\mu_{s,sph}$ and $\mu_{s,cyl}$ are the scattering coefficients for micro-spheres and cylinders respectively. The scattering coefficient of micro-spheres is constant once the size, concentration and refractive index are set. This is also the case for the randomly oriented fibers while it would depend on ζ for a structure where all fibers are aligned. [16] Before each scattering event a decision of which particle the photon will interact with is considered. The probability of a scatter by a sphere is described by $P_{sph} = \mu_{s,sph}/(\mu_{s,sph} + \mu_{s,cyl})$.

As a scattering event occurs, a new Stokes vector S_{new} has to be calculated using

$$S_{new} = M(\zeta, \theta) R(\beta) S_i, \tag{3}$$

where $M(\zeta, \theta)$ is the Mueller matrix for the incident angle ζ, and scattering angle θ. $R(\beta)$ is the rotational matrix that rotates the original polarization state S_i to its reference plane. The reference plane is calculated slightly different for micro-spheres [7] and cylinders. [19]

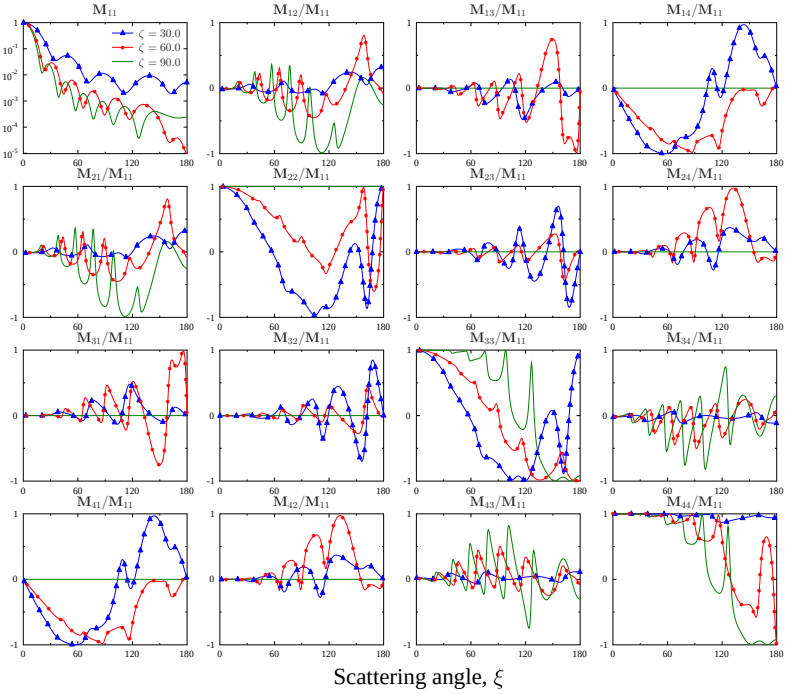


Figure 4: The angular distribution of the Mueller matrix elements for the scattering on a infinitely long cylinder with size factor $x = 10.0$ and relative refractive index $m = 1.1$ at incident angles ζ equal to 30 (-▲-), 60 (-●-) and 90 (-) degrees. All elements are normalized by M_{11} except for the element itself which is normalized and plotted on a logarithmic scale. For the incident angle $\zeta = 90$, one can notice the same symmetric block-diagonal structure as in Mie theory. For other incident angles, however, one notices that all elements are nonzero and that ten are independent.

The shape of the Mueller matrix is quite different compared to that determined from Mie theory. In fact, it is only for perpendicular incident photons, $\zeta = 90$ degrees that scattering from a cylinder resembles Mie theory. All other incident angles generate 16 nonzero elements in the Mueller matrix where 10 of them are unique as follows:

$$\begin{pmatrix} M_{11} & M_{12} & M_{13} & M_{14} \\ M_{12} & M_{22} & M_{23} & M_{24} \\ -M_{13} & -M_{23} & M_{33} & M_{34} \\ M_{14} & M_{24} & -M_{34} & M_{44} \end{pmatrix}. \tag{4}$$

This can be compared to six independent and eight zero elements for spheres. The

scattering elements seen in Fig.4 for incident angles $\zeta = 30$, 60, and 90 degrees are normalized using the M_{11} element, which is the scattering intensity for unpolarized light along any given direction. Element M_{11} is normalized by its maximum value and plotted on a logarithmic scale.

To validate the code, we compared the simulation results for spheres with a previously published model [7] with excellent agreement. To our knowledge no one has published simulation results for randomly oriented cylinders but a visual comparison of the reflectance was made with published results on oriented structures. [17, 19] We got good agreement when the cylinder density was chosen to $C_a = 7.1 \cdot 10^9 cm^{-2}$.

A few conclusions can be drawn from the Mueller elements plotted in Fig.4. Most interesting is the M_{14} which is the element that affects circular polarization for plain unpolarized light. The initial dip toward left-circular polarization combined with the phase function which primarily scatters light within this region causes photons to become left-circularly polarized. Elements that affect linear polarization start off by being close to zero for small deflection angles and then tend to alternate around zero.

3 Results

We observed that the anisotropy factor varies little for different sizes and relative refractive indexes of the randomly oriented cylinders, which also has been indicated in experiments reported by Zhao [21]. Scattered intensity is not solely affected by the phase function. The dominant effect instead stems from the scattering orientation by the half-angle ζ toward the cylinder axis. The effect of this is that the transmitted intensity through a volume where two sets of fibers with equal scattering coefficients but different diameters would vary little in comparison to spheres. The reflectance is also affected by the orientation towards the cylinder. In fact, only when the incident light towards the cylinder is larger than 45 degrees, reflectance by a single scatter can occur.

All parameters where fixed except the proportions of spheres and cylindrical scatterers in the Monte Carlo simulations. The relative refractive index was chosen to be $m = 1.1$, the total extinction coefficient $\mu_t = 1.0\ cm^{-1}$, and the thickness of the sample was set to $3.0\ cm$. Absorption was set to zero in these simulations for simplicity. It is, however, true that fibers and fines absorb light and that the absorption differs between each constituent. The size factor of the spheres where $x_{sph} = 1.0$ which results in an anisotropy factor of $g = 0.17$. The size factor of the cylinders was chosen to be $x_{cyl} = 70.0$ which would represent a fiber diameter of approximately 10 μm.

Figure 5 shows transmission, reflection and state of polarization for simulations with initially linearly polarized light. The x-axis shows the proportion of scatterers from only scattering by cylinder to only scattering by spheres. The total transmitted linear depolarization seems to be similar for both spheres and cylinders with this setup. However, this is not the case for the reflected part which is completely depolarized for cylindrical scatterers. This is probably due to the orientation dependence, the polarized axis and the random orientation of the cylinder axis evens out the polarization. Combined with the fact that scattered photons in a randomly oriented cylinder rarely scatter just once

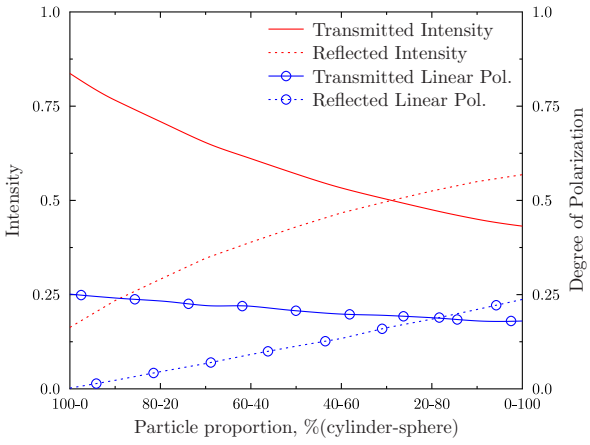


Figure 5: Totally transmitted and reflected intensities and polarization states for Monte Carlo simulations with initially linearly polarized light. All parameters are fixed except for the proportion of cylindrical and spherical scatterers.

before reflecting.

In another setup we choose to increase the size of the spheres to be in the same order as the cylinders. Both particles now have a size factor of $x = 70.0$. Figure 6 shows that the transmission and reflection stays nearly the same for the different particle proportions but the transmitted depolarization is affected strongly. The main reason for this difference is that the phase function for small incident angles have a higher probability of large scattering angles. These scattering angles, however, are the scattering angles in the cone around the cylinder, so the photon direction will not change significantly.

Another aspect that has to be considered is how efficiently the different particles scatters the light. Figure 7 shows how the scattering efficiency Q_{sca} varies as the size of the particle increases. One can clearly see that spheres do scatter the light much more efficient than cylinders averaged over all scattering angles. The scattering efficiency for perpendicularly incident light toward a cylinder behaves very similar as for the spherical particles but the average value as for a random mixture is lower.

4 Conclusions

The effect on polarization, transmission and reflection of light was investigated due to scattering by two fundamentally geometrically different particles. It was found that the

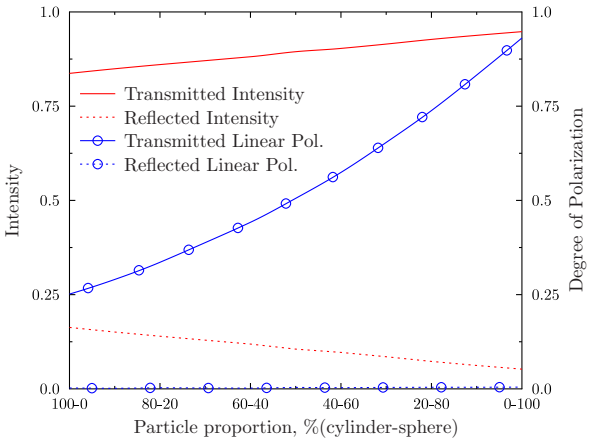


Figure 6: Transmission, reflection and degree of polarization for simulations with initially linearly polarized light. The size factor for both spheres and the cylinders are $x = 70.0$.

state of polarization is quite different for both constituents. The degree of polarization in back scattered light, for example, is completely depolarized after scattering only on cylinders. This is probably due to a combination of the random orientation of the fibers and the strong forward scattering.

It was found that photons that interact with a cylinder whose longitudinal axis is close to parallel to the photons incident direction can change their state of polarization completely while the propagation direction nearly stays the same. This effect does not occur for spherical scatters and the behavior is explained by the shape and nature of the phase function and the Mueller matrix for the scattering events, respectively. This effect can theoretically be used to determine particle composition where elongated particle shapes, like fibers, do not depolarize the light as efficiently as spherical objects.

Even unpolarized light can, due to the conically shaped scattering probability function and the fact that no element in the Mueller matrix is zero, achieve some degree of polarization upon exiting the simulation volume. Transmitted linear polarization seems to be the strongest characteristic feature that distinguishes the scattering properties between the scattering constituents.

The findings on the difference in depolarization and its spatial distribution opens for techniques that enables the relative proportions of fibers and fines in pulp to be determined. Even though wood fibers are far from perfectly symmetric objects it is likely that their behavior is more similar to infinitely long cylinders than that of spheres.

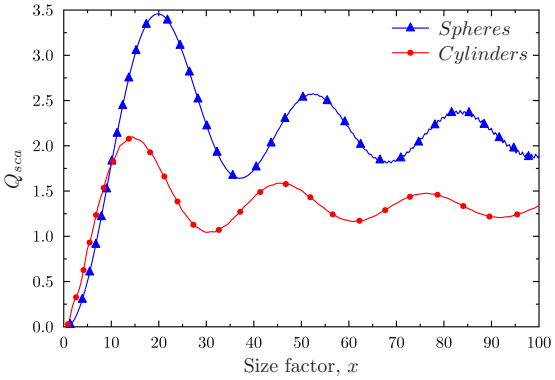


Figure 7: Scattering efficiency variation with increasing particle size for spheres and randomly oriented cylinders.

Future investigation can include both experiments on scattering by single fibers and also that of mixtures of different concentrations of fibers.

5 Acknowledgements

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Paper B

Anisotropic light propagation in paper

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Anisotropic light propagation in paper

Tomas Linder and Torbjörn Löfqvist

Abstract

We investigate anisotropic light propagation in paper using both a theoretical model and experiments. The theoretical model utilizes the Monte Carlo method to solve the photon transport equation numerically. It is assumed that wood fibres are represented by infinitely long, homogeneous and straight cylinders. The layer-like microstructure and anisotropic orientation of the fibres is considered in the model. The conical scattering by cylindrical objects, the wood fibres, is argued as the main source of anisotropic scattering. Simulations revealed that laterally resolved transmittance exhibits directional dependence. Experiments on light transmitted through a standard kraft liner product confirmed that light in fact do propagate more in the machine direction than in the cross direction. Reasonably good agreement was obtained between experimentally and numerically obtained iso-intensity patterns.

1 Introduction

Deeper knowledge of light propagation in paper is important in order to understanding its optical appearance as well as for various industrial applications. For a correct description of the optical properties of paper its microstructure, often including the effect of fibre orientation, has to be considered since fibre orientation is believed to cause anisotropic optical properties. Paper is an engineered material and the degree of anisotropy of the fibre orientation is also an important factor when quantifying the mechanical properties and quality of the paper. The fibre orientation originates from the paper forming stage where the pulp flow onto the wire is controlled to create a specified fibre alignment distribution. Fibres are oriented differently in the xy-plane, or machine direction (MD) cross direction (CD) plane, and in the thickness, z-direction, of the paper as the fibres tend to end up in interconnected layers (Deng and Dodson 1994).

From an optical point of view, paper can be regarded as a three-dimensional stochastic network structure where the fibres are arranged in interconnected layers. Air is the background medium and the scattering takes place at the air-fibre interfaces. In addition to this, light also scatters from small particles, fines, fillers and pigments, interspersed randomly among the fibres. Both fibres and fines are regarded as highly scattering particles making paper a complex medium for optical measurements and simulations.

The study of optical properties of paper has been an area of interest over the years. Monte Carlo simulation of photon transport in paper has evolved since Carlsson's et al. (1995) work on time resolved measurements of scattering in paper. Their results were

analyzed using a Monte Carlo simulation model based on the Henyey-Greenstein phase function. A similar model has also been presented by Modric et al. (2009) to describe the origin of optical dot gain. Chen et al. (2008) compared the performance of the Henyey-Greenstein phase function with an alternative phase function for transmittance in paper. Light scattering by single wood fibres was investigated by Saarinen and Muinonen (2001) where a Gaussian random fibre shape model together with ray-optics was used. Green et al. (2000) used a multiple scattering model based on geometrical optics with a pre-defined fibre-network. A similar model has been presented by Coppel et al. (2009). None of the simulation models mentioned above take into account the wave nature of light or the alignment of fibres in paper.

In this study we illustrate anisotropic light propagation in paper by employing simulations based on analytic solutions of light scattering by small particles where both fibre alignment as well as intensity of the scattered light is accounted for. A regular chemical pulp fibre is curved, kinked and has a hollow, non-cylindrical cross-section. In this study we assume that fibres can be represented by straight, homogeneous, circular cylinders whose length is much larger than their diameter. The light scattering from the cylinders are described by an analytical solution of Maxwell's equation for scattering at a straight, infinitely long, cylinder (Bohren and Huffman 1983). In addition, other particles smaller than fibres, for example fines, fillers and pigments, are assumed spherical and their light scattering can be described by Mie theory. The simulation model then describes a sphere-cylinder media where the constituents can have different geometrical and optical properties (Linder and Löfqvist 2011). Since the model is based on analytical solutions to Maxwell's equations the Monte Carlo simulation code developed has capability to handle both intensity and polarization of the scattered light from both fibres and smaller particles using the different scattering mechanisms described above. However, since the scope of the presented study is focused on the effects on light scattering from fibre orientation the fine particles are excluded in this study to clarify the results.

The microstructure of paper is represented by employing a stochastic model for the fibre network where the fibre orientation is set by a two dimensional Gaussian distribution centered at a pre-defined main direction. The standard deviations are then used as a measure on the degree of fibre alignment. Using this micro-structural model for paper, we employ a Monte Carlo method to solve the photon transport equation numerically. Models similar to the one used in this work has successfully illustrated anisotropic light propagation in biological material of fibrous nature like arteries, tendon and wood (Kienle et al. 2003, 2004, 2007, 2008). To preliminary verify the Monte Carlo simulations described above we compare simulated results for the laterally resolved transmittance for a specific microstructure with light transmission measurements on a sheet of a standard kraft liner product. The experimental setup is based on a HeNe laser as a light source and an optical fibre arrangement guiding the light to the paper surface. The transmitted light is then depicted using a CCD-camera. These preliminary experimental results indicate that the Monte-Carlo simulations can be used to aid the interpretation of the features observed in light transmittance patterns in paper.

2 Method

The Monte Carlo method used is based on the simulation of a large number of possible random photon trajectories. A photon, entering a scattering medium, will propagate between different scattering events until it either leaves the medium or gets absorbed. The sequential random walk between scattering events is determined by a probability density function based on Beer–Lamberts law. The random step size S a photon makes in a non-absorbing medium is specified as

$$S = -\frac{\ln \eta}{\mu_s} \tag{1}$$

where μ_s is the scattering coefficient and η is a uniformly distributed random number between 0 and 1. In our model the random walk is also affected by the micro-structure, this will be discussed further in the next subsection.

Anisotropic light scattering is usually described by a probability distribution like the Henyey-Greenstein phase function. We have chosen to represent anisotropic light scattering with phase functions that are derived from objects with similar geometrical properties as a fibre. We assume that the fibres can be described as long, straight cylinders. This enables us to use a phase function derived from light scattering by an infinitely long cylinder (Bohren and Huffman 1983).

In our previous work (Linder and Löfqvist 2011) we proposed that the other particulate constituents in paper pulp, like e.g. fines, scatter light according to Mie theory. The same assumption can be made for these particles in paper as well. Both phase functions used in our model can be represented in a Mueller-formalism and therefore we are able to keep track of the state of polarization at each scattering event. This is an advantage over using other phase functions, like for example the Henyey-Greenstein phase function, which is not taking polarization into account. Since the scope of the presented work is on fibre orientation we exclude smaller particles from the modeling and only consider the cylinders as scattering objects.

A photon interacting with a cylinder will scatter in a cone around the longitudinal axis of the cylinder (Fig 1). The phase function describes in which direction in the cone the light scatters. Scattering by cylinders is, in contrast to scattering by spherical particles, orientation dependent. At each scattering event the phase function is calculated for the incident angle ζ toward the cylinder longitudinal axis. The orientation of the cylinder is randomly generated at each scattering event; this is discussed further in the next subsection. The new photon trajectory is then extracted from the phase function; a detailed description is presented by Yun et al. (2009). The input parameters employed to specify the optical properties, like for example the phase function of the cylinder, are the relative refractive index n_{rel} and the size factor which is specified as

$$x_{cyl} = \frac{2\pi n_{bg}}{\lambda} \tag{2}$$

where r is the radius of the cylinder, n_{bg} the refractive index of the background medium and λ the wavelength. The irregular shape and surface roughness of fibres as

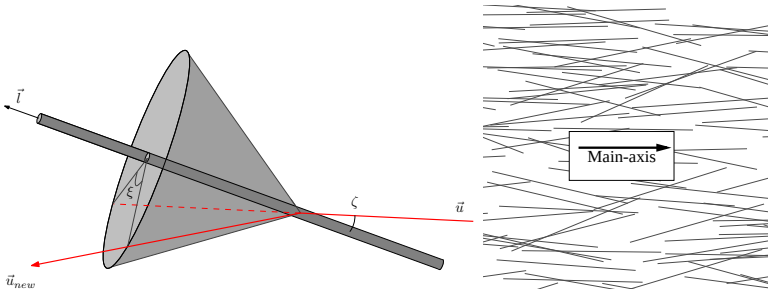


Figure 1: To left is a photon scattering on a cylinder at incident angle ζ. The new photon propagation vector $\vec{u}_{new}$ is derived from the scattering angle ξ and cylinder orientation $\vec{l}$. The right figure illustrates fibres aligned following a Gaussian distribution with standard deviation $\sigma_y = 10°$ in relation to the main-axis.

well as the lumen are not accounted for in the presented model. The assumption is also made that all fibres have a fixed average fibre diameter. The phase functions in Fig 2 reveals only a small variation in the scattered light from cylinders of similar sizes as wood fibres. This indicates that the size of the object, as a parameter, only has a minor effect on the results from multiple scattering.

The likelihood of light being scattered by a particle is often described by the scattering efficiency, Q_s. It specifies the ratio between the scattering cross-section, which is a hypothetical area describing the likelihood of light interaction and the geometrical cross-section of the particle. It is convenient to consider the scattering efficiency since it can be derived for both spheres and the infinitely long cylinders (Bohren and Huffman 1983). For a cylinder it varies with both particle size and with the incident angle towards. The scattering coefficient for a cylinder (Kienle 2003) is described as

$$\mu_s^{cyl}(\zeta) = Q_s^{cyl}(\zeta) d C_A \tag{3}$$

where d is the fibre diameter, C_A the density of cylinders and ζ the incident angle toward the cylinder. The cylinder density C_A can be defined as the total length of cylinder per unit volume.

2.1 Modelling paper structure

The anisotropic micro-structure of paper is modeled by a statistical approach in this study. The fibre micro-structure in paper is different in the plane (xy- or MD–CD-plane) and in the thickness direction (z-direction). We therefore choose different angle distributions in the thickness direction and in the xy-plane of the paper. A fibre orientation main axis in the microstructure is defined and is used for relating individual fibre orientations in the depth and plane. On each scattering event, the orientation vector, $\vec{l} = (l_x, l_y, l_z)$ of the cylinder is given as

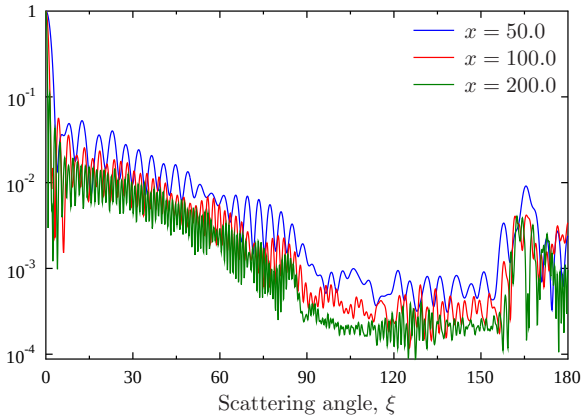


Figure 2: Phase functions for normal incident light ($\zeta = 90°$) at cylinders with size factors $x = 50$, 100, 200, see equation 2.2, and refractive index $n = 1.55$. The size factors corresponds to cylinder diameters of approximately 10, 20 and 40 μm which are the fibre diameters normally encountered in paper production.

$$\vec{l} = \begin{pmatrix} \cos(\varphi_y)\cos(\varphi_z) \\ \sin(\varphi_y)\cos(\varphi_z) \\ -\sin(\varphi_z) \end{pmatrix} \tag{4}$$

where φ_y and φ_z are the angular deflections from the x-axis (main axis) in the plane and depth respectively. They are generated based on distribution functions $G(\sigma_y)$ and $G(\sigma_z)$ which in this study is represented by Gaussian distributions with predetermined standard deviations σ_y and σ_z from the main axis.

Larger optical thickness is present perpendicular to the main axis compared to parallel to it. A photon propagating perpendicular to the fibre structures main axis will have a higher probability to scatter on a fibre than a photon propagating parallel to it. We model this by adjusting the scattering coefficient depending on the new photon trajectory relative the main fibre axis after each scattering event. Convolution between the scattering efficiency, $Q_s(\zeta)$, and the given fibre distribution function, $G(\sigma)$, describes how the optical thickness vary in the structure

$$Q_s^*(\sigma, \zeta) = \int_{-\infty}^{\infty} Q_s(\zeta) G(\sigma, \tau - \zeta) d\zeta. \tag{5}$$

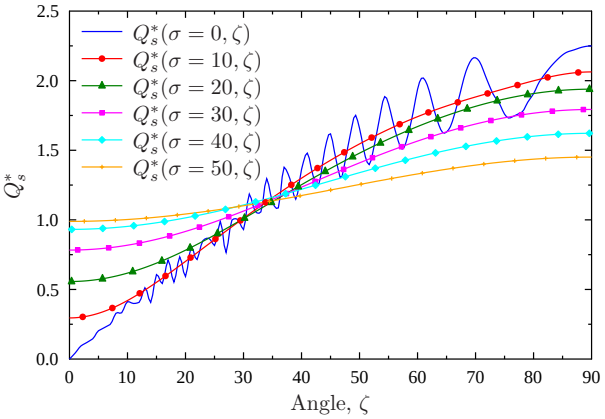


Figure 3: Scattering efficiency Q_s^* for different fibre alignments in relation to the main-axis by the angle ζ.

Fig 3 show the scattering efficiency for a photon propagating in relation to the main-axis by angle ζ for different Gaussian distributions with standard deviations $\sigma = 10°$, $20°$, $30°$, $40°$ and $50°$. Noticeable is how the scattering efficiency becomes more and more leveled out as the structural anisotropy increases. A completely random structure would obviously not have any optical thickness variations and hence a constant value of Q_s^* for all scattering angles. Note that Fig 3 only illustrates the change in optical thickness for one of the deviations from the main axis. The scattering efficiency differs with the angle distribution in the plane and depth of the paper. The total combined scattering efficiency for a given photon trajectory $\vec{u} = (u_x, u_y, u_z)$ is specified as

$$Q_s^{tot}(\zeta) = \frac{Q_s^*(\sigma_y, \zeta)|u_y| + Q_s^*(\sigma_z, \zeta)|u_z|}{|u_y| + |u_z|} \tag{6}$$

where $Q_s^*(\sigma_y, \zeta)$ and $Q_s^*(\sigma_z, \zeta)$ are the scattering efficiencies in the plane and depth of the paper. The scattering coefficient μ_s^{cyl} and step size S is updated after each scattering event as specified in Eq 2.1 and 2.3 to take into account the change in the optical thickness described above.

2.2 Experiment

In order to validate the simulation model we investigated the laterally resolved transmittance using a HeNe laser and a CCD camera, as shown in Fig 4. The optical fiber was mounted into a transparent plate of plexiglass. The fibre was aligned perpendicularly to the paper surface and positioned so that the fibre tip was in contact with the paper

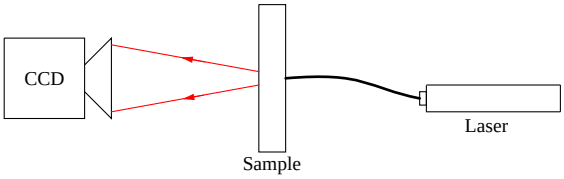


Figure 4: Experimental setup for spatially resolved transmittance of a single sheet of paper. Light is guided from the laser to the surface of the paper by an optical fiber and the multiple scattered transmitted light is detected on the opposite side by a CCD camera.

surface. The diameter of the fibre core was 9 μm and the wavelength of the HeNe laser used was 633 nm. The benefit of using an uncollimated source at the surface of the paper sample is that light to a greater extent will propagate into the plane of the paper. This enhances the anisotropic scattering pattern we want to observe as more of the light will scatter and propagate more in the plane of the paper. Chen et al. (2008) used a similar setup where the main difference compared to the present is that they used a free, unguided, laser beam. This way of introducing the light will limit the lateral spreading, as discussed above, and through lowered intensities hamper the detection of the laterally scattered light.

A thermoelectrically cooled CCD camera, a PCO SensiCam by PCO AG, Germany, was used to depict the light transmitted through the paper sheet. The resolution of the camera was 1376x1040 pixels with 12 bits dynamic range. The images sampled by a frame grabber card were stored in a computer for off-line analysis. The captured images were processed in Matlab, Mathworks, MA, USA, where the light intensities were normalized to the maximum intensity and iso-contours were plotted. The iso-contour levels were chosen low specifically to identify the low level light scattered in the paper. In these measurements we only used the light intensity and hence, only used a part of the information that could be retrieved through studying both intensity and polarization of the scattered light. The simulations described above include both light intensity as well as state of polarization.

The paper sample used in this study was a standard kraft liner product supplied from SCA Munksund, Piteå, Sweden, with grammage 175 g/m^2 and average thickness 0.20 mm. The raw material of the paper is pine and no fillers or other additives are present. In this study we use one paper sample since we only wanted to display the light scattering effects in paper and obtain an indication that the simulations and the experimental technique is working as intended. At this preliminary stage we are not seeking to determine quantitative results on fibre orientation or fibre alignment distribution.

The paper tensile strength orientation, or TSO value, and the tensile strength index,

or TSI value, were determined using a L&W TSO Tester, Lorentzen & Wettre AB, Sweden, based on ultrasonic measurement principles. The TSI MD/CD ratio as well as the TSO value gives an indication of the fibre orientation as well as the degree of anisotropy in the MD-CD plane of the paper since the tensile stiffness is closely related to the fibre orientation (Vahey et al. 2008, Lindblad and Fürst 2007).

3 Results

The parameters for the Monte Carlo simulations were chosen to match the paper sheet selected, the experimental setup and data on Swedish pine. We assume that the fibres have an average diameter of 25 μm, this value is valid for both Swedish pine and birch, see Fellers and Norman (1996). The refractive index of cellulose (1.55) was used for the fibre material (Saarinen and Muinonen, 2001). The background medium is assumed to be air with a refractive index of $n_{bg} = 1.00$. The thickness of the paper sheet was set as $t = 0.20\ mm$ and the incident laser source were modeled as a uniform circular dot with diameter 9 μm with an uncollimated beam profile at the surface. The cylinder density C_A can be estimated by using previously published data on the scattering coefficient where the mass scattering coefficient for paper sheets without fillers is stated to be about 40 m^2/kg, (Alince et al. 2002). Assuming that the diameter of the cylinder is 25 μm and the scattering efficiency Qs(90)= 2.0 along with a paper density of 750 kg/m^3, a derivation of equation 2.2 yields a cylinder density $C_A = 6 \cdot 10^8\ m^{-2}$.

The results of the simulations are displayed in Figs 5. The parameter changed between the simulated iso-intensity patterns in Figs 5:a-f is the fibre alignment deviation, σ_y, in the xy-plane. It ranges 10°, 30°, 50°, 70°, 90° to isotropic fibre alignment. The deviation in the depth or z-direction was arbitrary chosen as $\sigma_z = 5$ to account for the interconnection of the fibre layers. Both absorption and scattering by spherical particles was excluded from this simulation to simplify the problem setting and clarify the results on the impact of fibre alignment. The iso-intensity patterns in Fig 5:a-f show the relative intensity levels 0.5, 0.1, 0.01 and 0.001. From the figures it is seen that the iso-intensity curves show a distinct elliptical shape with its major axis along the main axis of fibre orientation. Note that the iso-contours for the lowest σ_y, Fig. 5:a, seems to be almost rhombic, a pattern also observed by Kienle et al. (2008). The ellipsoidal shape gradually becomes circular when the degree of fiber alignment anisotropy is reduced. The reason for the elliptical iso-contours is that the multiple scattered light propagates more in the machine direction, which is also the main direction of orientation for the fibres, than in the cross direction. This observation is supported by the finding that scattered light have a higher probability to scatter along a fibre than perpendicular to it (Kienle 2008).

Fig 6 shows experimentally obtained iso-intensity patterns from two different orientations of the paper sheet tested. Note that the paper sheet was rotated 90° and shifted in-plane between the figures. The machine direction is the x-axis in Fig 6:a and the y-axis in Fig 6:b. The light intensity pattern is shown to have a weak elliptical shape with its major axis along the x-direction. The measurements on the paper sheet using the L&W TSO Tester confirms the orientation of the fibres since the TSI MD/CD-ratio

of this sample was 2.4° ± 0.3° and the TSO angle was 0.9° ± 1.4°. A TSI MD/CD-ratio >1 and a TSO angle nearly zero indicates that the fibres are aligned along the machine direction (MD) (Lindblad and Fürst, 2007).

4 Discussion

The reason for the ellipsoidal shape of the iso-contours in Fig 5 lies in the effect that light that scatters on a fibre surface has a higher probability to scatter along the fibre than perpendicularly. Thus, photons propagating along the longitudinal axis of an aligned fibre structure tend to do so at a greater extent, thereof the origin of the ellipsoidal iso-contour of the transmitted light intensity. This effect is also enhanced by the modeled change in optical thickness by the scattering efficiency for different cylinder distributions.

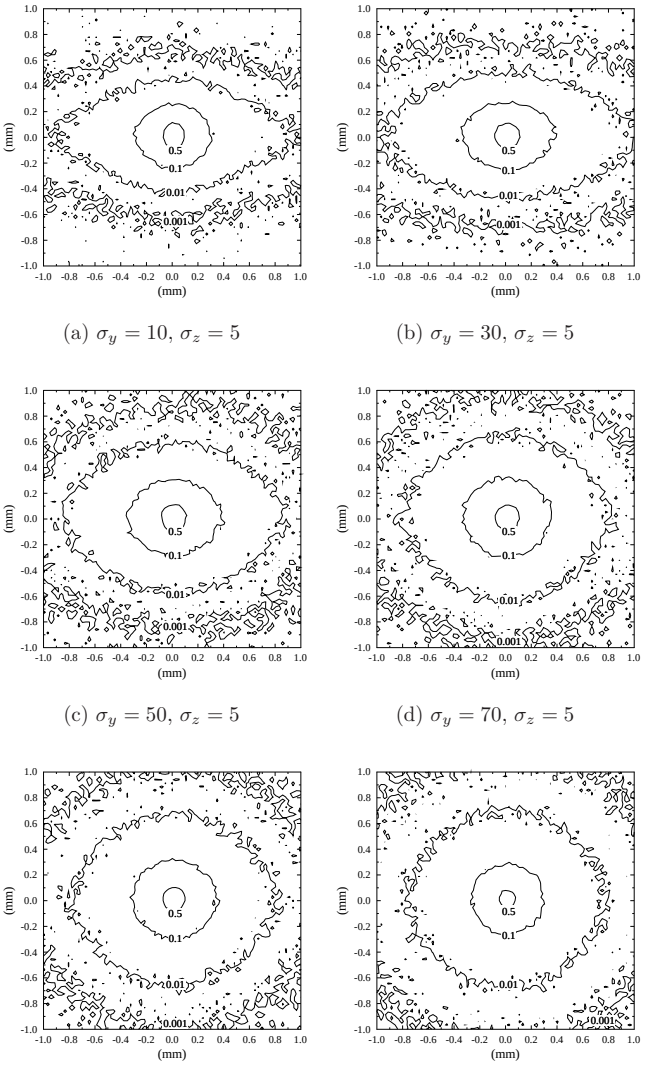
The parameters affecting the shape of the iso-contours of the transmitted light most significantly are found to be the density of cylinders and the degree of fibre alignment. The parameters used to achieve the phase function, like the refractive index, particle size and wavelength of the light was found to affect the result very little when tuned between the parameter values for regular paper products.

From the light transmission experiments we observe in Fig 6 that the major axis of the ellipsoidal iso-contour of the transmitted light is rotated as the paper sheet is rotated. The major axis of the iso-contour corresponds to the main orientation direction (MD) of the fibres in the paper, which is also shown in the TSI/TSO tests were the tensile strength orientation is along the MD, as discussed in section 3.

It is also shown in the figure that there is a slight differ-rence in size of the iso-intensity curves between Fig 6:a and Fig 6:b at the lowest intensity levels. This is probably due to local thickness and density variations in the paper since the spot where the light entered the paper was shifted between the measurements.

These preliminary results are promising and further studies, using paper samples of e.g. different thicknesses, grammages and fibre orientations needs to be performed in order to establish the connection between experiment-tally obtained scattering patterns and fibre alignment distributions.

The model used does not account for several aspects of both paper and wood fibres. Since scattering also can take place at the lumen-fibre interface, it is possible that the scattering efficiency can be influenced. The fibres are assumed to be cylindrical and straight while they in fact are partly compressed, curved and kinked. Effects from other irregularities, like surface roughness and micro defects, are assumed to be averaged out by the multiple scattering of the photons. Thickness and density variations are also ignored as we chose to focus this study the effects of fibre orientation. The approach in this study is different compared to models based on geometrical optics (e.g. Green et al. 2000, Coppel et al. 2009). They generate a pre-defined statistical fibre network prior to the light scattering simulation while the model used in this study generates the structure in parallel with each scattering event. Since our model is completely statistical, we do not model effects of flocculation and other local structural distortions that appear in paper. Therefore, our model for the paper microstructure is an ideal version of the



(a) $\sigma_y = 10$, $\sigma_z = 5$ (b) $\sigma_y = 30$, $\sigma_z = 5$

(c) $\sigma_y = 50$, $\sigma_z = 5$ (d) $\sigma_y = 70$, $\sigma_z = 5$

(e) $\sigma_y = 90$, $\sigma_z = 5$ (f) $\sigma_y = -$, $\sigma_z = 5$

Figure 5: Simulated iso-intensity patterns on transmitted light intensity. Patterns are normalized to the maximum intensity value. The standard deviation of the fibre alignment in the xy-plane, σ_y, is varying, all other parameters are kept constant. The x-axis, the main fibre orientation direction, is horizontal and y-axis vertical in the figures above. The numbers on the iso-curves are the relative intensity levels. Fig a-f shows how the scattered intensity is affected with increasing anisotropy of the fibre alignment, i.e. larger σ_y.

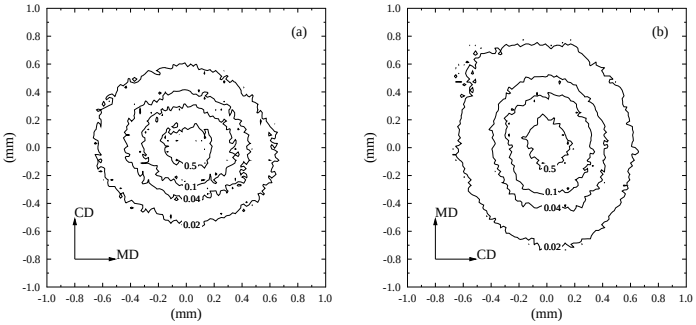


Figure 6: Experimentally determined iso-intensity patterns on transmitted light intensity through a paper sheet. The paper was rotated 90° and in-plane translated between the two measurements. The numbers on the iso-curves are the relative intensity levels.

fibre structure for a given spatial fibre distribution. This can be used when determining an undistorted scattering pattern when comparing simulations with experimental results. The results from both the experiments and the simulations in the present study concern the intensity of the scattered light only. Another aspect which can be investigated in later studies is the state of polarization of the transmitted light and its relation to fiber properties and orientation.

5 Conclusions

We propose a theoretical model based on the microstructure of paper to simulate light propagation in paper. The model has a completely stochastic representation of both the scatterers and fibre structure. It considers the layer-like nature of the paper along with the fibre orientation anisotropy. The directional dependent conical scattering by cylindrical objects is argued as the main source of the anisotropic laterally resolved transmittance. Simulations, as well as experiments showed that diffuse light travel more along the main axis of fibre orientation (MD) than perpendicular to it (CD) in a paper sheet.

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Paper C

Lateral light scattering in fibrous media

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Lateral light scattering in fibrous media

Tomas Linder, Torbjörn Löfqvist, Ludovic G. Coppel, Magnus Neuman and Per Edström

Abstract

Lateral light scattering in fibrous media is investigated by computing the modulation transfer function (MTF) of 22 paper samples using a Monte Carlo model. The simulation tool uses phase functions from infinitely long homogenous cylinders and the directional inhomogeneity of paper is achieved by aligning the cylinders in the plane. The inverse frequency at half maximum of the MTF is compared to both measurements and previous simulations with isotropic and strongly forward single scattering phase functions. It is found that the conical scattering by cylinders enhances the lateral scattering and therefore predicts a larger extent of lateral light scattering than models using rotationally invariant single scattering phase functions. However, it does not fully reach the levels of lateral scattering observed in measurements. It is argued that the hollow lumen of a wood fiber or dependent scattering effects must be considered for a complete description of lateral light scattering in paper.

1 Introduction

The underlying physics of light scattering in turbid media is important in a wide variety of fields. Light scattering in paper has a fundamental role as the optical appearance is one of its main functions. Lateral light scattering has a key role in the optical dot gain in paper, the so called Yule-Nielsen effect [1], where printed dots due to light scattering appear larger. It has also been suggested that monitoring scattering in paper can potentially be used in measuring fiber formation [2].

Proper models for lateral light scattering in turbid media are requried to correctly relate to the optical properties in materials such as paper. It has been shown by for example Neuman et al. [3] that both single scattering anisotropy and the medium mean free path has a big impact on the point spread function. Modulation transfer function (MTF) analysis is a convienient way of evaluating lateral light scattering [4]. Arney et al. [5] compared measured values of the inverse frequency at full width half maximum of the MTF, k_p, with a model based on the Kubelka-Munk (KM) scattering coefficient S. Recently Coppel et al. [6] compared these results with Monte Carlo simulations based on the Henyey-Greenstein phase function with two different asymmetry factors. They found that neither isotropic scattering nor strong forward scattering correctly could predict the large lateral scattering observed in the measurements. Since paper is highly anisotropic, based upon layers of cellulose fibers, phase functions that depend on the absolute direction of light within the structure are called for. Applying such phase functions to address

similar observations has been suggested for other turbid media. Anisotropic directional dependent scattering can for example be modelled by using phase functions derived from an analytical solution of electromagnetic scattering by an infinite cylinder [7]. This was first proposed by Kienle et al. [8, 9] who observed the anisotropic shapes of backscattered point spread functions (PSF) for different stochastic representations of the cylinder alignments. Similar Monte Carlo simulations have since then been used to model light scattering in anisotropic structures such as biological tissue [10, 11], softwood [12] and textile [13]. A detailed description of the implementation of such a Monte Carlo model can be found in Yun et al. [14]. One benefit of using analytical solutions is that the phase function is derived from material parameters like particle size and refractive index. This means that the asymmetry factor g is no longer required in the same way as in for example the Henyey-Greenstein phase function. Another benefit is that the scattering efficiency Q_s can be obtained, it can be used to define how scattering distance depend on the absolute direction of photons propagating in aligned fiber structures.

The purpose of the present work is to utilize the cylinder phase function through Monte Carlo simulations to test if it can be used to mimic scattering in paper. The structural anisotropy is considered by aligning the cylinders isotropically in the plane of the paper and with a small Gaussian distribution in the thickness direction. The simulated MTFs are then compared to both the measured [5] and simulated values [6] of k_p to test whether they can predict the amount of lateral scattering in the real paper samples.

2 Method

The Monte Carlo model in this work utilizes phase functions from an analytical solution of scattering by an infinite cylinder [7]. For perpendicular incidence ($\zeta = 90°$) light scatters in a disc around the longitudinal axis of the cylinder and for oblique incidence light scatters in a cone around the longitudinal axis where the phase function specifies the distributed intensity. The half angle of the scattering cone equals the incident angle ζ governing the directional dependencies observed by for example Kienle et al. [8]. μ_s is related to the density of cylinders C_a, cylinder diameter d and scattering efficiency Q_s for any incident angle ζ as

$$\mu_s(\zeta) = C_a d Q_s(\zeta). \tag{1}$$

The cylinder density is defined as the total length of cylinders per volume unit. It can be valid to test the performance of the model in relation to C_a as it is the concentration of particles, especially since the cylinder phase function only is valid for plane electromagnetic waves.

The scattering parameters are determined by matching the reflectance and transmittance factors to those of the previously published measurements for each of a set of 22 samples. The shape of the simulated MTFs are then compared to the Monte Carlo simulations using the Henyey-Greenstain phase function with isotropic single scattering ($g = 0.0$) and forward single scattering ($g = 0.8$) and the simulated values of k_p are also compared to the measured values.

2.1 Material parameters and parameter estimation

Paper is essentially an entangled network of fibers, however the model used in this work assumes that the fibers are not in contact with one another. Surface effects are not considered as the scattering only take place at the air-cylinder interfaces in the model. The cylinders are aligned isotropically in the plane of the simulated microstructure and are assumed to be following a Gaussian distribution with standard deviation $\sigma_z = 5^\circ$ in the thickness direction. A more detailed description of the cylinder alignment has been reported in [2]. The refractive index of cellulose ($n = 1.55$) is used for the cylinders and the surrounding medium is air ($n = 1.0$). The cylinder diameter is chosen to be 20 μm as typical wood fiber dimension [15]. The wavelength of the light was chosen to be $\lambda = 510$ nm as the experiments in Arney et al. used a green filter. We optimized C_a iteratively using a Gauss-Newton method in the $0^\circ/d$ geometry with the Monte Carlo simulation tool to match the measured total transmission and reflection for each of the paper samples with their given thicknesses. The obtained cylinder density C_a, scattering coefficient in the thickness direction $\mu_s(\overline{\mathbf{z}})$ and absorption coefficient μ_a are shown in Table 1.

2.2 Simulation of the edge response

The Monte Carlo simulation tool (available at *http://fibermc.sourceforge.net/*) is used to compute the point spread function (PSF) for each of the 22 sets of paper sheets. Both for a single paper sheet and for an opaque pad of paper sheets. The resolution was chosen to 10 μm and the incident trajectories of the photons was tilted with a 20° angle toward the surface normal to match the measurements. A 2D convolution between the simulated point spread function and a intensity distribution $i(x, y)$ was used to derive the edge response

$$ESF(x, y) = PSF(x, y) * i(x, y). \tag{2}$$

The intensity distribution is uniformly distributed and cut off by a sharp knife edge, i.e. a matrix consisting of 1's on one side and 0's on the other. A similar convolution was used by Ukishima et al. [16] who considered the intensity distribution of pencil light together with the point spread function. This approach is different compared to Coppel et al. who simulated the edge response directly by distributing each of the photons over an area. Since the statistical response of a PSF is more focused it greatly reduces the amount of simulated photons required to keep down the noise levels. It is therefore preferred to use the convolution as Monte Carlo simulations are very time consuming.

2.3 MTF

The line spread function (LSF) is obtained by taking the derivative of the edge spread function (ESF, see Fig. 1(a)). We then obtain the MTF by taking the Fourier transform of the LSF. The inverse frequency at full width half maximum of the MTF, k_p, is defined

Table 1: Sample thickness t, cylinder density C_a, scattering coefficient $\mu_s(\overline{\mathbf{z}})$, absorption coefficient μ_a and simulated values of k_p for a single sheet and an opaque pad (k_p^∞).

sample	t	C_a	$\mu_s(\overline{\mathbf{z}})$	μ_a	k_p	k_p^∞
	μm	$10^9\ m^{-2}$	mm^{-1}	mm^{-1}	mm	mm
1	42	0.71	32.2	0.13	0.250	0.794
2	71	1.39	63.1	0.0	0.346	0.718
3	45.5	0.78	35.4	0.0	0.266	0.807
4	76.5	1.45	65.9	0.0	0.360	0.693
5	97	4.73	214.4	0.295	0.224	0.254
6	109.5	4.795	217.3	1.40	0.200	0.208
7	95	9.03	409.3	0.332	0.142	0.153
8	98	3.62	164.1	0.0	0.275	0.372
9	89	0.895	40.6	0.0	0.453	0.771
10	130	0.43	19.5	0.0	0.641	0.881
11	84	2.87	130.1	0.0	0.291	0.455
12	65	7.155	324.3	0.372	0.153	0.180
13	89	8.06	365.3	0.479	0.149	0.163
14	122	2.965	134.4	0.0	0.333	0.441
15	126	4.30	194.9	0.158	0.259	0.279
16	99	3.83	173.6	0.105	0.263	0.318
17	124	5.80	262.9	0.139	0.213	0.239
18	104	3.77	170.9	0.0	0.275	0.370
19	110	1.07	48.5	0.0	0.470	0.757
20	293	5.92	268.3	0.268	0.218	0.229
21	93	8.51	385.7	0.312	0.146	0.161
22	130	6.29	285.1	0.275	0.193	0.203

as

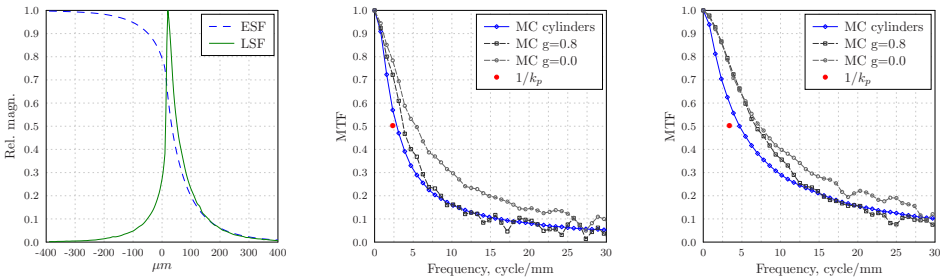
$$MTF(\frac{1}{k_p}) = 0.5. \tag{3}$$

Only measured values of k_p are compared to the simulations since Arney et al. only reported k_p. The full MTF would have been preferred over the metric k_p as it only indicates the amount of lateral light scattering and lose a lot of the information held by the MTF [6].

3 Results

The simulated MTFs of samples 2 and 17 plotted for the three phase functions are shown in Fig 1(b-c). The shape of the simulated MTFs is narrower at low frequencies for the simulations with the cylinder phase function compared to the previous Monte Carlo simulations but seems to agree well with simulations with anisotropic scattering

($g = 0.8$) at higher frequencies. The measured values of $1/k_p$ are presented by red dots in the figures. Figure 2 shows the simulated results for all the 22 samples for both single sheets and opaque pads of paper sheets in relation to the measurements and previous simulations. The general trend is that the simulated values of k_p are underestimated in comparison to measurements but they predict larger values than simulations with rotationally invariant phase functions.



(a) ESF and LSF of sample 2 (b) MTFs of sample 2 (c) MTFs of sample 17

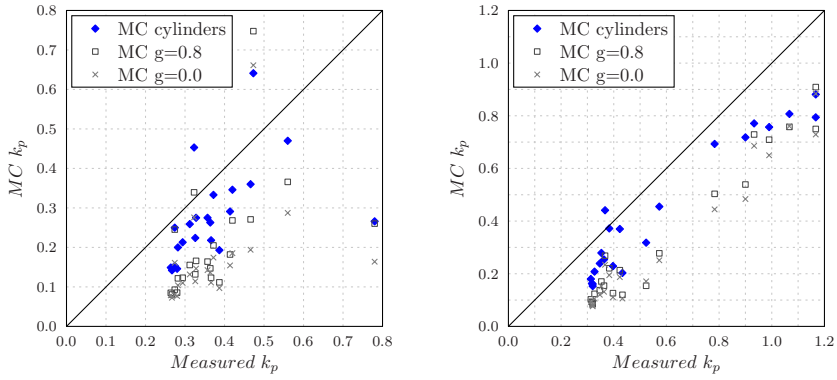
Figure 1: Simulated ESF and corresponding LSF in (a) and simulated MTFs for single paper sheet sample 2 in (b) and sample 17 in (c). The main difference between the samples is that sample 17 has a about four times larger scattering coefficient compared to sample 2. The red dots indicate the measured value of k_p presented by Arney et al.

Figure 3 shows the values of k_p in relation to C_a for a single sheet and an opaque pad of paper samples. At low concentrations, or low scattering coefficients, the cylinder model seems to agree better with the previous simulations than with the measurements. It appears that the best correlation to the measurements are found in the middle region where C_a lie between $3 \cdot 10^9$ and $5 \cdot 10^9$ m^{-2}. For larger values of C_a a decresing correlation can be observed.

4 Discussion and Conclusions

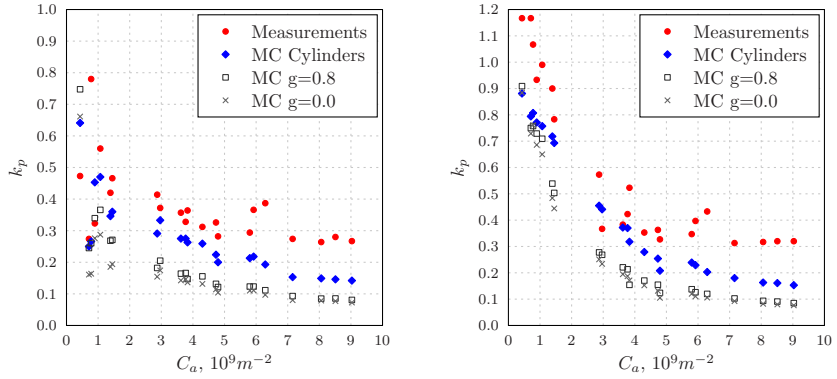
Simulations and measurements of lateral light scattering in paper was compared to a Monte Carlo model employing scattering by infinitely long homogeneous cylinders. The cylinder model clearly predicted a higher value of k_p than models using rotationally invariant isotropic and strong forward scattering. However, the cylinder model still underestimates k_p compared to the measured values. This means that the model predicting less lateral scattering than observed.

The conical scattering by a cylinder has a tendency to sustain photon propagation



(a) Single (b) Opaque

Figure 2: Measured values of k_p versus simulated values for a single sheet (a) and an opaque pad of sheets (b).



(a) Single (b) Opaque

Figure 3: Measured and simulated values of k_p for a single paper sheet (a) and an opaque pad of paper sheets (b) plotted against the density of cylinders C_a.

in the plane of the paper thus increasing the lateral scattering. We originally thought that low correlation between simulations and measurements for the samples with low optical thickness (Fig. 3) were due to photons only scattering once before leaving the medium. Increased lateral scattering can only be observed if the photons manage to scatter so that they start propagating in the plane. However, the trend remains for the

low concentrations in an opaque pad (Fig. 3(b)) and therefore contradicts this. The decreasing correlation at the larger values of the scattering coefficient, i.e. for very dense papers, is not suprising. The phase functions for scattering by cylinders is only valid for plane waves which means that the scatterers need to be in the far-field of one another. A possible explanation for the large values of the cylinder density is that the fibers in paper often are of banded or elliptical shapes. It shall be emphasized that it is a rough approximation to model a wood fiber as a homogenous cylinder. Additional simulations were made with adjusted values of the material parameters that affect the phase function, like e.g. the diameter of the scattering cylinders d and wavelength λ. We observed that this did not have any significant effect on the shape of the PSF and the resulting MTFs. This indicates that the model cannot predict more lateral light scattering for parameters within the natural parameter range of wood fibers. Arney suggested that the hollow lumen of a wood fiber could increase the lateral light scattering through a light-piping effect. This seem to be a reasonable explanation to why the model still predicts less lateral scattering than measured. Another reasonable explanation is that the concentration of particles is high, causing dependent scattering effects giving rise to interactions which are different from the interdependent scattering theory used in this work. Part of the lateral scattering can, however, be explained by the conical scattering by cylindrical objects. Compared to the Monte Carlo simulations using the Henyey-Greenstain phase function it roughly closes half the gap between earlier MC simulations and the experimental results. The cylinder phase function gives a good idea of how the lateral light scattering in fibrous materials is generated and has potential to better model for example optical dot gain.

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Paper D

Light scattering in fibrous media with different degrees of in-plane fiber alignment

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Light scattering in fibrous media with different degrees of in-plane fiber alignment

Tomas Linder, Torbjörn Löfqvist, Erik L. G. Wernersson and Per Gren

Abstract

Fiber orientation is an important structural property in paper and other fibrous materials. In this study we explore the relation between light scattering and in-plane fiber orientation in paper sheets. Light diffusion from a focused light source is simulated using a Monte Carlo technique where parameters describing the paper micro-structure were determined from 3D x-ray computed tomography images. Measurements and simulations on both spatially resolved reflectance and transmittance light scattering patterns show an elliptical shape where the main axis is aligned towards the fiber orientation. Good qualitative agreement was found at low intensities and the results indicate that fiber orientation in thin fiber-based materials can be determined using spatially resolved reflectance or transmittance.

1 Introduction

Deeper knowledge of light propagation in fibrous structures is fundamental in order to understand the optical appearance as well as for industrial applications such as fiber alignment measurements. The anisotropic behavior of light scattering in paper has been known for a long time even though the underlying physics is not well investigated, due to the difficulties of solving Maxwell's equations for such complex materials. Work on diffraction from paper sheets have shown that transmitted light scatters in an elliptical shape where the major axis is aligned perpendicular to the fiber orientation main axis [1]. Thicker sheets, where multiple scattering dominates, can instead scatter light in an elliptical shape where the major axis is aligned towards the main fiber orientation direction [2]. It is believed that the ellipticity for optically thicker materials originates from either multiple scattering by fibers or from light guided inside the fibers.

Paper can be regarded as a three-dimensional but mostly planar stochastic network of interconnected fibers. For light propagations simulations in such materials, it is possible to describe the cellulosic fibers as long cylinders. By using an analytical solution to Maxwell's equations for scattering by an infinitely long cylinder [3] it is possible to take both the wave nature of light and structural properties of the fiber network into consideration. It was originally suggested by Kienle et al. that phase functions from infinitely long cylinders can be used to describe multiple scattering in fiber-based materials [4, 5]. They based their simulations on the Monte Carlo (MC) method which is often used to solve the radiative transfer equation numerically. Similar models have since

Table 1: Paper sheet data, area density, tensile strength in the machine-direction, tensile strength in the cross-direction and the ratio between them, all these values where provided by the paper manufacturer.

Sample	P1	P2	P3	P4	P5
Area density (g/m^2)	102.0	102.9	102.0	103.1	102.6
Tensile strength, MD (kN/m)	10.30	11.60	14.10	14.40	15.70
Tensile strength, CD (kN/m)	7.55	6.57	4.70	4.12	3.80
Tensile strength MD/CD ratio	1.36	1.77	3.00	3.50	4.13

then successfully been used to model light scattering in anisotropic structures such as biological tissue [6–8], softwood [9] and textile [10]. We have previously used a similar model to compare scattering by spherical and cylindrical particles [11] as well modeling anisotropic scattering in paper [12, 13].

The focus of this work is to investigate the anisotropic light scattering effects in paper for both spatially resolved reflectance and transmittance. Light scattering patterns from a set of paper samples are obtained experimentally from papers with different degrees of in-plane fiber alignment. Experimental results are compared to simulations where multiple scattering and directional dependent scattering is considered. The scatterers, or fibers, are described as infinitely long cylinders and their orientation has been estimated from x-ray computed tomography images.

2 Method

2.1 Materials

A set of paper sheets was made by MoRe Research, Örnsköldsvik, Sweden, where fiber material, area density and thickness was kept constant and only the fiber alignment was varied. The degree of fiber alignment was varied from randomly oriented fibers, i.e., isotropic structural properties, to high fiber orientation along one axis. The sheets were made from unbleached chemical pulp from Swedish pine and in order to obtain a purely fibrous material, the pulp was filtered to remove any fine material. Furthermore, the sheets were not calendared or compressed in any way which leaves the fibers at their natural cross-section and therefore we assume that the fibers will retain a circular shape. The average sheet thickness is 0.19 mm and the average area density 102.5 g/m^2, which correspond to a sheet density of $\rho_s = 539.5\ kg/m^3$. This in turn corresponds to a porosity of 64% if the density for cellulose is assumed to be $\rho_c = 1500.0\ kg/m^3$. Measurements on the tensile strength of the paper sheets, which closely relate to the fiber orientation, where also provided by the manufacturer. Table 1 shows the differences in tensile strength in the machine-direction (MD) and cross-direction (CD) along with the ratio between them. The paper sheets span from nearly isotropic fiber orientation (sample 1) to highly aligned fiber orientation (sample 5).

2.2 Fiber orientation estimation using x-ray computed tomography

To be able to correlate the simulations to the experimental results we need to determine the papers three-dimensional micro-strucure. In the presented work we obtained three-dimensional images of the paper by using x-ray computed tomography (CT) where x-ray absorption is directly related to the local density. The CT-images were captured using a Bruker Skyscan 1172 at the division of applied mechanics, Uppsala University. The voxel size was $2 \times 2 \times 2$ μm^3 which enabled us to resolve the spatial extent of single fibers as well as their lumen.

Samples from each paper sheet were cut to roughly 3×10 mm^2 and then folded into cubes to fit the field of view of the CT-scanner. After imaging and reconstruction, the orientation in the samples was estimated. The procedure is based on the image gradients, which in turn are averaged using the structure tensor approach. Such methodology has previously shown to be suitable for cellulose fibers [14]. For each voxel, the orientation is "at last" found as the direction of the eigenvector corresponding to the smaller eigenvalue of the structure tensor. The orientation is finally summarized for regions of about $1.3 \times 1.3 \times 0.1$ mm^3.

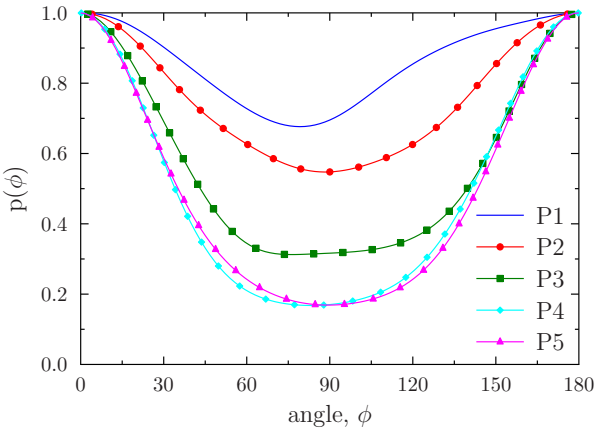
The distribution functions of the fiber orientation can be seen in Fig. 1(a) and a sample image from the reconstruction can be seen in Fig. 1(b) where pseudo-coloring is used to visualize the estimated fiber orientations.

2.3 Light scattering measurements

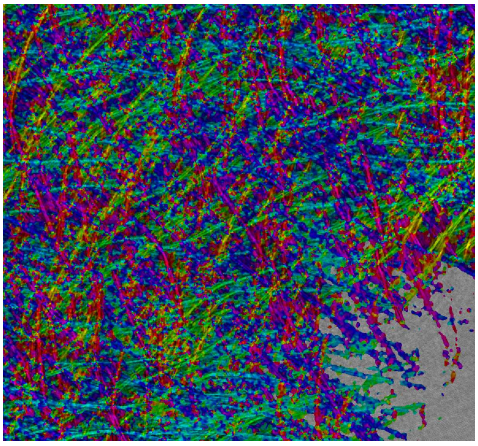
To record light scattering patterns we used a 10 mW 633 nm HeNe laser source focused on the surface of the paper using a lens system, see Fig. 2. For both the transmittance and reflectance measurements a numerical aperture of $NA = 0.005$ was used for the incident laser source. The focal length was set to 200 mm, which provides enough room for a beam splitter in the reflectance measurement along with a theoretically calculated beam waist of 81 μm and a depth of focus of 16 mm. The spatially resolved transmittance and reflectance is detected using a 16 bit CMOS camera, a PCO Edge 4.2 with a Micronikkor 55 mm lens. The exposure time was adjusted for each measurement point to utilize the full dynamic range of the camera and each recorded image was normalized with the maximum pixel value. The intensity patterns were measured on each of the five paper sheets by capturing 12 points along a 12 mm line which was placed at six arbitrary positions, totaling 72 measurement points for each paper sheet.

2.4 Monte Carlo Simulations

The simulation tool used in this work is based on the Monte Carlo method. It is a stochastic numerical method based on the assumption that the electromagnetic field can be regarded as a large number particles that move between scattering events until they all have left the medium or been absorbed. Phase functions from an analytical solution of scattering by infinite cylinders are utilized to describe the scattering behavior of the fiber



(a) Fiber distribution functions



(b) Fiber structure

Figure 1: Distribution functions of the fiber alignment estimated with x-ray computerized tomography for each paper sheet in (a). High resolution $2.6 \times 2.4\ mm^2$ color image of sample 3 from the reconstructed micro-tomography in (b), can be zoomed on a computer screen.

structure. A detailed description of the implementation of the MC simulation technique with infinite cylinders as scatterers can be found in Yun et al. [15].

A cylinder illuminated at perpendicular incidence will scatter light in the shape of a disc around the longitudinal axis of the cylinder. Oblique incident light scatters in

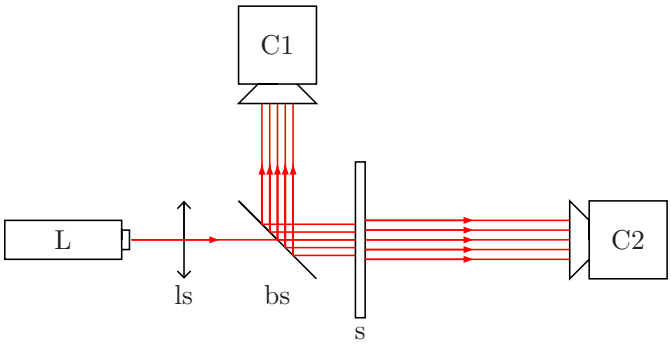


Figure 2: Illustration of the experimental setup where light from the laser, L, is focused through the lens system, ls, and beam splitter, bs, onto the sample, s. The spatially resolved reflected and transmitted light is detected at camera positions C1 and C2 respectively.

the shape of a cone around the longitudinal axis of the cylinder and the phase function describes the intensity distribution in these discs or cones. The cones half-angle toward the longitudinal axis of the cylinder equals that of the incident light [3]. This behavior is the origin of a directional dependency for multiple scattering observed in structures where the fibers are aligned [4].

The sequential random walk between scattering events follow the Beer-Lambert law and directly relate to the concentration of scatterers in the medium. As the paper sheets where custom made, with a very low amount of small scattering particles, we assume that the only scatterer is the cellulose fibers. Hence, the cylinder density is calculated as

$$C_a = \frac{\rho_s}{A\rho_c}, \tag{1}$$

where ρ_s is the density of the paper sheet, ρ_c the density of cellulose and A is the cross-sectional area the cellulose fibers. The density of the paper sheets is $\rho_s = 539.5\ kg/m^3$ and the density for cellulose is $\rho_c = 1500.0\ kg/m^3$. A typical dimension of a cellulose fiber from pine has an outer diameter of 35 μm and a lumen diameter of 30 μm [16]. This would correspond to a cylinder density of $C_a = 1.41 \cdot 10^9\ m^2$ which is consistent with the values for C_a achieved in our previous work [13] where C_a was estimated iteratively using a Gauss-Newton method to match total transmission and reflectance values.

The refractive index of the fibers was assumed to be $n = 1.55$ [17] and the background medium air is $n = 1.0$. The parameters used to calculate the phase functions and scattering efficiency Q_s of the cylinders were the diameter $d = 35.0\ \mu m$, the refractive

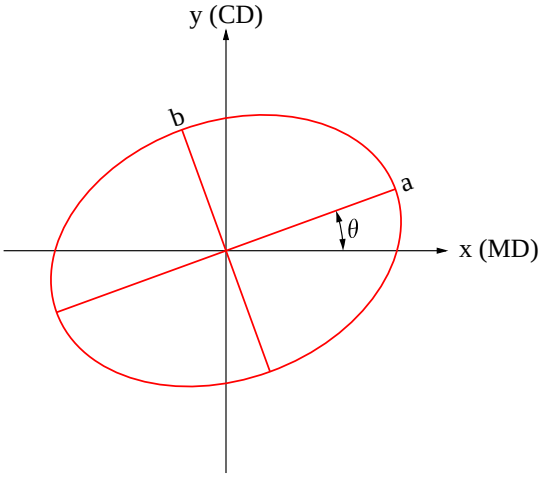


Figure 3: Ellipse where a and b are the major and minor axis respectively and θ is the angle between the ellipse major axis and the machine-direction.

index missmatch $n = 1.55$ and the wavelength $\lambda = 633\ nm$. The scattering coefficient is computed as

$$\mu_s(\zeta) = C_a d Q_s(\zeta), \tag{2}$$

where Q_s the scattering efficiency which depends on the incident angle toward the cylinder ζ. The dependence of ζ means that the scattering coefficient will be orientation dependent in aligned fiber structures in addition to the phase function. The absorption coefficient was for simplicity set to $\mu_a = 0$ as absorption in paper sheets usually is weak. The structure in the xy-plane was modeled with the fiber orientation distribution functions $p(\phi)$ obtained from the CT-scanning, see Fig. 1(a). The main fiber axis, $\phi = 0°$, was set to follow the x-direction. The distribution in the depth, z-direction, was modeled following a Gaussian distribution with standard deviation $\sigma_z = 10°$ to relate to the interconnectivity between fiber layers. Further it is assumed that each fiber scatters light independently meaning that the mean free path between scatterers is assumed to be much longer that the wavelength.

The obtained phase functions and cylinder density C_a together with the fiber alignment distribution functions obtained from the CT-images are then used to simulate the spatial intensity for each of the measured samples. The light source was modeled as a circular Gaussian profile with width 80 μm and the thickness was set to 0.19 mm. Only scattered particles with a propagation direction within 1.2° of the z-direction are detected. This value is close to the angle detected by the camera which is estimated to be between 1.4–2° depending on the aperture.

2.5 Ellipticity of iso-contour patterns

The measured and simulated iso-intensity curves are evaluated by fitting intensity levels into ellipses using the least squares criterion. The equation of an ellipse is

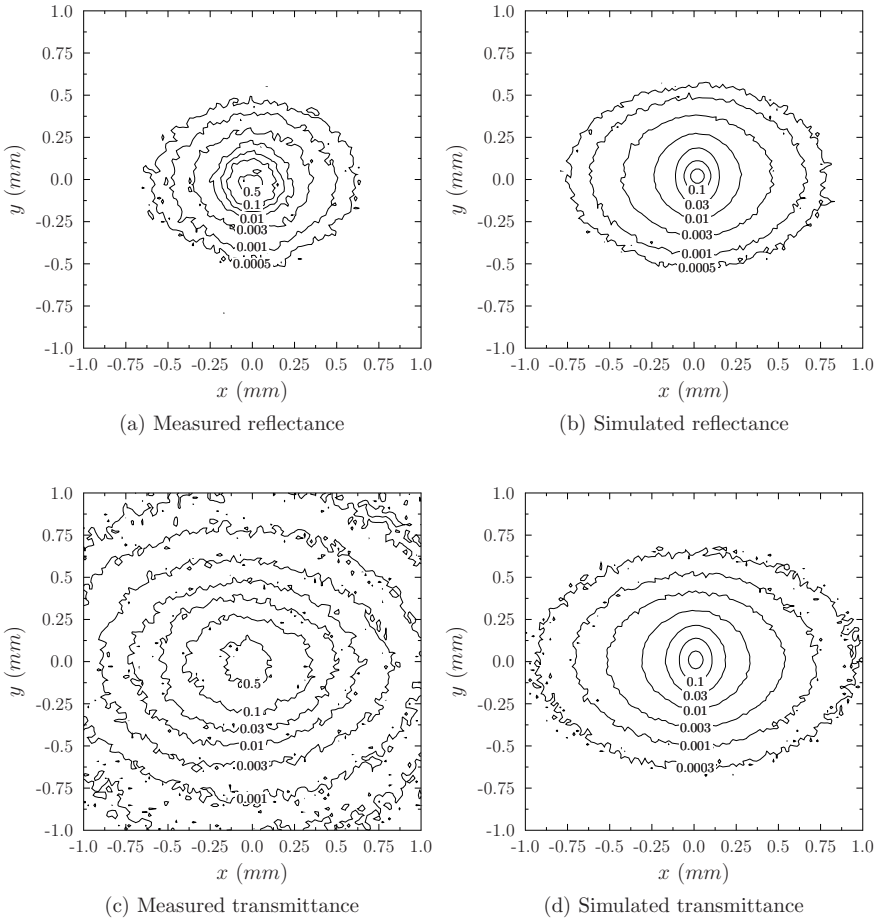
$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0, \tag{3}$$

where the coefficients A, B, C, D, E and F describes the size, shape and position. After solving the system it is possible to calculate the orientation and axes of the ellipse, see Fig 3. The ratio between major axis, a and the minor axis, b will be compared to the MD/CD-ratio of the tensile strength.

Another important property in for example paper is the amount of fiber flocs, or small-scale area density variations [18]. This can for example be measured by studying the amount of transmitted light through the paper sheet [16]. As the exposure time of the camera was adjusted for each measurement image, we chose to consider the area of the ellipses, $A_{ellipse} = \pi ab$, at the different relative intensity levels. The area of the ellipses thus provides an indication of the optical thickness of the material.

3 Results

Figure 4 shows the iso-intensity curves for both reflected and transmitted light for the paper sheet with the highest degree of fiber alignment, sample 5. Note that the main fiber direction, or machine-direction, is referred to as the x-direction and the cross-direction is referred to as the y-direction. An overall similar characteristic behavior can be observed in the scattering patterns for both the measurements and MC simulations. At high relative intensity levels the measured iso-intensity patterns exhibit a circular shape, while simulations have an elliptical shape elongated in the y-direction, perpendicular to the main fiber direction. The simulated elongation in the y-direction originates from incident light being scattered by cylinders whose orientation is distributed toward the x-direction. When the incident light is scattered for the first time the angle between the cylinder and light propagating direction has a high probability of being close to $\zeta = 90°$. So when the cylinders have an alignment toward the x-direction, scattering will be dominant in the y–z-plane due to the scattering characteristics of a cylinder. At lower intensity levels an increasing elliptical elongation toward the x-direction is observed in both the experimental and simulated intensity patterns. Simulated multiple scattered light propagates foremost in the direction of the main fiber direction, x-direction. This is due to light scattering in the shape of a cone around the cylinder. This aligns the light propagation along the cylinder axis, which in turn produces the elliptical iso-intensity patterns elongated in the x-direction at low relative intensity levels. Even though it is not specifically shown here it was found that lower values of $\sigma_z < 10°$ will enhance the elliptical shape in the y-direction at high intensity levels. This in turn slightly shifts the elliptical shape in the x-direction to occur at even lower intensity levels. For values $\sigma_z > 10°$ the elliptical shape in both the y-direction and x-direction will decrease. This is obvious as lateral diffusion will decrease if the alignment of the cylinders are not in



(a) Measured reflectance

(b) Simulated reflectance

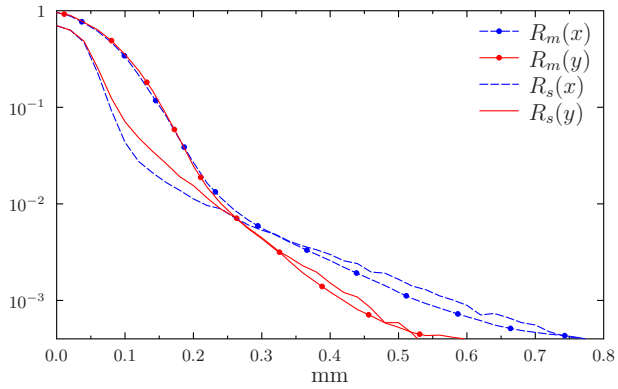
(c) Measured transmittance

(d) Simulated transmittance

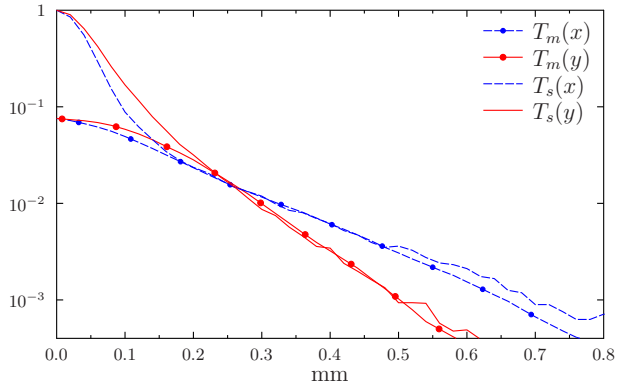
Figure 4: Iso-intensity curves for a measurement point on the paper sheet with the highest degree of fiber alignment, sample 5, in reflectance (a) and transmittance (c). Simulated iso-intensity curves for the corresponding sample is found in (b) and (d), respectively.

the plane, i.e., the scattering cones will not sustain propagation in the x–y-plane. While this parameter does affect the appearance of the scattering pattern it still does not do so to a very high degree and for values close to $\sigma_z = 10°$.

In order to compare experimentally determined and simulated scattering patterns more quantitatively the spatially resolved transmittance and reflectance along the x- and y-directions from the center of the light source is show in Fig. 5. The experimental



(a) Reflectance



(b) Transmittance

Figure 5: Spatially resolved reflectance (a) and transmittance (b) along the x- and y-directions for sample 5. The measured and simulated reflectance is denoted R_m and R_s and the measured and simulated transmittance is denoted T_m and T_s.

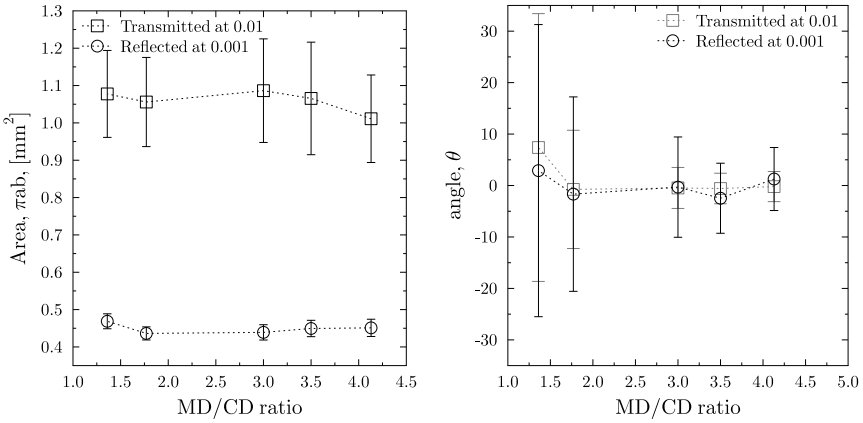
curves are averaged over all 72 measurement points and both the experimental and simulated reflectance and transmittance curves intensity maxima were normalized to enable comparisons with the lower intensity levels. The intensity of the measurements and simulations do not agree well at high relative intensity levels close to the light source. At small distances from the center of the source, between 0 and 0.2 mm, the reflectance de-

termined from simulations R_s is smaller than that of the corresponding experiment R_m. The results on the spatially resolved reflectance, Fig. 5(a), are very similar to work done by Kienle et. al. [9]. Note that their work considered solid wood (softwood) which has a much more prominent fiber alignment where the spatially resolved reflectance spanned several millimeters whereas the work presented here is limited to distances below one millimeter. Their simulations also predicted good agreement to measurements at lower intensity levels even though there were slight differences close to the laser source. They argued that this is due to reflections from the surface which was not considered in the simulations as the topology of that surface was unknown. This can possibly apply to this work as well, however the distance over which the theoretical and experimental data does not fit indicate that the diameter of the incident light source was larger than our estimations. Figure 5(b) show the relationship between measured and simulated results for the transmitted intensity. The simulated intensity T_s close to the light source, between 0 and 0.2 mm, is a lot larger than the measured intensity T_m. A reasonable explanation for this large difference is that the wavefront of the incident light source is not considered in the simulations. The small numerical aperture of the focusing lens induces small oblique angles for the incident photons, this would in turn greatly reduce the number of detected ballistic photons. Note also that in the measured transmittance in Fig. 5(b) it is possible to see indications of a elliptical shape in the y-direction at high intensity levels, between 0.1 and 0.2 mm from the source. It is however not nearly as apparent as in the simulated transmittance. The scattered spatial intensity at distance greater 0.2 mm from the center of the incident light source was found to agree very well between simulations and measurements. This indicates that the model correctly describes multiple scattering in the fiber structure.

Figure 6(a) shows the calculated area from the ellipse fitting in relation to the tensile strength MD/CD-ratios for the five different paper sheets. The center value is the average of 72 measurement points and the error bars are two standard deviations wide for each of the paper sheets. The area of the chosen intensity levels fluctuates a lot more for transmitted light than the reflected area. This is most likely due to transmitted light being more affected by thickness variations in the paper sheets. This was confirmed by measurements on two paper sheets stacked on top of each other (not specifically shown) where the size of the transmittance light intensity patterns increased in size while no apparent difference was found in the size of the reflectance light intensity patterns.

Figure 6(b) shows how the ellipse elongation angle θ vary in relation to the x-direction. This provides information about variations of the fiber alignment in the structure. A decrease in fluctuations of θ can be observed for samples with higher degree of fiber alignment. This is reasonable since a smaller elliptical elongation of the scattered intensity pattern is more sensitive when the ratio a/b is close to unity. This also partly explains why the standard deviation is lower in the transmittance measurements. Another explanation is that the reflectance measurements are more localized, fewer fibers scatter the light in comparison to the transmitted measurements.

The measured elliptical shape of the iso-intensity curves for each of the different paper sheets was found to show a strong correlation to the fiber alignment. Figure 7 show the



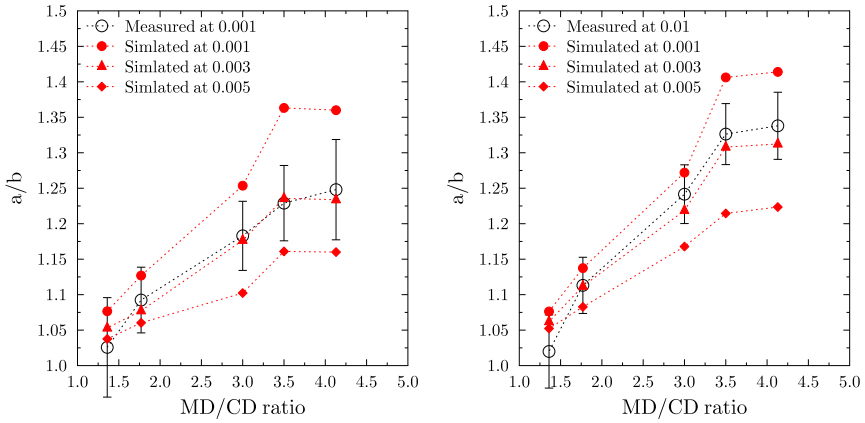
(a) Ellipse area (b) major axis angle

Figure 6: Area of the ellipses matched into iso-contour patterns at relative intensity levels 0.01 and 0.001 in (a) and the angle between the ellipse major axis and the machine-direction in (b). The area of the ellipse indicate sample thickness or density is and the major axis angle indicates the variability of the main fiber orientation.

measured and simulated a/b ratios in relation to the tensile strength MD/CD ratios. The analyzed relative intensity levels were 0.001 for the iso-contours in reflectance Fig. 7(a) and 0.01 for transmittance Fig. 7(b). The corresponding intensity levels obtained from simulations are different due to the differences in intensity near the source between simulations and experiments. This is most apparent in the transmittance measurements, however, the experimentally determined elliptical scattering patterns at relative intensity level of 0.01 and the simulated at relative intensity 0.003 have nearly the same area. The overall behavior of the experiments and simulations is very similar, a close to proportional relationship between tensile strength and elliptical shape is observed. Both measurements and simulations based on CT-images found little difference between the two samples with the highest degree of fiber alignment. This indicates that the tensile strength measurement of sample 4 possibly can be slightly off. Note however that the CT-images were restricted to small measurement volumes and can therefore posses a fairly large error in relation to the fiber orientation of the whole paper.

4 Discussion and Conclusion

Measurements on spatially resolved transmittance and reflectance patterns on paper sheets were found to have a strong anisotropic shape elongated toward the main fiber orientation at low intensity levels. The experimental results were compared to Monte



(a) Reflectance (b) Transmittance

Figure 7: The x-axis show the tensile strength MD/CD ratio and the y-axis show the a/b ratio of the ellipse fitting where the bar is two standard deviation wide. The ratio a/b of 72 measurement points for each of the five paper sheets is taken at relative intensity level 0.001 for reflectance (a) and at relative intensity level 0.01 for transmittance (b).

Carlo simulations that considered the micro-structure of the fiber material. In the theoretical model we consider fibers to be described by long cylinders distributed in the structure following probability distributions estimated using CT-images. By using the average diameter of a pine fiber and the refractive index of cellulose we calculated the scattering behavior of the cylinders with an analytical solution of Maxwell's equations. The only free parameters, that were not estimated in any way, are the fiber distribution in the thickness direction (z-direction) which was modeled as a Gaussian distribution following a standard deviation with $\sigma_z = 10°$ and the absorption coefficient which was set to $\mu_a = 0$.

Good agreement between experimental and simulated results was found for diffused light sufficiently far from the laser source. Large differences can however be observed close to the light source which at least partly are due to the profile and width of the incident laser beam. Measurements on the ellipticity of relative intensity patterns at level 0.01 in transmittance and 0.001 in reflectance show a strong elongation toward the main fiber axis. The iso-intensity contours followed the same overall characteristics in anisotropy both in measurements and simulations far from the laser source. These results indicate that anisotropic light diffusion in paper originates from multiple scattering between fibers and not from light guiding within the hollow lumen of the cellulose fibers. This has also been suggested for other fiber-based materials such as biomedical tissue [6] and wood [9].

The results can be considered good as paper is a relatively complex material and the model is relatively simple where for example effects from contactpoints between fibers and size variations of the fibers are ignored. Further it must also be mentioned that fibers are not ideal homogeneous cylinders, they can be curved, have irregular shapes or surface roughness likely to cause some of the scattering to occur outside the theoretical scattering cone. This is a reasonable explanation to why the elliptical intensity contour patterns in the y-direction are harder to observe in the measurements. Note that the paper sheets in this work had a porosity of about 64% and contained very low amounts of small scattering particles. The addition of small particles or other filler materials is expected to reduce the anisotropic diffusion observed in this work.

The small variation in area of the elliptical scattering pattern for the reflected light indicates that most scattering occur in the layers of fibers closest to the light source. This due to the diffused reflected light propagating foremost in the fiber layers close to the light source while the transmitted light propagates through all layers. Simultaneous measurements of both spatially resolved transmittance and reflectance from the same laser source therefore has a potential to detect fiber orientation variations between the top and bottom layers of the sheets. This would apply for both the ellipse axis ratio a/b and the angle between the ellipse major axis and the machine-direction, θ. Another extension to such a technique could be to use the degree of depolarization in reflectance measurements to retrieve information about variations in the different layers.

In conclusion, we have shown a strong relation between tensile strength and anisotropic shape in spatially resolved reflectance and transmittance patterns at small relative intensity levels. Both measurements and simulations agree well for multiple scattered light even though differences close to the light source were found. These findings indicate that parameters such as fiber orientation and variations in fiber orientation can be measured on-line by monitoring scattered light in various fiber-based materials using a camera with a high dynamic range. Also, as the reflected spatial intensity only starts to show the elliptical shape at very low relative intensity level, none or very little anisotropic influence should be expected for optical dot gain in paper. Further, the method of estimating the fiber orientation distribution functions from CT-images proved excellent to get valid input parameters for the simulations. A suggestion for a possible extension of the present work is to use CT-images to estimate fiber dimensions, cylinder density and fiber alignment in the thickness direction to validate light scattering models even further. It is also of interest to investigate the anisotropic light diffusion in fiber-based materials mixed with smaller particles.

5 Acknowledgments

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Paper E

Light scattering characteristics by infinitely long cylinders with a hollow core

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To be submitted.

Light scattering characteristics by infinitely long cylinders with a hollow core

Tomas Linder and Torbjörn Löfqvist

Abstract

There is a growing interest in finding more realistic phase functions for light scattering models utilizing particle level scattering. The light scattering behavior of cylindrical particles with a hollow core is therefore investigated and compared to homogeneous cylinders. The cylinder wall was found to generate a increasing amount of resonance modes with increasing particle size. Especially one large additional peak was observed for hollow cylinders with a large inner radius. In general the differences between the phase functions, scattering efficiencies and mueller matrix elements, are small when a hollow core is introduced. It can therefore be concluded that it is, in most cases when considering multiple scattering, a valid approximation to ignore the hollow core of the particle.

1 Introduction

Over the last decade a growing interest has unfolded in employing phase functions derived analytically from Maxwell's equations when modeling multiple scattering. These phase functions take into account the geometrical shape of the scattering particles and plays a key role when modeling light scattering phenomena. It is therefore of interest to consider different phase functions impact on the scattering behavior in a wide variety of fields.

Phase functions from infinitely long homogeneous cylinders [1] has proven an efficient way of modeling light propagation in anisotropic fiber-structures. Kienle et al. originally suggested that cylinders can be used to describe anisotropic scattering in fiber-structures using the Monte Carlo method [2, 3]. Yun et al. later described in detail how infinite cylinders can be implemented into a Monte Carlo based simulation to track both scattered intensity and polarization [4]. Since then cylinders have been used to describe scattering in a wide variety of different anisotropic structures, such as biological tissue [5,6], textiles [7], wood [8] and paper [9, 10].

Fibers often are of inhomogeneous nature, recently Peng et al. used cylinders with different radial and axial refractive index to account for scattering properties in textile materials [11]. They found that the change in scattered intensity hardly was noticeable compared to a homogeneous cylinder. However, both polarization and scattering efficiency were affected greatly. Fibers from cellulose and some types of polyester fibers have a hollow cores. It is therefore of interest to investigate how a cylinder with a lumen affect the scattering properties of cylindrical particles.

In this work we will employ an analytical solution of electromagnetic scattering by an infinitely long concentric cylinder to investigate how the scattering is affected by cylinders with different properties in the core and wall regions. The analytical solution of electromagnetic scattering by a concentric cylinder at perpendicular incidence was originally solved by Kerker and Matijevic [12]. The solution was later extended for oblique incidence by G. A. Shah [13]. Yousif et al. [14] investigated the differences in scattering between homogeneous cylinders and cylinders with hollow and metallic cores. They found that the Mueller matrix elements M_{11} and M_{12} did not differ much between homogeneous and hollow cylinders with a small inner radius, but more for cylinders with a metallic core. Even though these solutions were made a long time ago, in the 60's, little work has been done on how concentric cylinders scatter light intensity or change its polarization compared to homogeneous cylinders.

The topic of this paper is to investigate to what extent the phase function and Mueller matrix elements differ between a concentric cylinder and a single homogeneous cylinder. Since Yousif et al. [14] focused on cylinders with a small inner radius this work will focus on particles with dimensions similar to cellulose fibers or polyester fibers, i.e larger particles with fairly thin fiber walls [15].

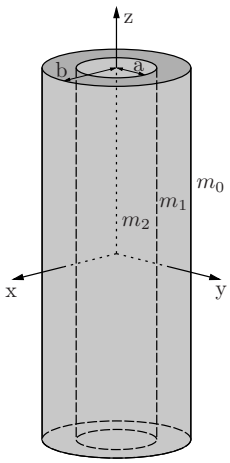


Figure 1: Concentric cylinder where a and b are the radii of the inner and outer cylinders. The refractive index are denoted m_0, m_1 and m_2 for the surrounding medium, outer cylinder and inner cylinder respectively.

2 Method

The geometry of the cylinder is shown in Fig. 1. It includes two infinitely long concentric cylinders of radii a and b, with refractive index m_2 and m_1 respectively. The region between the axis of the cylinder and a is the core while the region between a and b is the mantle of the cylinder.

A photon interacting with a cylinder will scatter in a cone around the longitudinal axis of the cylinder [1]. The scattering cone will have the half angle to the fiber longitudinal axis equal to the incident angle ζ. The phase function will describe in which directions in the cone the light will scatter for each incident angle. The solution of plane wave scattering by infinite concentric cylinders will in principle follow that of G. A. Shah [13]. The solution will be repeated for clarity in this paper with a few modifications to agree with the syntax of Bohren and Hoffman [1] and more recent work [4, 11].

2.1 Wave components in the particle

The scattered electric and magnetic fields u and v in the particle satisfies the scalar wave equation

$$\nabla^2\psi + k^2\psi = 0 \tag{1}$$

where $k = \frac{2\pi m_0}{\lambda}$ is the wavenumber. The scattering particles in this work are assumed to be non-magnetic, i.e. $\mu_1 = \mu_2 = 1.0$. The wave is resolved into two components, Case I which is the TM mode and Case II which is the TE mode. The potential functions in the different regions of the concentric cylinders are as follows:

TM polarization (Case I):

$$\left.\begin{aligned} u_I &= \sum_{-\infty}^{\infty} F_n(J_n(\eta r) - b_{nI} H_n(\eta r)) \\ v_I &= \sum_{-\infty}^{\infty} F_n(a_{nI} H_n(\eta r)) \end{aligned}\right\} \quad r > b \tag{2}$$

$$\left.\begin{aligned} u_I &= \sum_{-\infty}^{\infty} F_n(d_{nI}^{(1)} J_n(\eta_1 r) - b_{nI}^{(1)} H_n(\eta_1 r)) \\ v_I &= \sum_{-\infty}^{\infty} F_n(c_{nI}^{(1)} J_n(\eta_1 r) - a_{nI}^{(1)} H_n(\eta_1 r)) \end{aligned}\right\} \quad a < r < b \tag{3}$$

$$\left.\begin{aligned} u_I &= \sum_{-\infty}^{\infty} F_n d_{nI}^{(2)} J_n(\eta_2 r) \\ v_I &= \sum_{-\infty}^{\infty} F_n c_{nI}^{(2)} J_n(\eta_2 r) \end{aligned}\right\} \quad r < a \tag{4}$$

TE polarization (Case II):

$$\left.\begin{aligned} u_{II} &= \sum_{-\infty}^{\infty} F_n(-b_{nII} H_n(\eta r)) \\ v_{II} &= \sum_{-\infty}^{\infty} F_n(J_n(\eta r) - a_{nII} H_n(\eta r)) \end{aligned}\right\} \quad r > b \tag{5}$$

$$\left.\begin{aligned} u_{II} &= \sum_{-\infty}^{\infty} F_n(d_{nII}^{(1)} J_n(\eta_1 r) - b_{nII}^{(1)} H_n(\eta_1 r)) \\ v_{II} &= \sum_{-\infty}^{\infty} F_n(c_{nII}^{(1)} J_n(\eta_1 r) - a_{nII}^{(1)} H_n(\eta_1 r)) \end{aligned}\right\} \quad a < r < b \tag{6}$$

$$\left.\begin{aligned} u_{II} &= \sum_{-\infty}^{\infty} F_n d_{nII}^{(2)} J_n(\eta_2 r) \\ v_{II} &= \sum_{-\infty}^{\infty} F_n c_{nII}^{(2)} J_n(\eta_2 r) \end{aligned}\right\} \quad r < a \tag{7}$$

where the coefficients are

$$F_n = (-i)^n e^{-i\omega t - ihz + in\theta}, \tag{8}$$

$$h = k * \cos\zeta, \tag{9}$$

$$\eta = \sqrt{k^2 - h^2}, \tag{10}$$

$$\eta_1 = \sqrt{m_1^2 k^2 - h^2}, \tag{11}$$

$$\eta_2 = \sqrt{m_2^2 k^2 - h^2} \tag{12}$$

and the subscripts on the coefficients (1) and (2) denotes the mantle and core region respectively.

The fields outside the cylinder are due to to the superposition of the incident (first order Bessel function, J_n) and scattered (Hankel function, H_n) waves. For the field in the inner cylinder only a first order Bessel function is used to maintain the regularity at the origin. The field in the mantle region has a combination of the fields found in the core and surrounding. The potential function in the mante region was defined differently in Yousif et al. [14] where they used a second order Bessel function instead of the Hankel function. They will however produce the same constants in the scattered field after solving the system of equations.

2.2 Boundary conditions

The boundary conditions require that the tangential components of E and H fields are continuous at the interface between the core and mantle and the mantle and the surrounding. The derived tangential components of E and H for the θ and z components are as follows (se reference [13] for a complete derivation):

$$E_\theta = -\frac{\partial v}{\partial r} + \frac{inh}{mkr}u, \tag{13a}$$

$$H_\theta = m\frac{\partial u}{\partial r} + \frac{inh}{kr}v, \tag{13b}$$

$$E_z = \frac{m^2k^2 - h^2}{mk}u, \tag{13c}$$

$$H_z = (m^2k^2 - h^2)v. \tag{13d}$$

At the outer cylinder ($r = b$):

$$E_z^i + E_z^s = E_z^{(1)} \tag{14a}$$

$$E_\theta^i + E_\theta^s = E_\theta^{(1)}, \tag{14b}$$

$$H_z^i + H_z^s = H_z^{(1)}, \tag{14c}$$

$$H_\theta^i + H_\theta^s = H_\theta^{(1)}. \tag{14d}$$

At the inner cylinder ($r = a$):

$$E_z^{(1)} = E_z^{(2)}, \tag{15a}$$

$$E_\theta^{(1)} = E_\theta^{(2)}, \tag{15b}$$

$$H_z^{(1)} = H_z^{(2)}, \tag{15c}$$

$$H_\theta^{(1)} = H_\theta^{(2)}. \tag{15d}$$

Note that the incident electric field E_z^i equals zero in the TM case and the incident magnetic field H_z^i equals zero in the TE case. This can be seen in equations 2 and 5 where the incident first order Bessel function is missing in v_I and Hankel function in u_{II}. The application of the four boundary conditions at the two boundaries gives rise to a linear system consisting of eight independent algebraic equations for both the TM and TE cases. The unknowns are a_{nI}, b_{nI}, $a_{nI}^{(1)}$, $b_{nI}^{(1)}$,$c_{nI}^{(1)}$, $d_{nI}^{(1)}$, $c_{nI}^{(2)}$, $d_{nI}^{(2)}$ and a_{nII}, b_{nII}, $a_{nII}^{(1)}$, $b_{nII}^{(1)}$,$c_{nII}^{(1)}$, $d_{nII}^{(1)}$, $c_{nII}^{(2)}$, $d_{nII}^{(2)}$ for the TM and TE case respectively. This is sufficient in order to solve the constants in the scattered fields a_{nI}, b_{nI}, a_{nII} and b_{nII}.

2.3 Far-field scattering solution

The coefficients a_{nI}, b_{nI}, a_{nII} and b_{nII} are then used to determine the scattering matrix in the far-field scattering region of the particle. In the far-field region the Hankel functions can be replaced by their asymptotic expressions [1]. This can be used to derive the scattering matrix elements (T's) as follows:

$$T_1 = \sum_{-\infty}^{\infty} b_{nI} e^{-in\Theta} = b_{0I} + 2\sum_{n=1}^{\infty} b_{nI}\cos(n\Theta), \tag{16a}$$

$$T_2 = \sum_{-\infty}^{\infty} a_{nII} e^{-in\Theta} = a_{0II} + 2\sum_{n=1}^{\infty} a_{nII}\cos(n\Theta), \tag{16b}$$

$$T_3 = \sum_{-\infty}^{\infty} a_{nI} e^{-in\Theta} = -2i\sum_{n=1}^{\infty} a_{nI}\sin(n\Theta), \tag{16c}$$

$$T_4 = \sum_{-\infty}^{\infty} b_{nII} e^{-in\Theta} = -2i\sum_{n=1}^{\infty} b_{nII}\sin(n\Theta). \tag{16d}$$

where $\Theta = \pi - \phi$, where ϕ is the scattering angle. They relate to the scattered electric field parallel and perpendicular to the scattering plane in matrix form as

$$\begin{pmatrix} E_{\parallel s} \\ E_{\perp s} \end{pmatrix} = e^{i3\pi/4}\sqrt{\frac{2}{\pi kr\sin\zeta}}e^{ik(r\sin\zeta - z\cos\zeta)}\begin{pmatrix} T_1 & T_4 \\ T_3 & T_2 \end{pmatrix}\begin{pmatrix} E_{\parallel i} \\ E_{\perp i} \end{pmatrix} \tag{17}$$

The scattering elements can also be used to derive the Mueller matrix elements which describe the behavior of both scattered intensity and polarization. Even though the Mueller matrix has 16 elements some symmetry relations reduces the number to 10 and the change in polarization in Stokes formalism can be written as

$$\begin{pmatrix} I_s \\ Q_s \\ U_s \\ V_s \end{pmatrix} = \begin{pmatrix} M_{11} & M_{12} & M_{13} & M_{14} \\ M_{12} & M_{22} & M_{23} & M_{24} \\ -M_{13} & -M_{23} & M_{33} & M_{34} \\ M_{14} & M_{24} & -M_{34} & M_{44} \end{pmatrix}\begin{pmatrix} I_i \\ Q_i \\ U_i \\ V_i \end{pmatrix} \tag{18}$$

2.4 Verification

Scattering by concentric cylinders was verified with previously published data on both homogeneous cylinders and concentric cylinders. Note that the summations in the potential functions and scattering matrix elements where terminated after $n \leq NTB$ where $NTB = kb + (kb)^{\frac{1}{3}} + 2$.

The phase function, efficiency in scattering and extinction shows excellent agreement with calculations on homogeneous cylinders when parameters $a = b$ or $m_1 = m_2$. It may be important to note that this solution reduces to a homogeneous cylinder when the parameters are set to $a = b$ or $m_1 = m_2$. Similarly the solution reduces to Kerker's solution at normal incidence when $\zeta = 90.0°$.

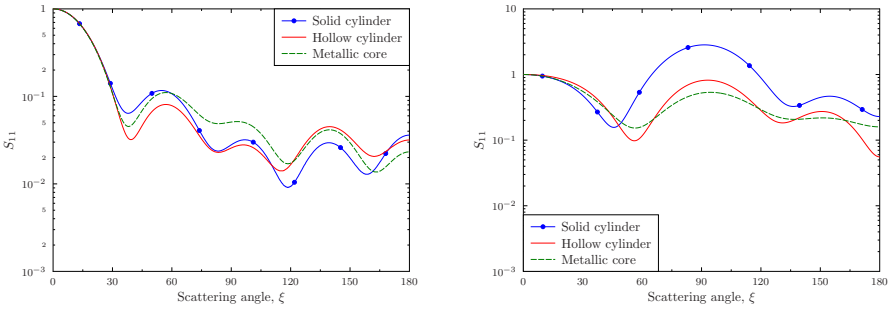
Generated numerical results where also compared to that of Evans et al. [16] who calculated scattering efficiencies for hollow cylinders at normal incidence. We where able to reproduce their tabulated values down to the third decimal. Excellent agreement with the data of extinction for scattering light at oblique incidence presented by G. A. Shah was also found [13]. Perfect agreement was also found comparing Mueller matrix elements s_{11} (see Fig. 2) and s_{12} of concentric cylinders with hollow and metallic cores presented by Yousif et al. [14].

2.5 Numerical simulation tool

The code for scattering by a concentric cylinder has been included into the FiberMC package available at (http://fibermc.sourceforge.net/). It is a Monte Carlo simulation tool designed to calculate multiple scattering by turbid fibrous media. Additional solutions to both homogeneous spheres and infinite homogeneous cylinders are also available.

2.6 Particle size distributions

Media consisting of scattering particles rarely have a uniform distribution of particle shapes and sizes. Apart from that, scattering resonances in Mie theory are also present in



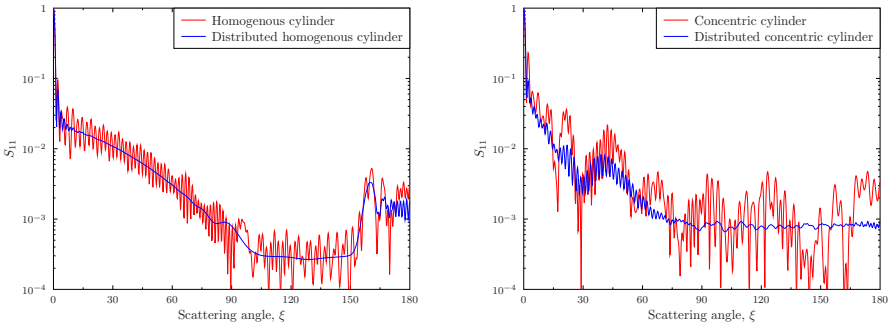
(a) $\zeta = 90.0$ (b) $\zeta = 30.0$

Figure 2: Mueller matrix element s_{11} for three different cylindrical particles; a homogeneous cylinder ($m_1 = m_2 = 1.6$), a hollow cylinder ($m_1 = 1.6, m_2 = 1.0$) and a cylinder with a metallic core ($m_1 = 1.6, m_2 = 1.4-50i$). Reproduction of the results in reference [14] where the particle dimensions are $ka = 0.8$, $kb = 4.0$.

scattering by both cylinders and concentric cylinders. This makes it difficult to observe differences in the scattering for larger particles and additional resonance modes that will be introduced by the fiber wall. We therefore look at the generated scattering by averaging over some fairly narrow particle size distribution. This is done by defining a center radius c which lie in between the inner radius a and outer radius b where the fiber wall thickness $\tau = b - a$ is following a Gaussian distribution with standard deviation σ_t. Figure 3 illustrates the phase functions of a homogeneous cylinder and a hollow cylinder with and without a size distribution.

3 Results

The influence of a fiber wall on the scattering phase function at $\zeta = 90°$ and wavelength $\lambda = 633$ nm for four different fibers can be seen in Fig. 4. The three hollow cylinders all have the same outer radius but different wall thicknesses. While the resonances for a homogeneous cylinder are almost completely smoothened out when the particle size is distributed that is not the case for a hollow cylinder. The presence of a fiber wall introduces complex resonances that vary with the thickness of the fiber wall producing additional scattering lobes. A large scattering lobe is related to the wall thickness, except when the inner radius either is very large or very small. The phase function also becomes more narrow the thinner the wall thickness is, meaning that particles with a larger inner radius in general will have a stronger forward scattering. This includes the position of the larger scattering lobe observed. For example, in the simulated case in Fig 4 the peak of the lobe moves from scattering angle $\xi \approx 50°$ for a fiber with wall thickness $t = 5$ μm



(a) Homogeneous cylinder (b) Hollow cylinder

Figure 3: Phase functions for homogeneous cylinder 3a and hollow cylinder 3b. The incident angle is $\zeta = 90°$, wavelength $\lambda = 633$ nm with size parameters $a = 8.5$ and $b = 11.5$ for the single particle size; and $c = 10$, $t = 3$ and $\sigma_t = 0.5$ for the size distribution.

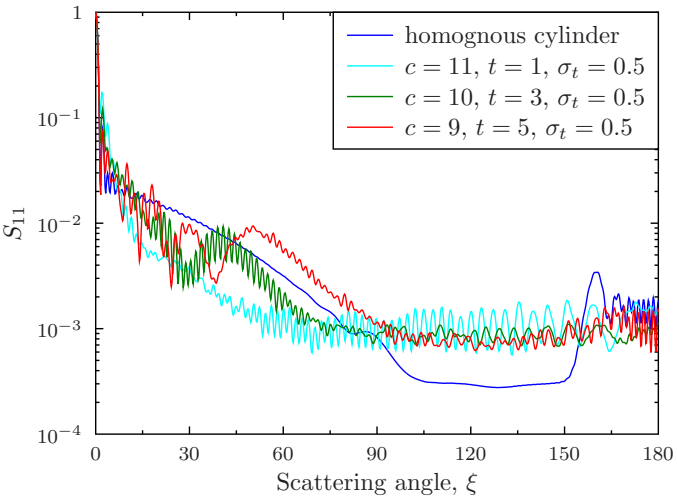
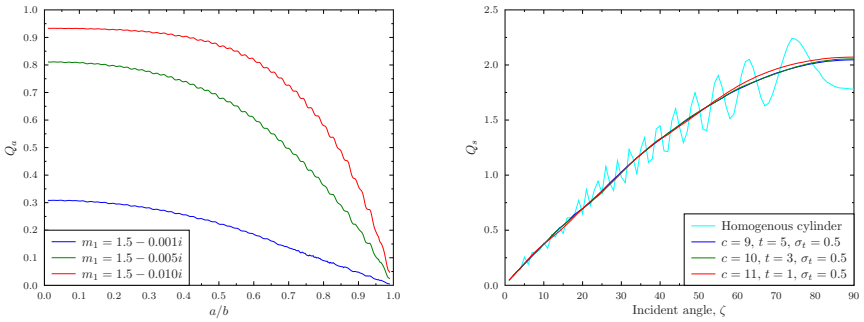


Figure 4: Phase functions for four different hollow cylinders.

to $\xi \approx 40°$ when the fiberwall is $t = 3\ \mu m$. One interesting aspect on larger particles is that the backscattering effects (fog bow and glory scattering) are not present for hollow particles.

Fig 5a shows the difference in absorption efficiency when introducing a complex part in the refractive index for hollow cylinders with fixed outer radius $b = 10$. The amount of absorbed energy by the scatterer largely depends on the thickness of the fiberwall, the thinner the less energy is absorbed. This is obvious as the waves propagates through less absorbing material.

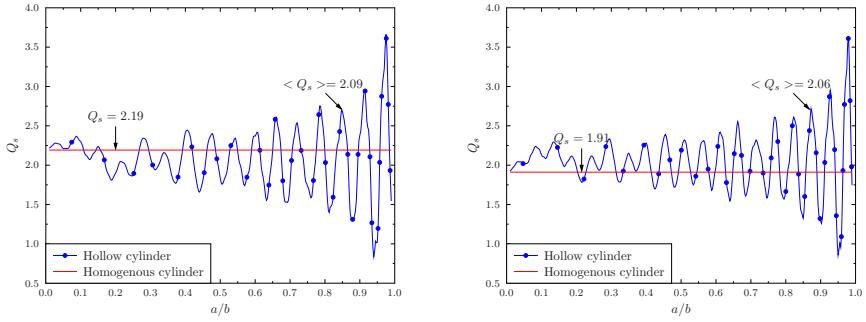
Fig 5b shows how the scattering efficiency varies with the incident angle for a few different fibers. No difference can be observed for distributed particle size regardless of the size of the hollow core. However, the scattering efficiency is also investigated by having outer radius b fixed while the inner radius a is varied. The influence from the fiber wall cause the scattering efficiency to alternate between scattering more and less efficient than a homogeneous cylinder. However, as the scattering efficiency for particle size distributions was found to vary on the second decimal. This means that in a general non-monodisperse medium no variation in scattering efficiency is expected between homogeneous and hollow cylinders.



(a) Q_a (b) Q_s

Figure 5: The change in absorption efficiency in 5a when the outer radius is fixed and the inner radius is variable and the change in scattering efficiency for different incident angles 5b.

Fig 7 shows the Mueller elements for the same fiber parameters as in Figure 4, except at different incident scattering angles. It can be seen that there are slight differences in the way the polarization is affected. The same scattering lobe as in the phase function can be seen in mueller elements $M12$, $M33$, $M34$ and $M44$. When comparing the Mueller elements to a homogeneous cylinder (not specifically shown) we also noticed that some of the elements at the diagonal ($M11$, $M22$, $M33$, $M44$) where all slightly narrower for a hollow particle.



(a) $b = 10$ (b) $b = 12$

Figure 6: Scattering efficiencies Q_s showing the oscillations from the fiber wall for two cylinder with different outer radii.

4 Discussion and Conclusion

Light scattering from a concentric cylinder with a hollow core was compared to that of a homogeneous cylinder. By distributing the particle thicknesses it is possible to identify the general difference in how the particles scatter light and compare to theory of homogeneous cylinders.

It was found that the fiber wall generates a large amount of additional resonance modes, especially when the wavelength is a lot smaller than the particle size. Also, the forward scattering was found to increase as the fiberwall thickness decreased. This indicates that it often can be a valid approximation to use homogeneous cylinders when modeling multiple scattering, especially if the wall thickness is large. It shall also be noted that the conical scattering by a cylindrical particle also enhances forward scattering and thereby reducing the effects of the phase function further. Considering that larger fibers often have other inhomogeneities in particle shape, surface roughness and refractive index, none of which are considered in this work, also motivate such an approximation.

The scattering efficiency was also found to be affected by the introduction of a hollow core. It also resonances with the fiber wall making the particle scattering more or less depending on the fiber wall thickness. However when a distribution of particle sizes and fiber wall thickness's are considered no significant difference will appear in scattering efficiency compared to a homogeneous cylinder. The absorption is however affected to a larger extent depending on the refractive index. It is quite obvious that a fiber with a larger core radius will absorb less as there is less material where absorption can take place.

The shape of the Mueller matrix elements all follow the same behavior as a homogeneous cylinder but there are differences. The same scattering lobe as in the phase function can be seen in mueller elements $M12$, $M33$, $M34$ and $M44$. Elements at the diagonal more narrow causing a quicker depolarization than a homogeneous cylinder.

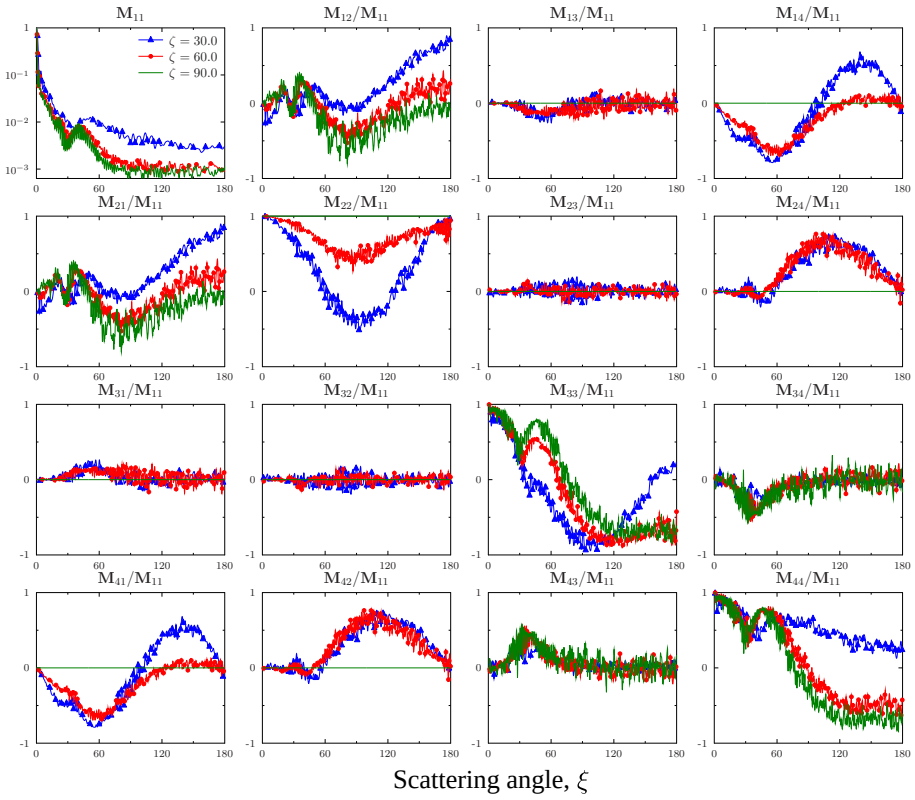


Figure 7: Mueller matrix elements

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